

# LARGE ERDŐS–STRAUS WITNESSES II: THE MULTIPLIER MODULUS IS INFINITELY OFTEN $\exp((\log p)^{1/4-o(1)})$

ANONYMOUS

**ABSTRACT.** For a prime  $p$  let  $W(p)$  be the least modulus  $M \equiv 3 \pmod{4}$  of a “multiplier” congruence that yields a solution of  $4/p = 1/x + 1/y + 1/z$ . We prove  $W(p) \geq \exp(c(\log p)^{1/4}(\log \log p)^{-1/4})$  for infinitely many primes  $p$  in Mordell’s hard classes, assuming Gallagher’s prime number theorem for the family of all characters and the Nair–Tenenbaum bound for short sums of multiplicative functions of polynomials; more precisely, the least hard prime with  $W(p) > T$  is at most  $\exp(O((\log T)^4 \log \log T))$ . The proof quarantines  $n$  to a *random square class* modulo a growing modulus (the residues of the multiplier congruences are Jacobi non-residues, so no congruence is ever forced), controls the cost by a  $\beta$ -weighted local lemma and a supermartingale bookkeeping, and transfers the resulting Haar density to primes through the one-sided  $\ell^2$  sandwich of Bazzi and Razborov, an energy bound for avoidance indicators of cylinder events, and a linear transfer theorem based on Gallagher’s theorem. We then describe the surrounding picture. The profinite density  $\delta^*(T)$  of the integers with  $W(n) > T$  satisfies  $(\log T)^3 \ll \log(1/\delta^*(T)) \ll (\log T)^3(\log \log T)^5$  (upper bound modulo Nair–Tenenbaum; lower bound via a Janson-type inequality for one-hot product spaces with a  $1/\log \log T$  loss, and without loss using the atoms of a companion note on the exceptional set). Typical primes have  $\#\{p \leq x : W(p) > T\} \ll \pi(x)e^{-c(\log T)^3}$  for  $\log T \leq c_1(\log x)^{1/4}$ . The exponent  $1/4$  is the limit of the minorant architecture: on every fibre  $n \equiv r \pmod{Q}$  with  $\log Q \leq T^{0.05}$ , no minorant of  $\mathbf{1}[W(n) > T]$  of level  $\exp(c(\log T)^4/\log \log T)$  (or  $\exp(c(\log T)^4)$ , using the companion note) has positive Haar mean. This follows from a planting argument, the dual of a lower-bound sieve of dimension  $\asymp (\log T)^3$ , and the planted law is exponentially pseudorandom against classes and characters of every modulus. Hence linear certificates with full-orbit uniform accounting cannot pass  $1/4$  whatever the prime input (Bombieri–Vinogradov, Elliott–Halberstam, GRH below  $x$ , Siegel-model main terms). Conversely, a Linnik-type hypothesis for sifted sets gives  $W(p) \geq \exp((\log p)^{1/3-o(1)})$  infinitely often, the exponent suggested by the Haar density, while a Hardy–Littlewood-type hypothesis for product sets gives only  $(\log p)^{2+o(1)}$ . All results are proved modulo the cited theorems listed in the text; the companion note is internally proved but not externally refereed.

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*Date:* Draft v5, October 2026.

## 1. INTRODUCTION

The Erdős–Straus conjecture asserts that  $4/n = 1/x + 1/y + 1/z$  is solvable in positive integers for every  $n \geq 2$ . It suffices to treat primes. It has been verified for  $n \leq 10^{18}$  ([29]; [39] for  $10^{17}$ ), and Mordell’s identities settle every prime outside the six classes  $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$  [28] (cf. [34]). We call the primes in these classes *hard*. For the exceptional set see Vaughan [45] and [35]; for the average number of solutions see [11].

For  $M \equiv 3 \pmod{4}$  put  $A_M = (M+1)/4$ . If  $uvw = A_M$  and  $n \equiv -uv^{-1} \pmod{M}$ , then  $s = (nv + u)/M$  is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw} \quad (1)$$

(direct verification; this is the classical multiplier family, cf. [45, 11]). Define

$$W(n) = \min\{M \equiv 3 \pmod{4} : n \equiv -uv^{-1} \pmod{M} \text{ for some } uvw = A_M\}, \quad (2)$$

with  $W(n) = +\infty$  if no such  $M$  exists. Thus  $W(p) \leq T$  provides an explicit representation with modulus at most  $T$ . Every prime  $p \equiv 1 \pmod{\text{lcm}\{M \leq T\}}$  has  $W(p) > T$  (Lemma 2.3), so Linnik’s theorem gives  $W(p) \gg \log p$  infinitely often. In [2] we proved that for every fixed  $k$ ,  $W(p) \geq (\log p)^k e^{-C_k \log \log p / \log \log \log p}$  for infinitely many hard primes. Here we prove a sub-exponential rate.

**Theorem 1.1** (Proved modulo Theorems 5.2 and 3.1). *There is an absolute constant  $c > 0$  such that for infinitely many hard primes  $p$*

$$W(p) \geq \exp\left(c \frac{(\log p)^{1/4}}{(\log \log p)^{1/4}}\right).$$

*More precisely, for every sufficiently large  $T$  there is a hard prime  $p$  with  $p > T$ ,  $W(p) > T$  and  $\log p \leq C(\log T)^4 \log \log T$ .*

Theorem 5.2 is Gallagher’s prime number theorem for the family of all primitive characters of conductor at most  $Q_G$  [14], quoted from [26, Thm. 28.19] together with the standard facts on exceptional zeros (Theorem 5.1). No hypothesis on Siegel zeros is needed. Theorem 3.1 is the Nair–Tenenbaum bound [30] for sums of multiplicative functions of several polynomials, quoted in the form of Henriot [16]; it is used once, to bound Haar-weighted divisor sums over the moduli  $M = 4n - 1$  (Lemma 3.2). The Elsholtz–Tao estimates used in the third version are not needed. The constants in Theorem 1.1 are effective, except that we have not checked whether the implied constants in [30] are; the results proved modulo [42] below have ineffective constants. We make no numerical claim.

*Typical size.* Theorem 1.1 is about the extreme primes. For the typical prime,  $W(p)$  is small: large values have a cubic-exponential tail.

**Theorem 1.2** (Proved modulo [42] and Page’s theorem). *There are absolute  $c, c_1 > 0$  such that, uniformly for  $x \geq 3$  and  $\log T \leq c_1(\log x)^{1/4}$ ,*

$$\#\{p \leq x : W(p) > T\} \ll \pi(x) e^{-c(\log T)^3}.$$

This is Theorem 11.2. It uses the atom family of the companion note [42] (internally proved, not externally refereed; inputs Bombieri–Vinogradov, Brun–Titchmarsh and Shiu) and, for small  $T$ , the prime number theorem in progressions with a Landau–Page term. At  $\log T \asymp (\log x)^{1/4}$  it is the prime case of the exceptional-set bound  $E(N) \ll Ne^{-c(\log N)^{3/4}}$  of [42]. Together with Theorem 1.1 it describes both ends of the range: for  $\log T \leq c_1(\log x)^{1/4}$  the primes  $p \leq x$  with  $W(p) > T$  have relative frequency at most  $e^{-c(\log T)^3}$ , and once  $\log x \geq C(\log T)^4 \log \log T$  such primes exist.

*The Haar exponent.* Let  $\delta^*(T)$  be the Haar measure of  $\{n \in \hat{\mathbb{Z}}^\times : W(n) > T\}$  relative to the class  $n \equiv 1 \pmod{24}$  (§3).

**Theorem 1.3** (Haar exponent 3; proved modulo Theorem 3.1 and [42], see below). *For large  $T$ ,*

$$c(\log T)^3 \leq \log \frac{1}{\delta^*(T)} \leq C(\log T)^3 (\log \log T)^5.$$

*The upper bound is proved modulo Theorem 3.1 (Theorem 3.3). The lower bound is proved modulo [42] (Proposition 11.1); with  $(\log T)^3 / \log \log T$  in place of  $(\log T)^3$  it is proved modulo the fundamental lemma of sieve theory alone (Theorem 4.4).*

*The ceiling.* The exponent of  $\log p$  in Theorem 1.1 is governed not by the Haar density but by the level of the minorant used to transfer it to primes, and this level cannot be lowered.

**Theorem 1.4** (Proved modulo Theorems 5.1, 5.2 and the fundamental lemma; sharpened modulo [42]). *There are absolute  $c, T_0 > 0$  such that for  $T \geq T_0$ , every modulus  $Q$  with  $\log Q \leq T^{0.05}$  and every unit  $r$  modulo  $Q$ : if  $B$  is a linear combination of functions  $n \mapsto \phi(n \bmod qQ)$  with  $\log q \leq c(\log T)^4 / \log \log T$ , and  $B(n) \leq \mathbf{1}[W(n) > T]$  for all units  $n \equiv r \pmod{Q}$ , then the Haar mean of  $B$  on this class is  $\leq 0$ . Modulo [42], the same holds with  $\log q \leq c(\log T)^4$ .*

This is Theorem 10.6 and Proposition 11.3. Consequently (Corollary 10.7) every argument that certifies a hard prime with  $W(p) > T$  through a minorant of positive Haar mean and a transfer theorem costing  $\log x \gg \log(\text{level})$ , as ours does, has  $\log x \gg (\log T)^4 / \log \log T$  (and  $\gg (\log T)^4$  modulo [42]): such arguments cannot give  $W(p) \geq \exp((\log p)^{1/4+\delta})$ , and Theorem 1.1 is optimal among them up to a factor  $(\log \log p)^{1/2}$  in  $\log W(p)$ , or  $(\log \log p)^{1/4}$  modulo [42]. It also shows that the energy bound used for the sandwich cannot be sharpened for this purpose (Corollary 10.8). The level  $(\log T)^4$  is the product of the log-size  $\log T$  of the moduli and the dimension  $(\log T)^3$  of the sieve formed by the events with one large prime factor. The same product, read at the budget  $\log N$  of an integer sieve instead of the budget  $\log x$  of a prime transfer, gives the exceptional-set exponent 3/4 of [42] (Remark 11.4).

*Beyond the ceiling.* The planted law behind Theorem 1.4 is exponentially pseudorandom against residue classes, Dirichlet characters and additive

characters of *every* modulus (Lemma 12.1). So every minorant  $B$  of any level has  $\mathbb{E}B \leq e^{-c(\log T)^4/\log \log T} \|B\|_\times$ , where  $\|B\|_\times$  is the  $\ell^1$  norm of its coefficients in classes and characters (Theorem 12.2). As a consequence, linear certificates with full-orbit uniform error accounting cannot beat the exponent  $1/4$ , whatever true prime input they use, including Bombieri–Vinogradov, Elliott–Halberstam, GRH restricted to moduli  $\leq x$ , and Siegel-model main terms (Corollary 12.3, Proposition 12.4). If primes behaved like a set of Haar density  $\delta^*(T)$ , Theorem 1.3 would suggest  $\log W(p) \approx (\log p)^{1/3}$  for the extremal primes. We isolate a hypothesis that gives this.

**Theorem 1.5** (Conditional on Hypothesis LS; proved modulo Theorem 3.1). *Assume that every system of unit-class congruences  $n \not\equiv a_E \pmod{m_E}$  with  $m_E \leq T$  whose solution set has Haar density  $\delta > 0$  contains a prime  $p > T$  with  $\log p \leq C(\log T + \log(1/\delta))$  (Hypothesis LS, “Linnik’s theorem for sifted sets”). Then  $W(p) \geq \exp(c_C(\log p)^{1/3}(\log \log p)^{-5/3})$  for infinitely many hard primes  $p$ .*

This is Theorem 13.1. For a single residue class LS is Linnik’s theorem. It is not reachable by the linear certificates above (§12), and the analogous hypothesis restricted to CRT product sets (a Hardy–Littlewood-type singular series) gives only  $W(p) = (\log p)^{2 \pm o(1)}$  (Proposition 13.2). Altogether, with  $\mathcal{L} = \log T$ : the extremal primes have  $\mathcal{L} \geq (\log p)^{1/4 - o(1)}$  (proved), the minorant method stops at  $(\log p)^{1/4 + o(1)}$  (proved, within its scope), the Haar heuristic and LS give  $(\log p)^{1/3 - o(1)}$ , and typical primes have  $W(p) > T$  with relative frequency at most  $e^{-c\mathcal{L}^3}$ .

*Versions.* The first version of this note had exponent  $1/14$ ; it used the Thorner–Zaman theorem [43] through the transfer theorem of [2]. The second had exponent  $1/7$ : it replaced Thorner–Zaman by the linear transfer theorem (Theorem 6.1 here), and bounded the approximation error by Håstad’s switching lemma. The third had exponent  $1/5$ : it replaced the switching lemma by the energy bound (§7) and used a graded class-of-one quarantine, whose cost was bounded with four results of Elsholtz and Tao. The fourth version reached  $1/4$  by a new quarantine: a random square class, a  $\beta$ -weighted local lemma and a supermartingale bookkeeping (§2) with Haar masses bounded by the Nair–Tenenbaum theorem (§3); it added the Haar lower bound (§4) and the ceiling (§10). The present fifth version leaves the proof of Theorem 1.1 unchanged and adds: the typical size, the Haar lower bound without logarithmic loss and the ceiling at level  $(\log T)^4$ , all via the atoms of [42] (§11); the Wiener-norm barrier and its consequences for linear certificates (§12); and the conditional exponent  $1/3$  (§13).

**The method.** Fix  $T$ . We restrict  $n$  to a class  $n \equiv r \pmod{Q}$ , with  $Q = 8 \prod_\ell \ell^{a_\ell}$  and  $r$  a quadratic residue modulo every prime of  $Q$ . Then  $W(n) > T$  whenever  $n$  avoids a finite family of congruence *events*  $n \equiv -4D \pmod{m}$ , one for each surviving pair  $(M, D)$  with  $M \leq T$ ,  $D \mid A_M^2$ , where  $m$  is the part of  $M$  not absorbed by  $Q$ . The key observation (Lemma 2.2) is

that the Jacobi symbol  $(\frac{-4D}{M})$  is always  $-1$ ; so if every prime of  $M$  divides  $Q$ , the event is excluded, and no event is ever forced, whatever square class  $r$  we choose. For  $r = 1$  this is the class-of-one quarantine of [2]; Mordell's hard classes are the case  $Q = 840$ .

We build  $(Q, r)$  by a random process (Lemma 2.9). While the weighted mass  $\tilde{w}_\ell = \sum_{E \ni \ell} \beta^{|\text{supp } E|} \mathbb{P}(E)$  at a prime  $\ell \leq Y = \mathcal{L}^{O(1)}$  exceeds  $\eta = \frac{3}{4} \log \beta$ , we reveal one more digit of  $n$  at  $\ell$ , uniformly among the squares. The  $\beta$ -weighted local lemma (Lemma 2.8) needs exactly  $\tilde{w}_\ell \leq \eta$  at every coordinate. With  $\beta = 1 + 1/\log \mathcal{L}$  the threshold is  $\asymp 1/\log \mathcal{L}$  at every prime and every level, instead of  $\asymp \log \ell / \mathcal{L}$  in the previous version. The masses form supermartingales (a step at a new prime at most doubles the mean of an event's probability, and this is paid by a factor  $2^{\omega_Y(M)}$ ), so by optional stopping the expected cost is a Haar-weighted divisor sum,  $\ll \mathcal{L}^3 (\log \mathcal{L})^5$  by the Nair–Tenenbaum theorem. Primes  $\ell > Y$  are never quarantined; a second-moment potential for pairs of events shows that their masses are small with high probability. The class of one, used before, inflates the masses by  $\gcd(M, 4D + 1)$ , which costs a factor  $\mathcal{L}$  and requires the Elsholtz–Tao estimates.

To find a *prime* in the avoiding set we need a function  $B = \sum_i c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$  with  $B \leq \mathbf{1}[\text{avoid}]$  pointwise, mean  $\gg \delta$  (the Haar density of the avoiding set on the fibre), Haar  $\ell^1$  norm  $\mathbb{E}|B|$  comparable to its mean, and small moduli  $d_i$ . The linear transfer theorem (Theorem 6.1) then gives a prime  $p \equiv r \pmod{Q}$  with  $\log p \ll \log Z$ , where  $Z$  is  $Q$  times the largest  $d_i$ . It is Linnik's theorem with weights: the function  $B(n) \mathbf{1}[n \equiv r \pmod{Q}]$  is expanded in characters, every coefficient is bounded by the Haar  $\ell^1$  norm of  $B$ , whatever the number of cells, and Gallagher's theorem bounds the contribution of all characters of conductor at most a power of  $Z$  at once. Since  $r$  is a square, every real character that factors through  $(\mathbb{Z}/Q)^\times$  is 1 at  $r$ , so the exceptional-zero analysis is the same as for the class of one.

As in the previous versions,  $B$  is the one-sided  $\ell^2$  sandwich of Bazzi [3], in the form of Razborov [37] (Lemma 8.2). For events  $E_1, \dots, E_m$  with indicators  $A_i$  and arbitrary functions  $u_j$ ,

$$B = 1 - \sum_i A_i \left( 1 - \sum_{j < i} A_j u_j \right)^2 \leq \mathbf{1}[\text{no } E_i \text{ occurs}],$$

and, if each  $u_j$  depends only on the coordinates not fixed by  $E_j$ ,  $\mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$ , where  $F^{(j)}$  is the indicator that no earlier event occurs, conditioned on  $E_j$ . The  $u_j$  are truncations of the Efron–Stein decomposition of  $F^{(j)}$ , and the error is an Efron–Stein tail, which we bound by the energy bound of the previous version (Theorem 7.5): if  $F$  is the avoidance indicator of any family of cylinder events on a finite product probability space, and  $\lambda_v \geq 1$  are weights with  $\prod_{v \in \text{supp } E} \lambda_v \leq 2$  for every

event  $E$ , then

$$\sum_U \left( \prod_{v \in U} \lambda_v \right) \|F^{=U}\|_2^2 \leq 1.$$

Applied to the base- $\ell$  digits of the residues, with weights adapted to the moduli (Theorem 7.9), it shows that truncating  $F^{(j)}$  to moduli at most  $e^\tau$  costs  $2^{-\tau/(2 \log T)}$ . So moduli of size  $\log d_i \ll \mathcal{L} \cdot (S + \mathcal{L})$  suffice, where  $S \ll \mathcal{L}^3 \log \mathcal{L}$  is the residual mass. Altogether  $\log Z \ll \mathcal{L}^4 \log \mathcal{L}$ . The ceiling (Theorem 10.6) shows that the level  $\mathcal{L}S$  is forced up to a factor  $(\log \mathcal{L})^2$ : conditioned on the small primes, the events with one large prime factor form a sieve of dimension  $\asymp \mathcal{L}^3 / \log \mathcal{L}$  on the primes above  $T^{0.6}$ , and a planting lemma (Lemma 10.1), the dual of a lower-bound sieve, shows that no function of lower level can be a minorant with positive mean.

**Relation to the literature.** *The energy bound.* The nearest prior art known to us is Lecomte and Tan [23], who bound the Fourier coefficients of a DNF on  $\{\pm 1\}^n$  by the probability that a set  $S$  is covered by the terms satisfied at a random point; this is the device of our Lemma 7.2. Their cover counts are unsigned, and their degree concentration still comes from Håstad’s switching lemma, with an unspecified constant. In Lemma 7.2 the cover counts are signed, so that they vanish when a satisfied event misses the set; the polarization identity (Lemma 7.3) and the deletion–contraction bound (Lemma 7.4) then give the constant 1, on any finite product probability space. In Boolean language (Remark 7.8): every width- $k$  DNF  $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$  has  $\sum_{|S| > t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k}$ , under any product measure, while the switching-lemma route gives  $2 \cdot 2^{-t/(20k)}$  [31, §4.4]. The base  $2^{-1/k}$  is sharp over general product spaces (Remark 7.7). Theorem 7.5 and its corollaries are new to us, but our literature search was partial (we checked [23], [31] and some lecture notes; we did not search the literature on switching-lemma variants, Fourier growth or hypercontractivity with  $\rho > 1$  systematically), and we make no priority claim.

*The local lemmas.* The  $\beta$ -weighted local lemma (Lemma 2.8) is the asymmetric local lemma of [1, Lemma 5.1.1] with the choice  $x_E = \beta^{|\text{supp } E|} \mathbb{P}(E)$ ; the conditional form for events outside the family is standard (cf. [17]). Theorem 4.3 combines the lopsided local lemma of Erdős and Spencer [10] with the proof scheme of Janson’s inequality [22], which needs increasing events and Harris’ inequality; for atomic events on one-hot coordinates we use instead the correlation inequality of Lemma 4.1. We did not find this form in the literature, but did not search systematically. The planting lemma (Lemma 10.1) is the linear-programming dual of lower-bound sieves; constructions of  $k$ -wise independent laws with extremal all-zero or all-one probabilities of the same kind appear in [4, 36], and we claim no novelty for it. Its use to bound the level of minorants of a concrete arithmetic indicator, on every fibre of modulus  $\log Q \leq T^{0.05}$ , is new to us.

*The rest.* The sandwich (Lemma 8.2) is Razborov’s identity. To our knowledge tools from the theory of bounded independence (“polylogarithmic

independence fools DNF and  $AC^0$ ” [3, 37, 7]) have not been used before to construct sieve minorants for primes. Gallagher’s theorem is classically the engine of Linnik’s theorem on the least prime in a single progression. Here it serves as a transfer device for an arbitrary signed combination of progressions; the cost is governed by the Haar  $\ell^1$  norm of the minorant and its largest modulus, not by the number of cells. The nearest classical analogues are bounds for the least prime in a union of residue classes, or in a Chebotarev class of an abelian extension (Lagarias–Montgomery–Odlyzko [27]; Thorner–Zaman [44]); these concern nonnegative indicators. We have not searched systematically for an earlier use of Gallagher’s theorem in this role. Earlier uses of the switching lemma in number theory known to us run in the opposite direction: they show that the Möbius function is uncorrelated with bounded-depth circuits [15, 6]. The Haar side, a local-lemma lower bound for the density of the uncovered set of a congruence system, has relatives in the work on covering systems: Filaseta, Ford, Konyagin, Pomerance and Yu [12], who to our knowledge already use the local lemma for this purpose, Hough [19] and Balister, Bollobás, Morris, Sahasrabudhe and Tiba [5]. We do not use those results. The Jacobi-symbol property of the multiplier classes (Lemma 2.2) is implicit in the classical fact that the identities cover all non-residue classes; its use to quarantine to random square classes is ours. The divisor-sum bounds use Nair–Tenenbaum [30, 16] in a standard way. Hypothesis LS (§13) is a uniform version of Linnik’s theorem for sifted sets; we did not search the literature for it systematically and claim no priority for the formulation.

**Status conventions.** Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Cited* marks external statements; *Classical* marks standard results for which we include a proof or a precise textbook reference. The cited inputs are Gallagher’s theorem (Theorem 5.2, used together with the classical Landau–Page facts of Theorem 5.1), the Nair–Tenenbaum theorem (Theorem 3.1) and, for Theorems 4.4 and 10.6, the fundamental lemma of sieve theory. In §§11–13 three further inputs appear: the companion note [42], which is *internally proved but not externally refereed* (“proved modulo [42]” means proved here assuming its stated results, with their inputs Bombieri–Vinogradov, Brun–Titchmarsh and Shiu); the prime number theorem in progressions with a Landau–Page term (“Page’s theorem”, (12)); and the Barban–Davenport–Halberstam theorem. *Conditional on LS* marks a statement that assumes the conjectural Hypothesis LS of §13. In the labels, “modulo Theorem 5.2” includes Theorem 5.1. Classical results whose proof we include, or which are standard textbook facts (the local lemma, the Efron–Stein decomposition, Mertens’ theorem, the divisor bound, the Rosser–Schoenfeld bound for  $\vartheta(x)$ , mean values of non-negative multiplicative functions via Euler products), are not listed in the labels. *Proved implication*



marks a statement about a class of arguments. *Assessment* marks heuristic comparisons, which are not claims. Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large  $W(p)$  may well have other representations.

**Notation.**  $T$  is large and  $\mathcal{L} = \log T$ ;  $\log_2$  always denotes the binary logarithm (iterated logarithms are written out as  $\log \log$ ).  $\tau$  is the divisor function (and, in §§8–9, also a truncation parameter; no confusion arises).  $\omega(m)$  and  $\Omega(m)$  count prime factors without and with multiplicity;  $\omega_Y(m)$  counts the prime factors  $\leq Y$ , and  $m_Y$  is the  $Y$ -smooth part of  $m$ . A  $y$ -rough integer has all prime factors  $> y$ . For  $\lambda = (\lambda_v)$  and a finite set  $U$  we write  $\lambda^U = \prod_{v \in U} \lambda_v$ .

## 2. ATOMS, SQUARE CLASSES AND THE LOCAL LEMMA

For  $M \equiv 3 \pmod{4}$  put  $\mathcal{R}(M) = \{-4D \pmod{M} : D \mid A_M^2\}$ . Note that  $\gcd(A_M, M) = 1$  (as  $4A_M = M + 1$ ), so every element of  $\mathcal{R}(M)$  is a unit class.

**Lemma 2.1** (Proved; cf. [2]).  $\{-uv^{-1} \pmod{M} : uvw = A_M\} = \mathcal{R}(M)$ . Consequently, for every integer  $n$ ,  $W(n) > T$  if and only if  $n \pmod{M} \notin \mathcal{R}(M)$  for every  $M \leq T$  with  $M \equiv 3 \pmod{4}$ .

*Proof.* Since  $4A_M \equiv 1 \pmod{M}$ , from  $uvw = A_M$  we get  $v^{-1} \equiv 4uw$ , hence  $-uv^{-1} \equiv -4u^2w$  and  $D = u^2w$  divides  $A_M^2$ . Conversely let  $D \mid A_M^2$ . For each prime  $q$  with  $q^a \parallel A_M$  and  $q^d \parallel D$  ( $d \leq 2a$ ) put  $u_q = \max(0, d-a)$ ,  $w_q = d - 2u_q$ ,  $v_q = a - u_q - w_q$ ; these are non-negative, so  $u = \prod q^{u_q}$ ,  $v = \prod q^{v_q}$ ,  $w = \prod q^{w_q}$  satisfy  $uvw = A_M$  and  $u^2w = D$ .  $\square$

An *atom* is a pair  $(M, D)$  with  $M \leq T$ ,  $M \equiv 3 \pmod{4}$  and  $D \mid A_M^2$ . There are at most  $T^2$  atoms, since  $\tau(A_M^2) \leq 2A_M \leq T$ .

**Lemma 2.2** (Event classes are Jacobi non-residues; proved). *Let  $(M, D)$  be an atom and let  $d$  be the squarefree part of  $D$ . Then*

- (a) *every prime  $\ell \mid M$  satisfies  $\ell \nmid 2D$  and  $(\frac{-4D}{\ell}) = (\frac{-d}{\ell})$ ;*
- (b) *the Jacobi symbol  $(\frac{-4D}{M})$  equals  $-1$ .*

*Consequently, if  $r$  is an integer coprime to  $M$  that is a quadratic residue modulo every prime dividing  $M$ , then  $r \not\equiv -4D \pmod{M}$ .*

*Proof.* (a)  $\gcd(M, 4A_M) = 1$  and every prime of  $D$  divides  $A_M$ , so  $\ell \nmid 2D$ ; and  $-4D = -d \cdot (2t)^2$  for an integer  $t$ . (b) By (a),  $(\frac{-4D}{M}) = (\frac{-d}{M}) = (\frac{-1}{M})(\frac{d}{M}) = -(\frac{d}{M})$ , as  $M \equiv 3 \pmod{4}$ . Every prime  $p \mid d$  has  $v_p(D)$  odd and  $v_p(D) \leq 2v_p(A_M)$ , so  $p \mid A_M$  and  $M = 4A_M - 1 \equiv -1 \pmod{p}$ . For odd  $p$ , quadratic reciprocity, with  $(M-1)/2$  odd, gives  $(\frac{p}{M}) = (\frac{M}{p})(-1)^{(p-1)/2} = (\frac{-1}{p})(-1)^{(p-1)/2} = 1$ . If  $p = 2$ , then  $2 \mid A_M$ , so  $M \equiv 7 \pmod{8}$  and  $(\frac{2}{M}) = 1$ . Hence  $(\frac{d}{M}) = 1$ . For the consequence,  $(\frac{r}{M}) = \prod_{\ell \mid M} (\frac{r}{\ell})^{v_\ell(M)} = 1 \neq (\frac{-4D}{M})$ .  $\square$



We checked (a) and (b) by computer on all 2 401 032 atoms with  $M \leq 2 \cdot 10^5$  (no failures; this check is not used). The lemma is the mechanism behind two classical facts: the identities (1) cover the non-residue classes, and Mordell's hard classes modulo 840 are exactly the six square classes. For  $r = 1$  it contains Lemma 2.3 below.

**Lemma 2.3** (Class of one; proved).  $1 \notin \mathcal{R}(M)$  for every  $M \equiv 3 \pmod{4}$ .

*Proof.* Lemma 2.2 with  $r = 1$ .  $\square$

**Definition 2.4** (Square-class quarantine and residual system). For an odd prime  $\ell \leq T$  let  $f_\ell = \lfloor \mathcal{L} / \log \ell \rfloor \geq 1$  be the largest exponent with  $\ell^{f_\ell} \leq T$ . A *square-class quarantine* is a pair  $(Q, r)$ , where  $Q = 8 \prod_{\ell} \ell^{a_\ell}$  with integers  $0 \leq a_\ell \leq f_\ell$  for the odd primes  $\ell \leq T$ , and  $r$  is a unit modulo  $Q$  with  $r \equiv 1 \pmod{8}$  that is a quadratic residue modulo every odd prime dividing  $Q$ . The *coordinate primes* are the odd  $\ell \leq T$  with  $a_\ell < f_\ell$ ; write  $\mathcal{P}$  for their set. For  $\ell \in \mathcal{P}$  let

$$\Omega_\ell = \{x \in (\mathbb{Z}/\ell^{f_\ell})^\times : x \equiv r \pmod{\ell^{a_\ell}}\}$$

(all units if  $a_\ell = 0$ ), with the uniform measure, and let  $\mathbb{P}, \mathbb{E}$  be the product measure on  $\Omega = \prod_{\ell \in \mathcal{P}} \Omega_\ell$ , with coordinates  $X_\ell$ . An atom *survives*  $(Q, r)$  if  $-4D \equiv r \pmod{\gcd(M, Q)}$  and  $M \nmid Q$ . Its *support* is  $\text{supp}(M, D) = \{\ell : v_\ell(M) > a_\ell\}$ , a nonempty subset of  $\mathcal{P}$ , with  $s(M, D) = |\text{supp}(M, D)| \leq \omega(M)$ ; its *modulus* is  $\prod_{\ell \in \text{supp}} \ell^{v_\ell(M)}$ , and its *event* is

$$E_{M,D} = \{X_\ell \equiv -4D \pmod{\ell^{v_\ell(M)}} \text{ for all } \ell \in \text{supp}(M, D)\}.$$

For  $\beta \geq 1$  the *weighted fibre mass* at  $\ell \in \mathcal{P}$  and the weighted total mass are

$$\tilde{w}_\ell = \sum_{\substack{(M,D) \text{ surviving} \\ \ell \in \text{supp}(M,D)}} \beta^{s(M,D)} \mathbb{P}(E_{M,D}), \quad S_\beta = \sum_{\substack{(M,D) \\ \text{surviving}}} \beta^{s(M,D)} \mathbb{P}(E_{M,D}),$$

where surviving atoms are counted individually (two atoms may give the same event). For an integer  $n$  we put  $X_\ell(n) = n \bmod \ell^{f_\ell}$ , and  $F(n) = 1$  if  $n$  lies in no event of a surviving atom,  $F(n) = 0$  otherwise; on  $\Omega$ ,  $F$  is the indicator that no event  $E_{M,D}$  occurs.

Every event is nonempty:  $-4D$  is a unit modulo  $\ell$ , and  $-4D \equiv r \pmod{\ell^{a_\ell}}$  by survival. The modulus is at most  $M \leq T$ . The class-of-one quarantines of [2] and of the previous versions are the case  $r = 1$ .

**Lemma 2.5** (Reduction and weights; proved). *Let  $(Q, r)$  be a square-class quarantine.*

- (i) *If  $n \equiv r \pmod{Q}$  and  $F(n) = 1$ , then  $W(n) > T$ .*
- (ii) *For every surviving atom,*

$$\mathbb{P}(E_{M,D}) = \frac{\gcd(M, Q)}{M} \prod_{\ell \in \text{supp}(M,D), a_\ell=0} \frac{\ell}{\ell-1}.$$

- (iii) *If  $105 \mid Q$ , then  $r \bmod 840$  is one of Mordell's six hard classes, so every prime  $p \equiv r \pmod{Q}$  is hard.*

*Proof.* (i) Let  $M \leq T$ ,  $M \equiv 3 \pmod{4}$ , and suppose  $n \equiv -4D \pmod{M}$  with  $D \mid A_M^2$ . Reducing modulo  $\gcd(M, Q)$  gives  $-4D \equiv r$ . If  $M \mid Q$ , then  $r \equiv -4D \pmod{M}$ , and every prime of  $M$  divides  $Q$ , so  $r$  is a quadratic residue modulo it; this contradicts Lemma 2.2. Otherwise the atom survives, and  $n$  lies in its event, so  $F(n) = 0$ . Conclude with Lemma 2.1. (ii) Let  $\ell \in \text{supp}$  and  $v = v_\ell(M)$ . If  $a_\ell = 0$ , the class of  $-4D$  modulo  $\ell^v$  has probability  $1/\varphi(\ell^v) = \ell^{-v} \cdot \ell/(\ell-1)$  in  $\Omega_\ell$ . If  $a_\ell \geq 1$ , then  $|\Omega_\ell| = \ell^{f_\ell - a_\ell}$ , and the class (which lies in  $\Omega_\ell$  by survival) has  $\ell^{f_\ell - v}$  elements, so probability  $\ell^{a_\ell - v}$ . As  $M$  is odd,  $\gcd(M, Q) = \prod_{\ell \mid M} \ell^{\min(v_\ell(M), a_\ell)}$ , and  $\min(v_\ell(M), a_\ell) = v_\ell(M)$  for  $\ell \notin \text{supp}$ . (iii) A unit modulo 840 that is  $\equiv 1 \pmod{8}$  and a square modulo 3, 5 and 7 is a square modulo 840, and there are  $1 \cdot 1 \cdot 2 \cdot 3 = 6$  such classes; they are the six listed classes.  $\square$

**Remark 2.6** (The graded class-of-one quarantine). In the previous version  $r = 1$ , and the exponents  $a_\ell$  were raised while the fibre mass exceeded  $c(a_\ell + 1) \log \ell / \mathcal{L}$ . With  $r = 1$  a surviving atom has  $\gcd(M, Q) \mid g = \gcd(M, 4D + 1)$ , so by (ii) its weight is up to  $g/M$  instead of the Haar weight  $1/\varphi(M)$ . Summed over the atoms, the Haar weights give  $O(\mathcal{L}^3 \log \mathcal{L})$  (Lemma 3.2), while for the inflated weights we could only prove  $O(\mathcal{L}^4)$ , and that needed the Elsholtz–Tao divisor-sum estimates [11]. The threshold  $\propto \log \ell / \mathcal{L}$  cost a further factor  $\mathcal{L}$  in the quarantine. A *random* square class removes the inflation in expectation (Lemma 2.9), Lemma 2.2 guarantees that no event is ever forced, and the  $\beta$ -weighted local lemma (Lemma 2.8) replaces the log-weighted threshold by a constant one. The graded class-of-one quarantine is therefore superseded, and we omit it.

### The local lemma.

**Lemma 2.7** (Local lemma with conditioning; classical). *Let  $\mathcal{A}$  be a finite family of events in a product space, each depending on a finite set of coordinates; write  $A' \sim A$  if they share a coordinate. Suppose  $x_A \in [0, 1]$  satisfy  $\mathbb{P}(A) \leq x_A \prod_{A' \sim A, A' \neq A} (1 - x_{A'})$ . Then  $\mathbb{P}(\bigcap_A \bar{A}) \geq \prod_A (1 - x_A)$ , and  $\mathbb{P}(A \mid \bigcap_{A' \in \mathcal{S}} \bar{A}') \leq x_A$  for every  $\mathcal{S} \subseteq \mathcal{A}$  with  $A \notin \mathcal{S}$ . Moreover, for every event  $B$  depending on a set  $J$  of coordinates and every  $\mathcal{S} \subseteq \mathcal{A}$ ,  $\mathbb{P}(B \mid \bigcap_{A \in \mathcal{S}} \bar{A}) \leq \mathbb{P}(B) \prod_{A \in \Gamma(B)} (1 - x_A)^{-1}$ , where  $\Gamma(B)$  is the set of  $A \in \mathcal{S}$  depending on some coordinate in  $J$ .*

*Proof.* The first two statements are [1, Lemma 5.1.1] and its proof. For the third (cf. [17, Thm. 2.1]),  $B$  is independent of  $\bigcap_{A \in \mathcal{S} \setminus \Gamma(B)} \bar{A}$ , so

$$\mathbb{P}\left(B \mid \bigcap_{A \in \mathcal{S}} \bar{A}\right) \leq \frac{\mathbb{P}(B) \mathbb{P}(\bigcap_{A \in \mathcal{S} \setminus \Gamma(B)} \bar{A})}{\mathbb{P}(\bigcap_{A \in \mathcal{S}} \bar{A})} = \frac{\mathbb{P}(B)}{\mathbb{P}(\bigcap_{A \in \Gamma(B)} \bar{A} \mid \bigcap_{A \in \mathcal{S} \setminus \Gamma(B)} \bar{A})},$$

and the chain rule with the second statement bounds the denominator below by  $\prod_{A \in \Gamma(B)} (1 - x_A)$ .  $\square$

**Lemma 2.8** ( $\beta$ -weighted local lemma; proved). *Let  $\mathcal{A}$  be a finite family of events on a product probability space, each determined by the coordinates*

in a nonempty set  $\text{supp } A$ , and put  $s(A) = |\text{supp } A|$ . Let  $1 < \beta \leq e^{1/3}$ ,  $\eta = \frac{3}{4} \log \beta$ , and  $\tilde{w}_v = \sum_{A \ni v} \beta^{s(A)} \mathbb{P}(A)$ , the sum over the  $A$  with  $v \in \text{supp } A$ . Suppose that  $\tilde{w}_v \leq \eta$  for every coordinate  $v$ . Then

$$\mathbb{P}\left(\bigcap_A \bar{A}\right) \geq \exp\left(-\frac{4}{3} \sum_{A \in \mathcal{A}} \beta^{s(A)} \mathbb{P}(A)\right),$$

and for every event  $B$  determined by a coordinate set  $J$  and every  $\mathcal{S} \subseteq \mathcal{A}$ ,  $\mathbb{P}(B \mid \bigcap_{\mathcal{S}} \bar{A}) \leq \beta^{|J|} \mathbb{P}(B)$ .

*Proof.* Put  $x_A = \beta^{s(A)} \mathbb{P}(A)$ . Every  $A$  has a coordinate  $v \in \text{supp } A$ , so  $x_A \leq \tilde{w}_v \leq \eta \leq 1/4$ . For  $0 \leq x \leq 1/4$ ,  $-\log(1-x) \leq x/(1-x) \leq \frac{4}{3}x$ . For any coordinate set  $J$ , each  $A'$  sharing a coordinate with  $J$  is counted in some  $\tilde{w}_v$ ,  $v \in J$ , so

$$\prod_{A' \text{ meets } J} (1 - x_{A'}) \geq \exp\left(-\frac{4}{3} \sum_{v \in J} \tilde{w}_v\right) \geq \exp\left(-\frac{4}{3}|J|\eta\right) = \beta^{-|J|}. \quad (3)$$

With  $J = \text{supp } A$  this gives  $\mathbb{P}(A) = x_A \beta^{-s(A)} \leq x_A \prod_{A' \sim A} (1 - x_{A'})$ , the hypothesis of Lemma 2.7. Its conclusions give  $\mathbb{P}(\bigcap \bar{A}) \geq \prod (1 - x_A) \geq \exp(-\frac{4}{3} \sum x_A)$  and, with (3), the conditional bound.  $\square$

The criterion is per coordinate, with a threshold that depends neither on the coordinate nor on the level of its quarantine; the price is the weight  $\beta^{s(A)}$ . We shall take  $\beta = 1 + 1/\log \mathcal{L}$ , so that  $\beta^{\omega(M)}$  is  $O(1)$  on average, and  $\eta \asymp 1/\log \mathcal{L}$ .

**The square-class process.** Fix a parameter  $Y \geq 7$  and  $\beta = 1 + 1/\log \mathcal{L}$ ,  $\eta = \frac{3}{4} \log \beta$ . The *square-class process* is the following random sequence of square-class quarantines  $(Q_i, r_i)$ . Start from  $Q = 8$ ,  $r = 1$ . Then make three *forced* steps at  $\ell = 3, 5, 7$  with  $a = 0$ , and afterwards, while some odd prime  $\ell \leq Y$  with  $\ell \in \mathcal{P}$  has  $\tilde{w}_\ell > \eta$  (computed for the current quarantine), make a step at the least such  $\ell$ . A *step* at  $(\ell, a)$ , where  $a = a_\ell < f_\ell$  is the current exponent, replaces  $a_\ell$  by  $a + 1$  and  $r$  by a class  $r' \bmod Q\ell$  with  $r' \equiv r \pmod{Q}$ , chosen uniformly among the classes of the current fibre at  $\ell$  that are squares modulo  $\ell$ : if  $a = 0$  this is uniform among the  $(\ell - 1)/2$  nonzero squares modulo  $\ell$ , and if  $a \geq 1$  among the  $\ell$  lifts of  $r \bmod \ell^a$  to  $\ell^{a+1}$ , all of which are squares. Each step raises some  $a_\ell < f_\ell$  with  $\ell \leq Y$ , so the process stops after at most  $\sum_{\ell \leq Y} f_\ell$  steps.

Every  $(Q_i, r_i)$  is a square-class quarantine. After the forced steps,  $840 \mid Q$  and  $r \equiv 1 \pmod{24}$ , since 1 is the only nonzero square modulo 3. Primes  $\ell > Y$  are never stepped. By Lemma 2.2, no atom ever has all its coordinates quarantined and matched: the final system contains no event of probability one.

*Notation.* At stage  $i$  let  $\mu_i$  be the uniform probability on the units  $n$  with  $n \equiv r_i \pmod{Q_i}$  (on  $\hat{\mathbb{Z}}^\times$ , or modulo any multiple of  $\text{lcm}(Q_i, M)$ ), and for an atom  $E = (M, D)$  put  $p_i(E) = \mu_i(n \equiv -4D \pmod{M})$ . So  $p_i(E) = \mathbb{P}(E_{M,D})$  if the atom survives  $(Q_i, r_i)$ , and  $p_i(E) = 0$  otherwise (if  $M \mid Q_i$  the class

is excluded by Lemma 2.2). Let  $P_H(E) = 1/\varphi(M)$  be the Haar probability,  $s_i(E)$  the support size at stage  $i$ , and  $u_i(E) = \#\{\ell \leq Y : \ell \mid M, a_\ell(i) = 0\}$ . Write  $\omega_Y(M)$  for the number of primes  $\ell \leq Y$  dividing  $M$ , and  $M_Y$  for the  $Y$ -smooth part of  $M$ .

**Lemma 2.9** (Bookkeeping for the square-class process; proved). *Consider the square-class process with parameter  $Y$ .*

- (a) *For every fixed  $\phi(E) \geq 0$ ,  $G_i = \sum_E p_i(E) 2^{u_i(E)} \phi(E)$  is a supermartingale, and  $G_0 = \sum_E P_H(E) 2^{\omega_Y(M)} \phi(E)$ .*
- (b) *With  $Q_{\text{end}}$  the final modulus,*

$$\mathbb{E}[\log Q_{\text{end}}] \leq \log 840 + \frac{1}{\eta} \sum_E P_H(E) 2^{\omega_Y(M)} \beta^{\omega(M)} \log M_Y.$$

- (c)  $\mathbb{E}[\sum_E \beta^{s_{\text{end}}(E)} p_{\text{end}}(E)] \leq S_H^\beta := \sum_E P_H(E) 2^{\omega_Y(M)} \beta^{\omega(M)}$ .
- (d) *For every prime  $\ell > Y$ ,  $\mathbb{E}[\tilde{w}_\ell(\text{end})^2] \leq B_2(\ell)$ , where*

$$B_2(\ell) = \sum_{\substack{E, E' \\ \ell \mid M, \ell \mid M'}} \beta^{\omega(M) + \omega(M')} 2^{\omega_Y(M) + \omega_Y(M')} P_H(E) P_H(E') \\ \times \varphi\left(\frac{\gcd(M, M')}{\ell^{\min(v_\ell(M), v_\ell(M'))}}\right).$$

$$\text{Hence } \mathbb{P}(\exists \ell > Y : \tilde{w}_\ell(\text{end}) > \eta) \leq \eta^{-2} \sum_{\ell > Y} B_2(\ell).$$

*All sums are over atoms  $E = (M, D)$ ,  $E' = (M', D')$ .*

*Proof.* (a) Consider a step at  $(\ell, a)$ . Atoms with  $\ell \nmid M$ , or with  $v_\ell(M) \leq a$ , are unaffected: their class modulo  $\ell^{v_\ell(M)}$  is already determined by the fibre, and the step only refines it at higher levels. Let  $v_\ell(M) > a$ . If  $a \geq 1$ , the new class is uniform among the  $\ell$  classes of the fibre at level  $a+1$ , so  $\mathbb{E}[p_{i+1}(E) \mid \text{past}] = p_i(E)$ , and  $u$  is unchanged. If  $a = 0$ , the conditional probability of the event becomes  $\ell^{1-v}$  (with  $v = v_\ell(M)$ ) if  $r' \equiv -4D \pmod{\ell}$  and 0 otherwise; the class  $r' \bmod \ell$  is uniform on the  $(\ell-1)/2$  squares, so

$$\mathbb{E}[p_{i+1}(E) \mid \text{past}] = p_i(E) \varphi(\ell^v) \ell^{1-v} \frac{2}{\ell-1} \mathbf{1}\left[\left(\frac{-4D}{\ell}\right) = 1\right] \\ = \left(1 + \left(\frac{-d}{\ell}\right)\right) p_i(E) \leq 2p_i(E)$$

by Lemma 2.2(a), while  $u$  drops by one if  $\ell \leq Y$  (forced steps and all other steps have  $\ell \leq Y$ ). So each term  $p_i(E) 2^{u_i(E)} \phi(E)$  is a supermartingale. At the start  $Q = 8$ ,  $r = 1$  and  $M$  is odd, so  $p_0(E) = P_H(E)$ ; and  $u_0(E) = \omega_Y(M)$ .

(b) For a pair  $(\ell, a)$  with  $\ell \leq Y$  let  $\tau_{\ell,a}$  be the stage at which the (unforced) step at  $(\ell, a)$  is taken,  $\infty$  if it is never taken. At that stage  $\tilde{w}_\ell > \eta$ , so

$$\log \ell \cdot \mathbf{1}[\tau_{\ell,a} < \infty] \leq \frac{\log \ell}{\eta} G_{\tau_{\ell,a}}^{(\ell,a)} \mathbf{1}[\tau_{\ell,a} < \infty],$$

$$G_i^{(\ell,a)} = \sum_{E: v_\ell(M) \geq a+1} p_i(E) 2^{u_i(E)} \beta^{\omega(M)},$$

since at stage  $\tau_{\ell,a}$  the atoms with  $\ell \in \text{supp } E$  are those with  $v_\ell(M) \geq a+1$ , and  $\beta^s p \leq \beta^\omega 2^u p$ . The stopping times are bounded, and optional stopping for the nonnegative supermartingale  $G^{(\ell,a)}$  (part (a) with  $\phi = \beta^{\omega(M)} \mathbf{1}[v_\ell(M) \geq a+1]$ ) gives  $\mathbb{E}[G_{\tau}^{(\ell,a)} \mathbf{1}[\tau < \infty]] \leq G_0^{(\ell,a)}$ . Summing over  $\ell \leq Y$  and  $a < v_\ell(M)$ , and using  $\sum_{\ell \leq Y} v_\ell(M) \log \ell = \log M_Y$ , gives the claim; the forced steps cost  $\log 105$ , and  $\log Q \leq \log 8 + \log 105 + \sum_{\text{steps}} \log \ell$ .

(c) Part (a) with  $\phi = \beta^{\omega(M)}$ , and  $s_{\text{end}} \leq \omega$ .

(d) The prime  $\ell$  is never stepped, so for  $\ell \mid M$ ,  $p_i(E) = p_i(E^{(\ell)})/\varphi(\ell^{v_\ell(M)})$ , where  $E^{(\ell)}$  is the event with its condition at  $\ell$  removed, and

$$\tilde{w}_\ell^2 \leq \sum_{E, E'} \beta^{\omega(M) + \omega(M')} p(E) p(E'),$$

the sum over atoms with  $\ell \mid M$ ,  $\ell \mid M'$ . For two events  $F, F'$  (conditions at sets of primes, with given residues) put  $\Pi_i = p_i(F) p_i(F') R_i$ , where  $R_i = \prod_{\ell'} \rho_i(\ell')$  over the primes  $\ell'$  constrained by both, and  $\rho_i(\ell')$  is the number of classes modulo  $\ell'^j$  in the current fibre,  $j \leq \min(v_{\ell'}(F), v_{\ell'}(F'))$  being the depth to which the two prescribed residues agree ( $\rho_i(\ell') = 1$  once  $a_{\ell'} \geq j$ ). Then  $p_i(F) p_i(F') \leq \Pi_i$ , and  $\Pi_i 2^{u_i(F) + u_i(F')}$  is a supermartingale. Indeed, consider a step revealing level  $a+1$  at  $\ell'$ . If neither event constrains  $\ell'$  at this level, nothing changes. If only one does, its factor behaves as in (a), and its  $\rho$  is already 1. If both do and they agree at this level, both probabilities are multiplied by the same factor  $N \cdot \mathbf{1}[\text{match}]$ , where  $N$  is the number of classes of the current fibre at this level ( $N = \ell' - 1$  if  $a = 0$ ,  $N = \ell'$  if  $a \geq 1$ ), and  $\rho$  is divided by  $N$ ; so  $\Pi \mapsto N \mathbf{1}[\text{match}] \Pi$ , whose mean is  $\Pi$  if  $a \geq 1$  and  $(1 + (\frac{-d}{\ell'})) \Pi \leq 2\Pi$  if  $a = 0$ , in which case both  $u$ 's drop. If both constrain  $\ell'$  and disagree at this level, they cannot both match, so  $\Pi \mapsto 0$ . At the start  $R_0 \leq \prod \varphi(\ell'^j) \leq \varphi(\gcd(M_F, M_{F'}))$ , so  $\mathbb{E}[\Pi_{\text{end}}] \leq 2^{\omega_Y(M_F) + \omega_Y(M_{F'})} P_H(F) P_H(F') \varphi(\gcd(M_F, M_{F'}))$ . Apply this to  $F = E^{(\ell)}$ ,  $F' = E'^{(\ell)}$ , multiply by  $\varphi(\ell^{v_\ell(M)})^{-1} \varphi(\ell^{v_\ell(M')})^{-1}$  and sum. The last claim is Chebyshev's inequality and a union bound.  $\square$

Part (d) is the reason for the random class: the second moment of the mass at a prime that is never quarantined is controlled by a Haar pair-correlation sum, a deterministic divisor sum in which  $\gcd(M, 4D+1)$  does not occur.

### 3. HAAR MASSES AND THE UPPER BOUND ON THE HAAR SIDE

The only analytic input of this section is the following theorem. For fixed  $k \geq 1$ ,  $A, B \geq 1$  and  $\varepsilon > 0$ , let  $\mathcal{M}_k(A, B, \varepsilon)$  be the class of non-negative functions  $F$  of  $k$  variables with

$$F(a_1 b_1, \dots, a_k b_k) \leq \min(A^{\Omega(a_1 \cdots a_k)}, B(a_1 \cdots a_k)^\varepsilon) F(b_1, \dots, b_k)$$

whenever  $\gcd(a_1 \cdots a_k, b_1 \cdots b_k) = 1$ .

**Theorem 3.1** (Nair–Tenenbaum [30, Thm. 1]; cited in the form of [16, (1.1)]). *Let  $Q_1, \dots, Q_k \in \mathbb{Z}[X]$  be pairwise coprime irreducible polynomials,  $Q = Q_1 \cdots Q_k$  of degree  $g$ , without fixed prime divisor. Let  $\rho_{Q_j}(n)$  and  $\rho(n)$  be the numbers of zeros of  $Q_j$  and of  $Q$  modulo  $n$ . Let  $0 < \alpha, \delta < 1$ ,  $A, B \geq 1$ ,  $0 < \varepsilon \leq \alpha\delta/(12g^2)$  and  $F \in \mathcal{M}_k(A, B, \varepsilon)$ . Then, uniformly in  $x \geq c_0 \|Q\|^\delta$  and  $x^\alpha < y \leq x$ ,*

$$\begin{aligned} \sum_{x < n \leq x+y} F(|Q_1(n)|, \dots, |Q_k(n)|) &\ll y \prod_{p \leq x} \left(1 - \frac{\rho(p)}{p}\right) \\ &\times \sum_{n_1 \cdots n_k \leq x} F(n_1, \dots, n_k) \frac{\rho_{Q_1}(n_1) \cdots \rho_{Q_k}(n_k)}{n_1 \cdots n_k}, \end{aligned}$$

where  $c_0$  and the implied constant depend only on  $k, g, A, B, \varepsilon, \alpha, \delta$  and the discriminant of  $Q$ .

We use it with  $k = 2$ ,  $Q_1(n) = n$ ,  $Q_2(n) = 4n - 1$ : these are coprime and irreducible,  $Q = 4n^2 - n$  has discriminant 1 and no fixed prime divisor ( $Q(1) = 3$ ,  $Q(2) = 14$ ),  $\rho(2) = 1$ ,  $\rho(p) = 2$  for odd  $p$ ,  $\rho_{Q_1} \equiv 1$  and  $\rho_{Q_2}(m) = 1[m \text{ odd}]$ . We take  $y = x$ ,  $\alpha = \delta = 1/2$  and  $\varepsilon = 1/200$ . (Henriot notes that NT's own class  $\mathcal{M}_k$  requires the inequality whenever  $\gcd(a_i, b_i) = 1$  for each  $i$ , so the class above is slightly larger and Theorem 3.1 is Henriot's reading of NT's proof. Our functions  $F$ , products of multiplicative functions of separate variables, lie in NT's original class, and NT's own range ( $\varepsilon < 1/(8g^2)$ ,  $x^{4g^{2\varepsilon}} \leq y \leq x$ ) covers  $\varepsilon = 1/200$ ,  $g = 2$ ,  $y = x$ ; so for our use the citation rests on [30, Thm. 1] alone.)

Throughout,  $\beta = 1 + 1/\log \mathcal{L}$ ,  $2 \leq Y \leq T$ ,  $H(M) = \beta^{\omega(M)} 2^{\omega_Y(M)}$ , and

$$w(M) = \tau(A_M^2) H(M) \frac{M}{\varphi(M)} \quad (M \equiv 3 \pmod{4}), \quad w(M) = 0 \text{ otherwise.}$$

Summing over  $D \mid A_M^2$  gives  $\sum_D H(M) P_H(E_{M,D}) = w(M)/M$ . Sums over  $M$  run over  $M \leq T$ .

**Lemma 3.2** (Haar masses; proved, (A) modulo Theorem 3.1). *Let  $2 \leq Y \leq T$  and  $\beta = 1 + 1/\log \mathcal{L}$ .*

(A) We have

$$S_H^\beta = \sum_M \frac{w(M)}{M} \ll \mathcal{L}^3 \log Y,$$

$$\sum_E P_H(E) 2^{\omega_Y(M)} \beta^{\omega(M)} \log M_Y \ll \mathcal{L}^3 (\log Y)^4.$$

(B)  $\sum_{\ell > Y} \text{prime} B_2(\ell) \leq \frac{\mathcal{L}+1}{Y} \Xi$ , where  $\Xi = \sum_M w(M)^2 \tau(M) \omega(M)/M \ll \mathcal{L}^{C_0}$  for an absolute constant  $C_0$ , uniformly in  $2 \leq Y \leq T$ .

*Proof.* (A) Write  $n = A_M$ , so  $M = 4n - 1$ , and  $f_2(m) = H(m)m/\varphi(m)$ , a multiplicative function with  $f_2(p^k) = \beta 2^{[p \leq Y]} p/(p-1) \leq 6$  for odd  $p$  (as  $\beta \leq 2$ ). The function  $F(n_1, n_2) = \tau(n_1^2) f_2(n_2)$  satisfies  $F(a_1 b_1, a_2 b_2) = \tau(a_1^2) f_2(a_2) F(b_1, b_2)$  when  $\gcd(a_1 a_2, b_1 b_2) = 1$ , and  $\tau(a_1^2) f_2(a_2)$  is at most  $8^{\Omega(a_1 a_2)}$  and at most  $B(a_1 a_2)^{1/200}$  for an absolute  $B$  (divisor bound). So  $F \in \mathcal{M}_2(8, B, 1/200)$ , uniformly in  $T$  and  $Y$ . Theorem 3.1 on  $x < n \leq 2x$  gives

$$\sum_{x < n \leq 2x} \tau(n^2) f_2(4n-1) \ll \frac{x}{(\log x)^2} \sum_{n_1 \leq x} \frac{\tau(n_1^2)}{n_1} \sum_{n_2 \leq x} \frac{f_2(n_2)}{n_2}$$

$$\ll \frac{x}{(\log x)^2} (\log x)^3 (\log x)^\beta (\log Y)^\beta,$$

by the Euler products  $\prod_{p \leq x} (1 + 3/p + \dots)$  and

$$\prod_{p \leq Y} \left(1 + \frac{2\beta}{p-1} + O(p^{-2})\right) \prod_{Y < p \leq x} \left(1 + \frac{\beta}{p-1} + O(p^{-2})\right).$$

Since  $(\log x)^{\beta-1}, (\log Y)^{\beta-1} \leq \mathcal{L}^{1/\log \mathcal{L}} = e$ , this is  $\ll x(\log x)^2 \log Y$ . As  $\tau(A_M^2) H(M)/\varphi(M) = \tau(n^2) f_2(M)/M$  with  $M \asymp x$ , the block contributes  $\ll \mathcal{L}^2 \log Y$ ; the  $O(\mathcal{L})$  dyadic blocks (and the bounded range  $x < c_0$ ) give the first bound. For the second, use  $\log M_Y \leq \log Y \cdot \Omega(M_Y) \leq \log Y \cdot \tau(M_Y)$  (as  $\sum k_i \leq \prod (k_i + 1) - 1$ ), and repeat the argument with  $f_2(m) \tau(m_Y)$  in place of  $f_2$ . Its values at  $p^k$  are at most  $6(k+1) \leq 12^k$  and  $\ll_\varepsilon p^{k\varepsilon}$ , so the class condition holds uniformly; its Euler factor at  $p \leq Y$  is  $1 + 4\beta/(p-1) + O(p^{-2})$ , so  $\sum_{n_2 \leq x} f_2(n_2) \tau((n_2)_Y)/n_2 \ll (\log x)(\log Y)^3$ , and the bound  $\mathcal{L}^3 (\log Y)^4$  follows.

(B) For an atom pair with  $\ell \mid M$ ,  $\ell \mid M'$  write  $M = \ell m$ ,  $M' = \ell m'$ . Then  $\gcd(M, M')/\ell^{\min(v_\ell(M), v_\ell(M'))}$  divides  $\gcd(m, m')$ , so its  $\varphi$  is at most  $\gcd(m, m') = \sum_{e \mid (m, m')} \varphi(e)$ . Summing over  $D, D'$ ,

$$B_2(\ell) \leq \sum_e \varphi(e) V(e\ell)^2, \quad V(q) = \sum_{q \mid M} \frac{w(M)}{M} = \sum_{k \leq T/q} \frac{w(qk)}{qk}.$$

By Cauchy-Schwarz in  $k$ ,  $V(q)^2 \leq (\mathcal{L} + 1) \sum_k w(qk)^2/(q^2 k)$ . Put  $N = \ell k$ : the term  $\varphi(e) w(N)^2/(e^2 \ell^2 k)$  equals  $(w(N)^2/N) \varphi(e)/(e\ell) \leq (w(N)^2/N)/\ell$ ,



and for fixed  $\ell$  there are at most  $\tau(N)$  factorisations  $N = e\ell k$ . So

$$\sum_{\ell > Y} B_2(\ell) \leq (\mathcal{L} + 1) \sum_N \frac{w(N)^2 \tau(N)}{N} \sum_{\ell | N, \ell > Y} \frac{1}{\ell} \leq \frac{\mathcal{L} + 1}{Y} \Xi.$$

For  $\Xi$ , use  $\omega \leq \tau$ ,  $2^{\omega_Y} \leq 2^\omega$  and Cauchy–Schwarz with  $M = 4n - 1 \geq 3n$ :

$$\Xi \leq \left( \sum_{n \leq T} \frac{\tau(n^2)^4}{n} \right)^{1/2} \left( \sum_{m \leq 4T} \frac{g(m)^2}{m} \right)^{1/2},$$

$$g(m) = \beta^{2\omega(m)} 4^{\omega(m)} \left( \frac{m}{\varphi(m)} \right)^2 \tau(m)^2.$$

Both are mean values of non-negative multiplicative functions  $f$  with  $f(p^k) \leq C(k+1)^C$  for an absolute  $C$ ; bounding them by  $\prod_{p \leq 4T} (1 + f(p)/p + f(p^2)/p^2 + \dots)$  gives  $\ll \mathcal{L}^{C_0}$ , with  $C_0$  absolute (since  $\beta \leq 2$ ) and independent of  $Y$ .  $\square$

The proof of (A) is the only place where Theorem 3.1 is used. All masses here are Haar masses, so the Elsholtz–Tao estimates are not needed.

Let  $\delta^*(T)$  be the Haar measure of  $\{n \in \hat{\mathbb{Z}}^\times : W(n) > T\}$  relative to the class  $n \equiv 1 \pmod{24}$ , that is, the proportion, among the units modulo  $L = \text{lcm}(24, M \leq T)$  that are  $\equiv 1 \pmod{24}$ , of those with  $W(n) > T$  (the condition  $W(n) > T$  depends only on  $n \pmod{L}$ ). On all of  $\hat{\mathbb{Z}}^\times$  the proportion is exactly half of this, because the atom  $(3, 1)$  excludes  $n \equiv 2 \pmod{3}$ .

**Theorem 3.3** (Haar side, upper bound; proved modulo Theorem 3.1). *For large  $T$  there is a square-class quarantine  $(Q, r)$  with  $840 \mid Q$ ,  $r \equiv 1 \pmod{24}$ , all prime factors of  $Q$  at most  $Y := \mathcal{L}^{C_0+4}$  (with  $C_0$  from Lemma 3.2), and, with  $\beta = 1 + 1/\log \mathcal{L}$  and  $\eta = \frac{3}{4} \log \beta$ ,*

$$\tilde{w}_\ell \leq \eta \quad (\ell \in \mathcal{P}), \quad S_\beta \leq C \mathcal{L}^3 \log \mathcal{L}, \quad \log Q \leq C \mathcal{L}^3 (\log \mathcal{L})^5, \quad (4)$$

where  $C$  is absolute. Consequently  $\delta = \mathbb{E}F \geq \exp(-\frac{4}{3}S_\beta)$  and

$$\log \frac{1}{\delta^*(T)} \leq \log \varphi(Q) + \frac{4}{3} S_\beta \ll \mathcal{L}^3 (\log \mathcal{L})^5.$$

*Proof.* Run the square-class process with this  $Y$ . As  $1/\eta \asymp \log \mathcal{L}$  and  $\log Y \asymp \log \mathcal{L}$ , Lemma 2.9(b,c) and Lemma 3.2(A) give  $\mathbb{E}[\log Q_{\text{end}}] \ll \mathcal{L}^3 (\log \mathcal{L})^5$  and  $\mathbb{E}[S_\beta(\text{end})] \ll \mathcal{L}^3 \log \mathcal{L}$ . By Lemma 2.9(d) and Lemma 3.2(B),  $\mathbb{P}(\exists \ell > Y : \tilde{w}_\ell > \eta) \leq \eta^{-2} (\mathcal{L} + 1) \mathcal{L}^{C_0} / Y \leq 1/4$  for large  $T$ . By Markov's inequality each of the events  $\log Q_{\text{end}} > 4\mathbb{E}[\log Q_{\text{end}}]$  and  $S_\beta > 4\mathbb{E}[S_\beta]$  has probability at most  $1/4$ . So some realisation  $(Q, r)$  avoids all three. In it every coordinate  $\ell \leq Y$  has  $\tilde{w}_\ell \leq \eta$  because the process stopped, and every  $\ell > Y$  by the choice of realisation; this is (4). Lemma 2.8 gives  $\delta \geq \exp(-\frac{4}{3}S_\beta)$ . By Lemma 2.5(i), the units  $n \equiv r \pmod{Q}$  with  $F(n) = 1$  have  $W(n) > T$ . Among the units  $\equiv 1 \pmod{24}$  the class  $r \pmod{Q}$  has relative Haar measure  $\varphi(24)/\varphi(Q)$ , as  $r \equiv 1 \pmod{24}$ , and within it the residues  $n \pmod{\ell^{\ell_e}}$ ,  $\ell \in \mathcal{P}$ , are independent and uniform on  $\Omega_\ell$ . So  $\delta^*(T) \geq (\varphi(24)/\varphi(Q)) \delta$ .  $\square$

Together with Theorem 4.4 below, this determines the Haar exponent,  $\log \log(1/\delta^*(T)) = (3+o(1)) \log \mathcal{L}$ . The quarantine costs  $\mathcal{L}^3(\log \mathcal{L})^5$ ; the class-of-one route of the previous version gave  $\log(1/\delta^*(T)) \ll \mathcal{L}^5 \log \mathcal{L}$  modulo the Elsholtz–Tao estimates.

#### 4. A JANSON-TYPE INEQUALITY AND THE LOWER BOUND ON THE HAAR SIDE

In this section  $(X_v)$  are independent random variables with finite or countable ranges, and an *atomic event* is an event  $E = \{X_v = a_{E,v} \text{ for } v \in S_E\}$  with  $S_E$  finite and single values  $a_{E,v}$ . For a family  $\mathcal{F}$  put  $\text{Av}(\mathcal{F}) = \bigcap_{E \in \mathcal{F}} \bar{E}$ . Two atomic events *conflict* if they prescribe different values on some common coordinate (then  $E \cap E' = \emptyset$ ), and *share a bit* if  $S_E \cap S_{E'} \neq \emptyset$  and they agree on all common coordinates (then  $\mathbb{P}(E \cap E') > 0$ ). Harris' inequality is not available here: for one-hot variables  $\mathbb{P}(X = a \mid X \neq a') > \mathbb{P}(X = a)$ . The following lemmas replace it.

**Lemma 4.1** (Compatible events; proved). *Let  $A$  be atomic and let  $\mathcal{C}$  be a family of atomic events none of which conflicts with  $A$ . Then  $\mathbb{P}(A \cap \text{Av}(\mathcal{C})) \leq \mathbb{P}(A)\mathbb{P}(\text{Av}(\mathcal{C}))$ .*

*Proof.* Condition on  $A$ . For  $C \in \mathcal{C}$ ,  $C \cap A = A \cap C'$ , where  $C'$  is  $C$  with the coordinates in  $S_A$  deleted;  $C'$  is independent of  $(X_v)_{v \in S_A}$  and  $C' \supseteq C$ . So  $\mathbb{P}(\text{Av}(\mathcal{C}) \mid A) = \mathbb{P}(\bigcap \bar{C}') \leq \mathbb{P}(\bigcap \bar{C})$ .  $\square$

**Lemma 4.2** (Lopsided local lemma; proved, standard). *Let  $\mathcal{F}$  be a family of atomic events,  $\Gamma(E)$  the set of  $E' \in \mathcal{F}$  conflicting with  $E$ , and suppose  $x_E \in (0, 1)$  satisfy  $\mathbb{P}(E) \leq x_E \prod_{E' \in \Gamma(E)} (1 - x_{E'})$ . Then for every atomic event  $A$  and every  $\mathcal{S} \subseteq \mathcal{F}$ ,*

$$\mathbb{P}(A \mid \text{Av}(\mathcal{S})) \leq \mathbb{P}(A) \prod_{E' \in \mathcal{S} \text{ conflicting with } A} (1 - x_{E'})^{-1}.$$

*Proof.* By Lemma 4.1,  $\mathbb{P}(E \mid \text{Av}(\mathcal{S}')) \leq \mathbb{P}(E)$  whenever no event of  $\mathcal{S}'$  conflicts with  $E$ , so the lopsided local lemma of Erdős and Spencer [10] applies with the graph  $\Gamma$ ; its proof gives  $\mathbb{P}(E \mid \text{Av}(\mathcal{S})) \leq x_E$  for  $E \notin \mathcal{S}$ . For  $A$ , let  $\mathcal{S}_1$  be the events of  $\mathcal{S}$  conflicting with  $A$  and  $\mathcal{S}_2 = \mathcal{S} \setminus \mathcal{S}_1$ . Then  $\mathbb{P}(A \mid \text{Av}(\mathcal{S})) \leq \mathbb{P}(A \mid \text{Av}(\mathcal{S}_2)) / \mathbb{P}(\text{Av}(\mathcal{S}_1) \mid \text{Av}(\mathcal{S}_2))$ ; the numerator is at most  $\mathbb{P}(A)$  by Lemma 4.1, and the chain rule bounds the denominator below by  $\prod_{S_1} (1 - x_E)$ .  $\square$

**Theorem 4.3** (Janson-type inequality for atomic events; proved). *Under the hypothesis of Lemma 4.2 put  $K = \max_{E \in \mathcal{F}} \prod_{E' \in \Gamma(E)} (1 - x_{E'})^{-1}$ ,  $\mu = \sum_{E \in \mathcal{F}} \mathbb{P}(E)$ , and  $\Delta = \sum \mathbb{P}(E \cap E')$  over the unordered pairs of distinct events of  $\mathcal{F}$  that share a bit. Then*

$$-\log \mathbb{P}(\text{Av}(\mathcal{F})) \geq \mu - K\Delta, \quad -\log \mathbb{P}(\text{Av}(\mathcal{F})) \geq \min\left(\frac{\mu}{2}, \frac{\mu^2}{4K\Delta}\right).$$

*Proof.* Order  $\mathcal{F} = \{E_1, \dots, E_N\}$  and put  $B_i = \text{Av}(\{E_j : j < i\})$ , so  $-\log \mathbb{P}(\text{Av}(\mathcal{F})) = -\sum_i \log(1 - \mathbb{P}(E_i | B_i)) \geq \sum_i \mathbb{P}(E_i | B_i)$ . Let  $B_i^d = \text{Av}(\{E_j : j < i, S_j \cap S_i = \emptyset\}) \supseteq B_i$ . Then  $\mathbb{P}(E_i | B_i) \geq \mathbb{P}(E_i) \mathbb{P}(B_i | E_i) / \mathbb{P}(B_i^d)$ . Given  $E_i$ , the events  $E_j$  ( $j < i$ ) conflicting with  $E_i$  are impossible, those sharing a bit become  $E'_j$  ( $E_j$  with the coordinates of  $S_i$  deleted, so  $\mathbb{P}(E_i) \mathbb{P}(E'_j) = \mathbb{P}(E_i \cap E_j)$ ), and  $B_i^d$  is independent of  $E_i$ . Hence

$$\mathbb{P}(B_i | E_i) \geq \mathbb{P}(B_i^d) - \sum_{j < i, j \text{ shares a bit with } i} \mathbb{P}(E'_j \cap B_i^d).$$

$B_i^d$  is the avoidance event of a subfamily of  $\mathcal{F}$ , and every event conflicting with  $E'_j$  conflicts with  $E_j$ , so Lemma 4.2 gives  $\mathbb{P}(E'_j | B_i^d) \leq K \mathbb{P}(E'_j)$ . Therefore  $\mathbb{P}(E_i | B_i) \geq \mathbb{P}(E_i) - K \sum_{j < i, j \sim i} \mathbb{P}(E_i \cap E_j)$ , and summing over  $i$  gives the first bound. For the second, a subfamily  $\mathcal{F}'$  satisfies the hypothesis with the same  $x_E$  and the same or smaller  $K$ , and  $\mathbb{P}(\text{Av}(\mathcal{F})) \leq \mathbb{P}(\text{Av}(\mathcal{F}'))$ . For a random  $\mathcal{F}'$  with inclusion probability  $p$ ,  $\mathbb{E}\mu' = p\mu$  and  $\mathbb{E}\Delta' = p^2\Delta$ , so some  $\mathcal{F}'$  has  $\mu' - K\Delta' \geq p\mu - p^2K\Delta$ ; take  $p = \min(1, \mu/(2K\Delta))$ .  $\square$

$\Delta$  counts only compatible pairs; conflicting pairs, which carry the structure “one coordinate, many residues”, enter only through  $K$ . If  $x_E = 2\mathbb{P}(E)$ ,  $\mathbb{P}(E) \leq 1/8$  and  $\sum_{E' \in \Gamma(E)} \mathbb{P}(E') \leq 1/8$  for all  $E$ , the hypothesis holds and  $K \leq e^{1/3}$ . Theorem 4.3 was checked by exact enumeration on small random instances (not used). We did not find it in the literature in this form; the Janson inequality itself [22] needs positively correlated (increasing) events, and the lopsided local lemma alone gives upper bounds on  $\mathbb{P}(E | \cdot)$ , not lower bounds. We make no priority claim.

**The lower bound.** For the ES events the coordinates are  $X_\ell = n \bmod \ell^f$  ( $f$  large) for odd primes  $\ell$ , independent and uniform on units (on  $X_3 \equiv 1 \pmod{3}$  in the normalisation of  $\delta^*(T)$ ), and  $E_{M,D} = \bigcap_{\ell|M} \{X_\ell \equiv -4D \pmod{\ell^{v_\ell(M)}}\}$ . For squarefree  $M$  with  $3 \nmid M$  these are atomic events in the variables  $X_\ell \bmod \ell$ , and  $\mathbb{P}(E_{M,D}) = 1/\varphi(M)$ . Distinct pairs  $(M, -4D \bmod M)$  are distinct events.

**Theorem 4.4** (Haar side, lower bound; proved modulo the fundamental lemma). *There are absolute constants  $c > 0$  and  $T_0$  such that for  $T \geq T_0$*

$$\log \frac{1}{\delta^*(T)} \geq c \frac{\mathcal{L}^3}{\log \mathcal{L}}.$$

The only external input is the fundamental lemma of sieve theory [13, Lemma 6.3, Cor. 6.10] (cited), used for a lower bound for rough numbers in progressions to small moduli.  $T_0$  is astronomically large.

*The family.* Put  $y = \mathcal{L}^5$ ,  $N_0 = \mathcal{L}^2$ , and for an integer  $D$  write  $D^* = \prod_p p^{\lfloor v_p(D)/2 \rfloor}$ . Then  $D | A^2 \iff D^* | A$ , and each  $n \geq 1$  equals  $D^*$  for exactly  $2^{\omega(n)}$  integers  $D$  (write  $D = \kappa t^2$  with  $\kappa$  squarefree; then  $D^* = \kappa t$ ).

Let  $\mathcal{F}$  be the set of events  $E_{M,D}$  with

$$\sqrt{T} \leq M \leq T, \quad M \text{ squarefree with all prime factors } > y, \quad M \equiv 3 \pmod{4},$$

$$n := D^* \in [N_0, T^{1/10}], \quad n \mid A_M.$$

We use: (F1)  $D \leq n^2 \leq T^{1/5} < M/4$ , so different  $D$  give different residues  $-4D \pmod{M}$ , and  $\mathbb{P}(E_{M,D}) = 1/\varphi(M)$ ; (F2) for  $y$ -rough squarefree  $m \leq T$ ,  $\omega(m) \leq \mathcal{L}/\log y$  and  $\varphi(m) \geq m/2$ ; (F3)  $\sum_{n \leq x} 2^{\omega(n)}/n \leq (1 + \log x)^2$  and  $\sum_{n \leq x} 2^{\omega(n)} \leq x(1 + \log x)$ ; (F4) for a progression  $v \equiv c \pmod{d}$  in  $[v_0, V]$  with least element  $v_1 \geq v_0$ ,  $\sum 1/v \leq 1/v_1 + \log(V/v_0)/d$ ; (F5) if  $\omega(m)y^{-1} \leq 1$ , then  $\sum_{g|m, g>1, g \text{ squarefree and } y\text{-rough}} g^{-1} \leq (1 + 1/y)^{\omega(m)} - 1 \leq 2\omega(m)/y$ .

**Lemma 4.5** (Mass; proved modulo the fundamental lemma). *We have  $\mu = \sum_{E \in \mathcal{F}} \mathbb{P}(E) \geq c_1 \mathcal{L}^3 / \log y$ .*

*Proof.* By (F1),  $\mu \geq \sum_{N_0 \leq n \leq T^{1/10}} 2^{\omega(n)} \sum_M 1/M$ , over  $y$ -rough squarefree  $M \in [\sqrt{T}, T]$  with  $M \equiv -1 \pmod{4n}$ ; this congruence implies  $M \equiv 3 \pmod{4}$  and  $n \mid A_M$ . For a dyadic block  $(X, 2X] \subseteq [\sqrt{T}, T]$ , sift the progression (of length  $X/(4n) \geq X^{4/5}/4$ ) by the primes  $p < y$ ,  $p \nmid 2n$ , at level  $X^{1/2}$ ; as  $s = \log X/(2 \log y) \rightarrow \infty$ , the fundamental lemma gives at least  $c_2 X/(n \log y)$   $y$ -rough elements. Those divisible by  $p^2$  with  $p \geq y$  number at most  $X/(4n(y-1)) + \pi(\sqrt{2X})$ , which is negligible as  $n \leq X^{1/5}$ . So each of the  $\gg \mathcal{L}$  blocks contributes  $\geq c_2/(4n \log y)$ , and  $\sum_{N_0 \leq n \leq T^{1/10}} 2^{\omega(n)}/n \gg \mathcal{L}^2$ .  $\square$

**Lemma 4.6** (Local-lemma hypothesis; proved). *For  $T \geq T_0$ , every  $E \in \mathcal{F}$  has  $\mathbb{P}(E) \leq 1/8$  and  $\sum_{E' \in \Gamma(E)} \mathbb{P}(E') \leq 1/8$ . Hence Theorem 4.3 applies with  $x_E = 2\mathbb{P}(E)$  and  $K \leq e^{1/3}$ .*

*Proof.*  $\mathbb{P}(E) \leq 2/\sqrt{T}$ . Put  $w_q = \sum_{E' \in \mathcal{F}: q|M'} \mathbb{P}(E')$ . Events conflicting with  $E$  share a prime  $q \mid M$ , so  $\sum_{\Gamma(E)} \mathbb{P} \leq (\mathcal{L}/\log y) \max_{q>y} w_q$ . Writing  $M' = qv$  with  $qv \equiv -1 \pmod{4n'}$  and  $qv \geq \sqrt{T}$ , (F2) and (F4) give  $w_q \leq (2/q) \sum_{D'} (\mathcal{L}/(4n') + \min(1, q/\sqrt{T}))$ . By (F3) the first part is at most  $\mathcal{L}^3/(2q)$ , and the second at most  $4\mathcal{L}T^{-2/5}$ . So  $\sum_{\Gamma(E)} \mathbb{P} \leq (\mathcal{L}/\log y)(\mathcal{L}^3/(2y) + 4\mathcal{L}T^{-2/5}) = O(1/(\mathcal{L} \log \mathcal{L}))$ .  $\square$

**Lemma 4.7** (Pair sum; proved).  $\Delta = \sum_{\text{bit-sharing pairs}} \mathbb{P}(E \cap E') \leq C\mathcal{L}^2$ .

*Proof.* Let  $E = E_{M,D}$ ,  $E' = E_{M',D'}$  be distinct and share a bit; put  $g = \gcd(M, M') > 1$ ,  $M = gv$ ,  $M' = gv'$ . Then  $g$  is  $y$ -rough, so  $g > y$ , and compatibility means  $g \mid D - D'$ . Also  $M \neq M'$ : otherwise  $M \mid D - D'$  and  $E = E'$  by (F1). By (F2),  $\mathbb{P}(E \cap E') = 1/(\varphi(g)\varphi(v)\varphi(v')) \leq 8/(gvv')$ . For fixed  $g$  and  $D$  the admissible  $v$  satisfy  $gv \equiv -1 \pmod{4n}$  and  $gv \geq \sqrt{T}$ , so by (F4)

$$\sigma(g, n) := \sum_v \frac{1}{v} \leq \frac{\mathcal{L}}{4n} + \min\left(1, \frac{g}{\sqrt{T}}\right). \quad (5)$$

(a)  $D = D'$ . Then  $v \neq v'$  and  $v \equiv v' \equiv -g^{-1} \pmod{4n}$ . For fixed  $(v, n)$ ,  $g$  runs over a progression modulo  $4n$  with  $g > y$ , so  $\sum_g 1/g \leq 1/y + \mathcal{L}/(4n)$ ; and  $\sum_{v < v', v' \equiv v \pmod{4n}} 1/(vv') \leq (1 + \mathcal{L})^2/(4n)$ . So  $\Delta_a \leq 8 \sum_{N_0 \leq n \leq T^{1/10}} 2^{\omega(n)} (1/y + \mathcal{L}/(4n)) (1 + \mathcal{L})^2/(4n) \ll \mathcal{L}^4/y + \mathcal{L}^3 \log N_0/N_0 \ll \mathcal{L} \log \mathcal{L}$ .

(b)  $D \neq D'$ . Now  $g \mid m = |D - D'|$  with  $0 < m < T^{1/5}$ , so  $g < T^{1/5}$  and (5) gives  $\sigma(g, n) \leq \mathcal{L}/(4n) + T^{-3/10}$ . Summing over  $g$  with (F5),

$$\Delta_b \leq \frac{2\mathcal{L}}{y} \left[ \frac{\mathcal{L}^2}{2} \left( \sum_D \frac{1}{n} \right)^2 + 4T^{-3/10} \mathcal{L} \# \mathcal{D} \sum_D \frac{1}{n} + 8T^{-3/5} (\# \mathcal{D})^2 \right],$$

where  $\mathcal{D}$  is the set of  $D$  with  $D^* \leq T^{1/10}$ , so  $\sum_D 1/n \leq (1 + \mathcal{L})^2$  and  $\# \mathcal{D} \leq T^{1/10} (1 + \mathcal{L})$  by (F3). So  $\Delta_b \ll \mathcal{L}^7/y + T^{-1/10} \mathcal{L}^5 \ll \mathcal{L}^2$ .  $\square$

*Proof of Theorem 4.4.* The events of  $\mathcal{F}$  involve only primes  $> y > 3$ , so they are independent of  $n \pmod{24}$ , and  $\delta^*(T) \leq \mathbb{P}(\text{Av}(\mathcal{F}))$ . By Theorem 4.3 and Lemmas 4.5–4.7,  $\log(1/\delta^*(T)) \geq \mu - e^{1/3} \Delta \geq c_1 \mathcal{L}^3/(5 \log \mathcal{L}) - C \mathcal{L}^2$ .  $\square$

Two restrictions make the pair sum small:  $M \geq \sqrt{T} \geq (D^*)^5$  makes the first terms of the progressions in (5) harmless, and  $D^* \geq \mathcal{L}^2$  removes the small- $D$  hubs ( $D = 1$  has residue  $-4$  at every prime), whose same- $D$  pairs would make  $\Delta_a \asymp \mu$ . The factor  $\log \mathcal{L}$  is the price of  $y$ -roughness. A Monte Carlo estimate of  $\delta^*(T)$  for  $1023 \leq T \leq 32767$  fits  $\log(1/\delta^*(T)) \approx 0.07 \mathcal{L}^3/\log \mathcal{L}$  (evidence only, not used; at these  $T$  it cannot distinguish  $\mathcal{L}^3/\log \mathcal{L}$  from, say,  $\mathcal{L}^{2.5}$ ).

## 5. THE ANALYTIC INPUT: GALLAGHER'S THEOREM

For a Dirichlet character  $\chi$  put  $\vartheta(x; \chi) = \sum_{p \leq x} \chi(p) \log p$ , and let  $E_0(\chi) = 1$  if  $\chi$  is the trivial character modulo 1 and  $E_0(\chi) = 0$  otherwise.  $\sum_{\chi \bmod q}^*$  is a sum over primitive characters.

**Theorem 5.1** (Landau–Page; classical, [8, Ch. 14], [25, Cor. 11.10]). *There are absolute effective positive constants  $c_1, c_2$  such that, for every  $Y \geq 3$ ,  $\prod_{q \leq Y} \prod_{\chi \bmod q}^* L(s, \chi)$  has at most one zero in the region  $\Re s > 1 - c_1/\log Y$ ,  $|\Im s| \leq Y$ . If this zero  $\beta_1$  exists, it is real and simple, and its character  $\chi_1$  is quadratic. Moreover, every real zero  $\beta$  of  $L(s, \chi)$ , for  $\chi$  a real primitive character of conductor  $q \geq 3$ , satisfies  $1 - \beta \geq c_2^{-1} q^{-1/2} (\log q)^{-2}$ .*

The first part is the classical zero-free region with the Landau–Page theorem. The height restriction matters: without it the statement would be a zero-free region of width  $\asymp 1/\log Y$  uniformly in  $|\Im s|$ , which is not known. The “Exceptional Zero Statement” (28.61)–(28.62) of [26], which refers to [25, Cor. 11.10], omits the height restriction. The second part is the effective Page-type bound [8, Ch. 14], [26, (28.62)]. We use Theorem 5.1 only for *real* zeros: for the uniqueness of a real zero close to 1, for the quadraticity of its character, and for the lower bound for  $1 - \beta$ .

**Theorem 5.2** (Gallagher [14]; cited in the form of [26, Thm. 28.19]). *There are absolute constants  $c \geq 1$  and  $\kappa_0 \geq 3$  with the following property. Let  $\kappa \geq \kappa_0$  be fixed and  $1 < Q_G^{6c} \leq x$ . Then*

$$\sum_{q \leq Q_G} \sum_{\chi \bmod q}^* |\vartheta(x; \chi) - E_0(\chi)x| \ll_{\kappa} x \exp\left(-\frac{\log x}{\kappa \log Q_G}\right) + \left(\frac{\log x}{\log Q_G}\right)^2 \frac{x}{Q_G},$$

*unless  $\prod_{q \leq Q_G} \prod_{\chi \bmod q}^* L(s, \chi)$  has a real zero  $\beta_1$  with  $1 - \beta_1 < 1/(\kappa \log Q_G)$ . In that case the term of the exceptional character  $\chi_1$  on the left is replaced by  $|\vartheta(x; \chi_1) + x^{\beta_1}/\beta_1|$ , and the right-hand side by*

$$(1 - \beta_1)(\log x) \left( x \exp\left(-\frac{\log x}{\log Q_G}\right) + \frac{x \log x}{Q_G \log Q_G} \right).$$

*Source caveat.* We quote [26], an unpublished draft (third volume of Montgomery–Vaughan), archived by us. We have not seen Gallagher’s paper [14], so we cite it without a theorem number. The printed proof of [26, Thm. 28.19] has visible slips. Its case split is at  $1 - \beta_1 < 1/(3\kappa_0 \log Q_G)$ , so it establishes the statement for  $\kappa = 3\kappa_0$ . The statement for every larger  $\kappa$  follows from that case. Let  $\kappa \geq 3\kappa_0$ , and suppose a real zero satisfies  $1/(\kappa \log Q_G) \leq 1 - \beta_1 < 1/(3\kappa_0 \log Q_G)$ . Then  $x^{\beta_1}/\beta_1 \leq 2x \exp(-\log x/(\kappa \log Q_G))$ , and  $(1 - \beta_1)(\log x) x \exp(-\log x/\log Q_G) \ll x \exp(-\log x/(2 \log Q_G))$ . So the exceptional form of the bound implies the non-exceptional one with parameter  $\kappa$ . We therefore fix

$$\kappa := \max(3\kappa_0, 1/c_1).$$

An exceptional zero as in Theorem 5.2 is then a real zero in the region of Theorem 5.1 with  $Y = Q_G$ , so it is unique and  $\chi_1$  is real. Write  $C_G$  for the implied constant in Theorem 5.2 with this  $\kappa$ .

Gallagher’s original statement may carry a range condition of the form  $\exp(\sqrt{\log x}) \leq Q_G \leq x^b$ ; we have not checked this. Our use is robust to such a condition. In the proof of Theorem 6.1,  $\log x / \log Q_G = \kappa L \geq 6c$ , and  $\log x \geq (\kappa L)^2$  by the hypothesis of that theorem, so  $Q_G \geq \exp(\sqrt{\log x})$ . In Lemma 10.4,  $Q_G = 4T^\varepsilon$  and  $x \leq T^{0.7}$ , so again  $\exp(\sqrt{\log x}) \leq Q_G \leq x$ . The statement itself is the standard form of Gallagher’s prime number theorem for the whole family of characters of conductor  $\leq Q_G$ , which underlies the usual proofs of Linnik’s theorem.

## 6. A LINEAR TRANSFER THEOREM

**Theorem 6.1** (Linear transfer; proved modulo Theorems 5.1 and 5.2). *Let  $T \geq 1$  and  $Q \geq 2$  with  $8 \mid Q$ , and let  $r$  be a unit modulo  $Q$  with  $r \equiv 1 \pmod{8}$  that is a quadratic residue modulo every odd prime dividing  $Q$ . Let  $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$  be a finite real combination of consistent unit cells:  $\gcd(b_i, d_i) = 1$  and  $b_i \equiv r \pmod{\gcd(d_i, Q)}$ . Assume that  $B(n) \leq \mathbf{1}[W(n) > T]$  for every  $n \equiv r \pmod{Q}$  coprime to all  $d_i$ . Put  $D = \text{lcm}_i d_i$ ,  $N_0 = \text{lcm}(Q, D)$ ,  $H = \{n \in (\mathbb{Z}/N_0)^\times : n \equiv r \pmod{Q}\}$ , and let  $\mathbb{E}_H$  be the*

mean over  $H$ . Put

$$Z = Q \max_i d_i, \quad \mu = \mathbb{E}_H B, \quad A = \frac{\mathbb{E}_H |B|}{\mu},$$

and assume  $\mu > 0$ ,  $A \leq Z^{1/4}$ , and the twist condition: for every real primitive character  $\psi$  of conductor  $f > 1$  with  $\gcd(f, Q) = 1$  and  $f \mid d_i$  for some  $i$ ,

$$|\mathbb{E}_H [B\psi]| \leq \mu/4.$$

There is an absolute effective constant  $C_2$  such that, if

$$\log x \geq C_2(1 + \log A) \log Z,$$

then some prime  $p \leq x$  with  $p \equiv r \pmod{Q}$  and  $p \nmid D$  has  $W(p) > T$ .

In the second version  $r = 1$  and the cells were coprime to  $Q$ ; then  $\mathbb{E}_H$  is the mean over the units modulo  $D$ , and  $\mu = \sum_i c_i / \varphi(d_i)$ . Partial quarantines produce cells whose moduli share primes with  $Q$ ; consistency means that each cell meets the class  $r \pmod{Q}$ . The hypotheses on  $r$  are used only in (c) below: every real character that factors through  $(\mathbb{Z}/Q)^\times$  takes the value 1 at  $r$ , exactly as at 1.

Neither the  $\ell^1$  mass  $\sum_i |c_i| / \varphi(d_i)$  nor the number of cells enters; only the Haar quantity  $A \geq 1$  does.

*Proof. Parameters.* Put  $L = 2c + \log(400C_G(A+1))$  and  $Q_G = x^{1/(\kappa L)}$ . Since  $A \geq 1$  we have  $L \leq C_3(1 + \log A)$  for an absolute  $C_3$ . If  $C_2$  is large enough, the hypothesis gives the following, using  $A \leq Z^{1/4}$  and  $Z \geq 2$ :  $\log Q_G \geq (C_2/(\kappa C_3)) \log Z$ , hence  $Q_G \geq Z$  and  $Q_G \geq 10^4 C_G(A+1)(\kappa L)^2$ ;  $Q_G^{6c} \leq x$  (as  $\kappa L \geq 6c$ );  $\log x \geq 16$ ;  $\log x \geq (\kappa L)^2$  (as  $(1 + \log A)^2 \leq (1 + \log A)(1 + \frac{1}{4} \log Z)$ ); and  $x \geq C_4 A Z^4$ , with  $C_4$  as large as needed below.

*Character expansion.* The function  $f(n) = B(n) \mathbf{1}[n \equiv r \pmod{Q}]$  on units modulo  $N_0$  satisfies  $f(n) = \sum_{\chi \bmod N_0} c(\chi) \chi(n)$ . The reduction map  $(\mathbb{Z}/N_0)^\times \rightarrow (\mathbb{Z}/Q)^\times$  is onto, and  $H$  is the fibre over  $r$ , so  $|H| = \varphi(N_0)/\varphi(Q)$  and

$$c(\chi) = \frac{1}{\varphi(N_0)} \sum_{n \in H} B(n) \bar{\chi}(n) = \frac{\mathbb{E}_H [B\bar{\chi}]}{\varphi(Q)}.$$

Consequently:

- (a)  $c(\chi) = 0$  unless the conductor of  $\chi$  divides  $\text{lcm}(d_i, Q)$  for some  $i$ . Indeed, by consistency and the Chinese remainder theorem,  $\mathbf{1}[n \equiv b_i \pmod{d_i}] \mathbf{1}[n \equiv r \pmod{Q}]$  is the indicator of one unit class modulo  $\text{lcm}(d_i, Q)$ , and its  $\chi$ -coefficient vanishes unless  $\chi$  is trivial on  $\{n \equiv 1 \pmod{\text{lcm}(d_i, Q)}\}$ . So  $c(\chi) \neq 0$  implies that the conductor of  $\chi$  is at most  $Q d_i \leq Z \leq Q_G$ .
- (b)  $|c(\chi)| \leq \mathbb{E}_H |B| / \varphi(Q) = A\mu / \varphi(Q)$ , and  $c(\chi_0) = \mu / \varphi(Q)$  for the principal character  $\chi_0$ .
- (c) Let  $\chi$  be real with  $c(\chi) \neq 0$ , and let  $\psi^*$  be the real primitive character inducing it, of conductor  $f$ . Write  $f = f_1 f_2$ , where  $f_1$  is composed of primes dividing  $Q$  and  $\gcd(f_2, Q) = 1$ , and  $\psi^* = \psi_1 \psi_2$  accordingly.



A real primitive conductor is  $2^e f'$  with  $e \in \{0, 2, 3\}$  and  $f'$  odd and squarefree; as  $8 \mid Q$ ,  $f_1 \mid Q$ , and  $\psi_1$  is a product of a character modulo 8 and Legendre symbols modulo odd primes of  $Q$ . On  $H$  we have  $n \equiv r \pmod{Q}$ , so  $\psi_1(n) = \psi_1(r) = 1$ , as  $r \equiv 1 \pmod{8}$  and  $r$  is a quadratic residue modulo every odd prime of  $Q$ . So  $\chi = \psi_2$  on  $H$ . If  $f_2 = 1$ , then  $\chi = 1$  on  $H$  and  $c(\chi) = \mu/\varphi(Q)$ . If  $f_2 > 1$ , then  $c(\chi) = \mathbb{E}_H[B\psi_2]/\varphi(Q)$ . A cell with  $f_2 \nmid d_i$  has  $\mathbb{E}_H[\mathbf{1}[n \equiv b_i \pmod{d_i}]\psi_2] = 0$ : there is a prime  $p \mid f_2$  with  $p \nmid d_i Q$ ; on  $H$  the residue of  $n$  modulo  $p^{v_p(N_0)}$  is uniform among units and independent of  $n$  modulo  $N_0/p^{v_p(N_0)}$ , and the  $p$ -component of  $\psi_2$  is a nontrivial character modulo  $p$ . So  $f_2 \mid d_i$  for some  $i$ ,  $\gcd(f_2, Q) = 1$ , and the twist condition gives  $|c(\chi)| \leq \mu/(4\varphi(Q))$ .

Let  $\chi^*$  be the primitive character inducing  $\chi$ . For a fixed modulus the map  $\chi \mapsto \chi^*$  is injective, so by (a) at most  $\sum_{r \leq Z} \varphi(r) \leq Z^2$  characters have  $c(\chi) \neq 0$ .

*The prime sum.* Let  $S(x) = \sum_{p \leq x, p \nmid N_0} f(p) \log p = \sum_{\chi} c(\chi) \vartheta(x; \chi)$ . Since  $|\vartheta(x; \chi) - \vartheta(x; \chi^*)| \leq \log N_0 \leq \log Q + \log D \leq 2Z$  (as  $D \mid \text{lcm}(1, \dots, \max d_i)$ ), so  $\log D \leq \psi(\max d_i) < 1.04 \max d_i$  by Rosser–Schoenfeld [38]),

$$S(x) = \frac{\mu x}{\varphi(Q)} + \sum_{\chi: c(\chi) \neq 0} c(\chi) (\vartheta(x; \chi^*) - E_0(\chi^*)x) + R_1, \quad (6)$$

where  $|R_1| \leq Z^2 \cdot (A\mu/\varphi(Q)) \cdot 2Z = 2AZ^3\mu/\varphi(Q)$ . Here  $E_0(\chi^*) = 1$  only for  $\chi = \chi_0$ , and  $|R_1| \leq \mu x/(400\varphi(Q))$  as  $x \geq 800AZ^3$ .

*Case 0: no exceptional zero* in the sense of Theorem 5.2. The  $\chi^*$  in (6) are distinct primitive characters of conductor  $\leq Q_G$ . By (b) and Theorem 5.2, the middle sum is at most

$$\frac{A\mu}{\varphi(Q)} C_G x \left( e^{-L} + \frac{(\kappa L)^2}{Q_G} \right) \leq \frac{\mu x}{\varphi(Q)} \left( \frac{1}{400} + \frac{1}{10^4} \right) \leq \frac{\mu x}{200\varphi(Q)},$$

using  $\log x / \log Q_G = \kappa L \geq L$ . So  $S(x) > 0$ .

*Exceptional case:* a zero  $\beta_1$  with character  $\chi_1$  and  $1 - \beta_1 < 1/(\kappa \log Q_G)$ . Put  $u = (1 - \beta_1) \log x$ ; then  $u < \log x / (\kappa \log Q_G) = L$ , and  $\beta_1 > 1/2$ . The replaced right-hand side of Theorem 5.2 is  $C_G u x (e^{-\kappa L} + \kappa L / Q_G)$ . Since  $u \leq \min(u, 1) L$  (recall  $L \geq 2$ ), its product with  $A\mu/\varphi(Q)$  is at most

$$\frac{\mu x}{\varphi(Q)} \min(u, 1) \cdot C_G A L \left( e^{-\kappa L} + \frac{\kappa L}{Q_G} \right) \leq \frac{\mu x}{200\varphi(Q)} \min(u, 1).$$

Here  $C_G A L e^{-3L} \leq C_G A e^{-L} \leq 1/400$ , and  $C_G A \kappa L^2 / Q_G \leq 10^{-4}$ . If no  $\chi$  with  $c(\chi) \neq 0$  has  $\chi^* = \chi_1$ , this bounds the middle sum of (6), and  $S(x) > 0$  as in Case 0. Otherwise, by injectivity, exactly one such  $\chi$  exists. It is real, because  $\chi_1$  is; hence  $c(\chi)$  is real, as  $B$  and  $\chi$  are. Let  $f_2$  be as in (c). Its term is  $c(\chi)(\vartheta(x; \chi_1) + x^{\beta_1}/\beta_1) - c(\chi)x^{\beta_1}/\beta_1$ . The first part is covered by the bound just proved.

*Case A:  $f_2 = 1$ .* Then  $c(\chi) = \mu/\varphi(Q)$ , and the conductor  $q_1$  of  $\chi_1$  divides  $Q$ , so  $q_1 \leq Z$ . The main terms combine to  $\lambda \mu x / \varphi(Q)$  with  $\lambda = 1 - x^{\beta_1-1}/\beta_1$ .

Since  $1/\beta_1 \leq 1 + 2(1 - \beta_1)$ ,  $1 - e^{-u} \geq (1 - e^{-1}) \min(u, 1)$  and  $2(1 - \beta_1)e^{-u} = 2ue^{-u}/\log x \leq 2 \min(u, 1)/\log x$ ,

$$\lambda \geq 1 - e^{-u} - 2(1 - \beta_1)e^{-u} \geq (0.63 - 2/16) \min(u, 1) \geq \min(u, 1)/2.$$

So the error from Theorem 5.2 is at most  $\lambda\mu x/(100\varphi(Q))$ . By Theorem 5.1,  $u \geq 16(1 - \beta_1) \geq 16/(c_2 Z^{1/2}(\log Z)^2)$ , so  $1/\lambda \leq 2 + c_2 Z^{1/2}(\log Z)^2/8 \leq C_5 Z$ . Hence  $|R_1| \leq 2AZ^3\mu/\varphi(Q) \leq \lambda\mu x/(100\varphi(Q))$  once  $x \geq 200C_5AZ^4$ . Altogether  $S(x) \geq (1 - 2/100)\lambda\mu x/\varphi(Q) > 0$ .

*Case B:*  $f_2 > 1$ . By (c),  $|c(\chi)x^{\beta_1}/\beta_1| \leq (\mu/(4\varphi(Q))) \cdot 2x = \mu x/(2\varphi(Q))$ , and  $S(x) \geq (1 - 1/2 - 1/200 - 1/400)\mu x/\varphi(Q) > 0$ .

In all cases  $S(x) > 0$ . So some prime  $p \leq x$  with  $p \nmid N_0$  and  $p \equiv r \pmod{Q}$  has  $B(p) > 0$ , and then  $W(p) > T$ .  $\square$

**Remark 6.2** (Assessment). This is Linnik's theorem with weights. Every character coefficient is bounded by  $\mathbb{E}_H|B|/\varphi(Q)$ , not by the  $\ell^1$  mass of a cell representation. Theorem 5.2 controls all characters of conductor  $\leq Q_G$  simultaneously, so the number of moduli  $d_i$  never multiplies the error. In [2] and in the first version of this note, each progression was instead treated separately with the Thorner–Zaman theorem [43]. That costs  $\log x \gg K \max(\log Z, K)$  with  $K = 1 + \log(\sum |c_i|/\varphi(d_i)/\mu)$ , which here is of order  $(\log T)^{14}$  instead of  $(\log T)^7$  in the setting of the second version. No lower bound for  $1 - \beta_1$  is needed in Case B. In Case A we use the effective bound of Theorem 5.1 only to absorb the term  $R_1$ .

## 7. AN ENERGY BOUND FOR AVOIDANCE INDICATORS

This section is self-contained and does not use arithmetic.

*Setting.* Let  $\Omega = \prod_{v \in V_0} \Omega_v$  be a finite product of finite sets with a product probability measure  $\pi = \bigotimes_v \pi_v$ ; write  $\mathbb{P}, \mathbb{E}$  and  $\|\cdot\| = \|\cdot\|_{L^2(\pi)}$ . A *cylinder event* is a set  $E = \{X_v = \sigma_v \mid (v \in \text{supp } E)\}$  with  $\text{supp } E \subseteq V_0$  and  $\sigma \in \prod_{\text{supp } E} \Omega_v$ . For a finite family  $\mathcal{E}$  of cylinder events (repetitions allowed) let  $F = F_{\mathcal{E}} = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$  be its *avoidance indicator*.

*Efron–Stein decomposition* (classical; [9, 21], [31, §8.3]). For  $\varphi : \Omega \rightarrow \mathbb{R}$  and  $W \subseteq V_0$  let  $\mathbb{E}[\varphi \mid X_W]$  be the conditional expectation given the coordinates in  $W$ , and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid X_W].$$

Then  $\varphi = \sum_U \varphi^{=U}$ , the components are mutually orthogonal,  $\varphi^{=U}$  depends only on  $X_U$ , and  $\mathbb{E}_u \varphi^{=U} = 0$  for  $u \in U$ , where  $\mathbb{E}_u$  averages over the coordinate  $u$ . Put  $L_u = I - \mathbb{E}_u$  and  $L_V = \prod_{u \in V} L_u$ ; then  $L_V \varphi = \sum_{U \supseteq V} \varphi^{=U}$ . For a down-closed family  $\mathcal{D}$  of subsets of  $V_0$ , the function  $\varphi^{\mathcal{D}} = \sum_{U \in \mathcal{D}} \varphi^{=U}$  is a linear combination of the  $\mathbb{E}[\varphi \mid X_W]$ ,  $W \in \mathcal{D}$  (by Möbius inversion), and

$$\mathbb{E}(\varphi - \varphi^{\mathcal{D}})^2 = \sum_{U \notin \mathcal{D}} \|\varphi^{=U}\|^2. \quad (7)$$

For  $\mathcal{D} = \{U : |U| \leq t\}$  this is the *energy above level  $t$* ,  $\text{En}(\varphi; t)$ ; it is the least  $\mathbb{E}(\varphi - g)^2$  over sums  $g$  of functions of at most  $t$  coordinates.

**Definition 7.1** (Energy generating function). For weights  $\lambda_v \geq 1$  put  $\mu_v = \lambda_v - 1$  and

$$G_\varphi(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|\varphi^{=U}\|^2 = \sum_{V \subseteq V_0} \mu^V \|L_V \varphi\|^2.$$

The second form follows from the identity  $\lambda^U = \sum_{V \subseteq U} \mu^V$  and  $\|L_V \varphi\|^2 = \sum_{U \supseteq V} \|\varphi^{=U}\|^2$ . For every  $\tau$ ,

$$\sum_{U: \lambda^U > e^\tau} \|\varphi^{=U}\|^2 \leq e^{-\tau} G_\varphi(\lambda). \quad (8)$$

For a single event  $E$  with  $\pi_E = \mathbb{P}(E)$  one computes  $G_{1-\mathbf{1}_E} = 1 - 2\pi_E + \pi_E \prod_{v \in \text{supp } E} (\lambda_v - \mu_v \pi_v(\sigma_v))$ , so for small  $\pi_v(\sigma_v)$  the condition  $\lambda^{\text{supp } E} \leq 2$  below is necessary for  $G \leq 1$ .

*Hypergraphs.* A hypergraph  $\mathcal{H}$  is a finite set of subsets (*edges*) of  $V_0$ . For  $V \subseteq V_0$  put

$$N_{\mathcal{H}}(V) = \sum_{\mathcal{J} \subseteq \mathcal{H}: \bigcup_{J \in \mathcal{J}} (J \cap V) = V} (-1)^{|\mathcal{J}|}, \quad Q_\mu(\mathcal{H}) = \sum_{V \subseteq V_0} \mu^V N_{\mathcal{H}}(V)^2.$$

$N_{\mathcal{H}}(V)$  depends only on the traces  $J \cap V$ , and it is unchanged if edges are repeated or replaced by edges with the same trace: a class of  $m \geq 1$  edges with equal traces contributes  $\sum_{j \geq 1} \binom{m}{j} (-1)^j = -1$ , exactly like one edge. If some edge misses  $V$ , then  $N_{\mathcal{H}}(V) = 0$  (toggle that edge).

**Lemma 7.2** (Cover bound; proved). *In the Setting, for every  $V \subseteq V_0$ ,*

$$\|L_V F\|^2 \leq \mathbb{E}_x [N_{\mathcal{H}(x)}(V)^2], \quad \mathcal{H}(x) = \{\text{supp } E : E \in \mathcal{E}, x \in E\}.$$

*Consequently  $G_F(\lambda) \leq \mathbb{E}_x [Q_\mu(\mathcal{H}(x))]$  for all weights  $\lambda_v \geq 1$ .*

*Proof.* Write  $x = (\sigma, y)$  with  $\sigma \in \Omega_V$ ,  $y \in \Omega_{V_0 \setminus V}$ . The operator  $L_V$  acts on the  $V$ -coordinates only, so  $\|L_V F\|^2 = \mathbb{E}_y \|L_V g_y\|_{L^2(\Omega_V)}^2$ , where  $g_y = F(\cdot, y)$ . Now  $g_y$  is the avoidance indicator on  $\Omega_V$  of the restricted events: those  $E \in \mathcal{E}$  whose conditions off  $V$  hold at  $y$ , restricted to  $\text{supp } E \cap V$ . Expand  $g_y = \sum_{\mathcal{J}} (-1)^{|\mathcal{J}|} \mathbf{1}_{\mathcal{J}}$  over subfamilies  $\mathcal{J}$  of restricted events, where  $\mathbf{1}_{\mathcal{J}}$  is the indicator of their intersection, a cylinder on  $\Omega_V$  (or empty).  $L_V$  kills every function that does not depend on all coordinates of  $V$ , hence every cylinder whose support is a proper subset of  $V$ . A cylinder with support  $V$  is a point indicator  $\mathbf{1}_{\{\sigma\}}$ . So  $L_V g_y = L_V c_y$ , where

$$c_y(\sigma) = \sum_{\mathcal{J}: \text{supp } \mathbf{1}_{\mathcal{J}} = V, \mathbf{1}_{\mathcal{J}} = \mathbf{1}_{\{\sigma\}}} (-1)^{|\mathcal{J}|}.$$

As  $L_V$  is an orthogonal projection in  $L^2(\pi_V)$ ,  $\|L_V c_y\|^2 \leq \mathbb{E}_\sigma c_y(\sigma)^2$ . The subfamilies  $\mathcal{J}$  counted in  $c_y(\sigma)$  are exactly the subfamilies of events that hold at  $x = (\sigma, y)$  whose traces on  $V$  cover  $V$ . Two events that hold at  $x$

and have the same support are the same cylinder, so by the remark above  $c_y(\sigma) = N_{\mathcal{H}(x)}(V)$ . Taking  $\mathbb{E}_y$  gives the first claim. Multiply by  $\mu^V$  and sum over  $V$  for the second.  $\square$

The suppression is built in:  $N_{\mathcal{H}(x)}(V) = 0$  as soon as some event that holds at  $x$  misses  $V$ .

For a finite multiset  $\mathcal{C}$  of subsets of  $V_0$  (possibly empty ones) put

$$\Theta_\lambda(\mathcal{C}) = \sum_{\mathcal{J} \subseteq \mathcal{C}} (-1)^{|\mathcal{J}|} \lambda^{|\bigcup \mathcal{J}|} \quad (\Theta_\lambda(\emptyset) = 1).$$

**Lemma 7.3** (Polarization; proved). *Let  $\lambda_v \geq 1$ ,  $\mu_v = \lambda_v - 1$ , and let  $P \subseteq V_0$  be random with  $\mathbb{P}(v \in P) = \mu_v/\lambda_v$  independently. For every hypergraph  $\mathcal{H}$ ,*

$$Q_\mu(\mathcal{H}) = \mathbb{E}_P[\Theta_\lambda(\mathcal{H}_P)^2], \quad \mathcal{H}_P = \{J \in \mathcal{H} : J \cap P = \emptyset\}.$$

*Proof.* Let  $\tau(R) = \mathbf{1}[R \cap J \neq \emptyset \ \forall J \in \mathcal{H}]$  for  $R \subseteq V_0$ , and  $\hat{\tau}(V) = \sum_{R \subseteq V} (-1)^{|V \setminus R|} \tau(R)$ . Expanding  $\tau(R) = \sum_{\mathcal{J} \subseteq \mathcal{H}} (-1)^{|\mathcal{J}|} \mathbf{1}[R \cap \bigcup \mathcal{J} = \emptyset]$ , and using that the Möbius transform of  $R \mapsto \mathbf{1}[R \cap U = \emptyset]$  is  $V \mapsto (-1)^{|V|} \mathbf{1}[V \subseteq U]$ , we get  $\hat{\tau}(V) = (-1)^{|V|} N_{\mathcal{H}}(V)$ . Let

$$\mathcal{T}(p) = \sum_V \hat{\tau}(V) p^V = \sum_R \tau(R) \prod_{v \in R} p_v \prod_{v \notin R} (1 - p_v)$$

be the multilinear extension of  $\tau$ . If the  $p_v$  are independent with mean 0 and variance  $\mu_v$ , distinct monomials are orthogonal, and  $\mathbb{E}[\mathcal{T}(p)^2] = \sum_V \mu^V \hat{\tau}(V)^2 = Q_\mu(\mathcal{H})$ . Take  $p_v = 1$  with probability  $\mu_v/\lambda_v$  and  $p_v = -\mu_v$  otherwise (mean  $\mu_v/\lambda_v - \mu_v/\lambda_v = 0$ , variance  $\mu_v/\lambda_v + \mu_v^2/\lambda_v = \mu_v$ ), and  $P = \{v : p_v = 1\}$ . Only the  $R \supseteq P$  contribute; writing  $R = P \cup B$  with  $B \subseteq P^c$ ,

$$\mathcal{T}(p) = \sum_{B \subseteq P^c} (-\mu)^B \lambda^{P^c \setminus B} \tau(P \cup B).$$

Now  $\tau(P \cup B) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \mathbf{1}[B \cap \bigcup \mathcal{J} = \emptyset]$ , and for  $U \subseteq P^c$ ,

$$\sum_{B \subseteq P^c \setminus U} (-\mu)^B \lambda^{P^c \setminus B} = \lambda^U \prod_{v \in P^c \setminus U} (\lambda_v - \mu_v) = \lambda^U.$$

Hence  $\mathcal{T}(p) = \Theta_\lambda(\mathcal{H}_P)$ .  $\square$

**Lemma 7.4** (Matching bound; proved). *If every  $J \in \mathcal{C}$  has  $w_J := \lambda^J \leq 2$ , then for every matching  $\mathcal{M} \subseteq \mathcal{C}$  (pairwise disjoint members)*

$$|\Theta_\lambda(\mathcal{C})| \leq \prod_{J \in \mathcal{M}} (w_J - 1); \quad \text{in particular } |\Theta_\lambda(\mathcal{C})| \leq 1.$$

*Proof.* For  $v \in V_0$  let  $\mathcal{C} - v$  be the members not containing  $v$  and  $\mathcal{C}/v = \{J \setminus \{v\} : J \in \mathcal{C}\}$ . Splitting  $\mathcal{J}$  according to whether it contains a member through  $v$ ,

$$\Theta(\mathcal{C}) = \Theta(\mathcal{C} - v) + \lambda_v [\Theta(\mathcal{C}/v) - \Theta(\mathcal{C} - v)] = \lambda_v \Theta(\mathcal{C}/v) - \mu_v \Theta(\mathcal{C} - v).$$

Both families again have all weights at most 2. We induct on  $|\bigcup \mathcal{C}|$ . If  $\emptyset \in \mathcal{C}$ , then  $\Theta(\mathcal{C}) = 0$  (toggling the empty member pairs the terms), and the claim

holds. If  $\mathcal{C} = \emptyset$ , then  $\Theta = 1$  and  $\mathcal{M} = \emptyset$ . This covers the base case  $\bigcup \mathcal{C} = \emptyset$ . If  $\mathcal{M} = \emptyset$  and  $\mathcal{C} \neq \emptyset$ , it suffices to prove the bound for  $\mathcal{M} = \{J\}$  with any  $J \in \mathcal{C}$ , as  $w_J - 1 \leq 1$ . So let  $J_0 \in \mathcal{M}$  be nonempty,  $v \in J_0$ , and  $\Pi' = \prod_{J \in \mathcal{M} \setminus \{J_0\}} (w_J - 1)$ . In  $\mathcal{C}/v$  the family  $(\mathcal{M} \setminus \{J_0\}) \cup \{J_0 \setminus \{v\}\}$  is a matching, so  $|\Theta(\mathcal{C}/v)| \leq (w_{J_0}/\lambda_v - 1)\Pi'$ . In  $\mathcal{C} - v$  the family  $\mathcal{M} \setminus \{J_0\}$  is a matching, so  $|\Theta(\mathcal{C} - v)| \leq \Pi'$ . Therefore  $|\Theta(\mathcal{C})| \leq (w_{J_0} - \lambda_v)\Pi' + \mu_v \Pi' = (w_{J_0} - 1)\Pi'$ .  $\square$

**Theorem 7.5** (Energy bound; proved). (a) *If every edge of a hypergraph  $\mathcal{H}$  has  $\lambda^J \leq 2$ , then  $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} (\lambda^J - 1)$  for every matching  $\mathcal{M} \subseteq \mathcal{H}$ ; in particular  $Q_\mu(\mathcal{H}) \leq 1$ .* (b) *In the Setting, if  $\lambda_v \geq 1$  and  $\lambda^{\text{supp } E} \leq 2$  for every  $E \in \mathcal{E}$ , then*

$$G_F(\lambda) = \sum_U \lambda^U \|F^{=U}\|^2 \leq 1.$$

*Proof.* (a) By Lemma 7.3 and Lemma 7.4 applied to  $\mathcal{H}_P$  with the matching  $\mathcal{M} \cap \mathcal{H}_P$ ,  $Q_\mu(\mathcal{H}) \leq \mathbb{E}_P \prod_{J \in \mathcal{M} \cap \mathcal{H}_P} (\lambda^J - 1)^2$ . The members of  $\mathcal{M}$  are disjoint, so the events  $\{J \cap P = \emptyset\}$  are independent, with probability  $\prod_{v \in J} \lambda_v^{-1} = 1/w_J$ . Hence  $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} [(1 - 1/w_J) + (w_J - 1)^2/w_J] = \prod_{J \in \mathcal{M}} (w_J - 1)$ . (b) Lemma 7.2 and (a), since  $\mathcal{H}(x)$  consists of supports of events.  $\square$

**Corollary 7.6** (Energy tail; proved). *In the Setting, if every event of  $\mathcal{E}$  has  $|\text{supp } E| \leq k$  ( $k \geq 1$ ), then for  $f \in \{F, 1 - F\}$  and every integer  $t \geq 0$ ,*

$$\text{En}(f; t) = \sum_{|U| > t} \|f^{=U}\|^2 \leq 2^{-(t+1)/k}.$$

*Proof.* Theorem 7.5(b) with  $\lambda_v = 2^{1/k}$  and (8);  $(1 - F)^{=U} = -F^{=U}$  for  $U \neq \emptyset$ .  $\square$

**Remark 7.7** (Sharpness; proved). Over general product spaces the base  $2^{-1/k}$  cannot be improved. Take  $m$  events on pairwise disjoint sets of  $k$  coordinates, each fixing at every coordinate of its support a value of probability  $p$ ; put  $\pi = p^k$ . As the factors  $1 - \mathbf{1}_{E_i}$  are independent,  $F^{=U} = \prod_i (1 - \mathbf{1}_{E_i})^{=U \cap \text{supp } E_i}$ . For  $U$  the union of the supports of  $j$  of the events this gives  $\|F^{=U}\|^2 = [\pi(1 - p)^k]^j (1 - \pi)^{2(m-j)}$ , and  $|U| = jk$ . Hence

$$\text{En}(F; jk - 1) \geq \binom{m}{j} [\pi(1 - p)^k]^j (1 - \pi)^{2(m-j)} \rightarrow e^{-2s} \frac{s^j}{j!}$$

as  $p \rightarrow 0$  with  $m = \lfloor s/\pi \rfloor$ . With  $s = j/2$  and  $j! \leq e j^{j+1/2} e^{-j}$ , the limit is at least  $2^{-j}/(e\sqrt{j}) = 2^{-(t+1)/k}/(e\sqrt{j})$  for  $t = jk - 1$ . So no bound  $C\rho^{-(t+1)/k}$  with  $\rho > 2$  holds for all such systems (with alphabets of size about  $1/p$ ), and at most a factor of order  $\sqrt{t/k}$  could be gained. The condition  $\lambda^{\text{supp } E} \leq 2$  in Theorem 7.5 is necessary, by the single-event formula above.

**Remark 7.8** (Boolean form and comparison; proved, except the cited constant). A width- $k$  DNF  $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$  is  $1 - 2F$  for the family of its terms (a term with a literal set is a cylinder event). As  $g^{=U} = -2F^{=U}$  for

$U \neq \emptyset$ , Corollary 7.6 gives, for every product measure on  $\{\pm 1\}^n$  (Efron–Stein weights; Fourier weights for the uniform measure),

$$\sum_{|S| > t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k},$$

i.e.  $g$  is  $\varepsilon$ -concentrated up to degree  $k \log_2(4/\varepsilon)$ . The switching-lemma route of Linial, Mansour and Nisan gives  $\sum_{|S| \geq t} \hat{g}(S)^2 \leq 2 \cdot 2^{-t/(20k)}$  [31, §4.4], [24]. Also  $\lambda^{|U|} \geq 1 + |U| \ln \lambda$  gives total influence  $\sum_U |U| \|F^{=U}\|^2 \leq (k/\ln 2) \mathbb{P}(F = 0)$ . The same bounds hold for  $q$ -ary DNFs with literals  $x_v \in S_v$  (split each term into single-value cylinders on the same support;  $F$  and the supports do not change). We have not found Theorem 7.5 in the literature; see the introduction for what we checked.

*Digit filtrations.* For the application we need weights that grow along a filtration of each coordinate. Suppose that each coordinate  $X_\ell$  ( $\ell \in \mathcal{P}$ , a finite index set) is a tuple of independent *digits*  $(\delta_{\ell,i})_{i_0(\ell) \leq i < f_\ell}$ , with arbitrary distributions, and that the coordinates are independent; the Efron–Stein decomposition  $F = \sum_U F^{=U}$  below is taken over the digits, so  $U$  is a set of pairs  $(\ell, i)$ . For such  $U$  write  $U_\ell = \{i : (\ell, i) \in U\}$ . We say that an event  $E$  is a *digit-prefix event* if, for some integers  $v_\ell(E) \geq 0$ , it fixes exactly the digits  $\delta_{\ell,i}$  with  $i_0(\ell) \leq i < v_\ell(E)$ , for each  $\ell$  (to arbitrary values). Such an event is a cylinder event on the digits.

**Theorem 7.9** (Filtration energy bound; proved). *In this situation let  $\lambda_\ell \geq 1$  ( $\ell \in \mathcal{P}$ ), and suppose that every event of  $\mathcal{E}$  is a digit-prefix event with  $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$ . Then*

$$\sum_U \left( \prod_{\ell: U_\ell \neq \emptyset} \lambda_\ell^{1+\max U_\ell} \right) \|F^{=U}\|^2 \leq 1.$$

*Proof.* Write  $\Lambda_{\ell,i} = \lambda_\ell^{i+1}$ ,  $\Lambda_{\ell,i_0(\ell)-1} = 1$ , and  $\mu'_{\ell,j} = \Lambda_{\ell,j} - \Lambda_{\ell,j-1} \geq 0$  for  $i_0(\ell) \leq j < f_\ell$ . For such  $j$  let  $P_{\ell,j} = I - \mathbb{E}_{B_{\ell,j}}$ , where  $\mathbb{E}_{B_{\ell,j}}$  averages over the *tail block*  $B_{\ell,j} = (\delta_{\ell,i})_{i \geq j}$ . Averaging over a digit in  $U$  kills  $F^{=U}$ , and  $F^{=U}$  does not depend on digits outside  $U$ ; hence  $P_{\ell,j} F^{=U} = F^{=U}$  if  $U_\ell \neq \emptyset$  and  $\max U_\ell \geq j$ , and  $P_{\ell,j} F^{=U} = 0$  otherwise. Call a set  $V$  of pairs  $(\ell, j)$  a *selection* if it contains at most one pair for each  $\ell$ , and put  $L'_V = \prod_{(\ell,j) \in V} P_{\ell,j}$ . Then  $\|L'_V F\|^2 = \sum_{U \succ V} \|F^{=U}\|^2$ , where  $U \succ V$  means  $\max U_\ell \geq j$  for all  $(\ell, j) \in V$ . For fixed  $U$ , the sum of  $\mu'^V$  over the selections  $V \prec U$  factorises as  $\prod_\ell (1 + \sum_{i_0(\ell) \leq j \leq \max U_\ell} \mu'_{\ell,j}) = \prod_{\ell: U_\ell \neq \emptyset} \Lambda_{\ell, \max U_\ell}$ . So the left side of the claim equals  $\sum_V \mu'^V \|L'_V F\|^2$ , the sum over selections.

Fix a selection  $V$  and pass to the coarser product space whose coordinates are, for  $(\ell, j) \in V$ , the head  $(\delta_{\ell,i})_{i < j}$  and the tail block  $B_{\ell,j}$ , and, for the other  $\ell$ , the whole  $X_\ell$ . There  $L'_V$  is the operator  $L_{V'}$  of the Setting for the set  $V'$  of tail blocks. Each event  $E$  is a disjoint union of cylinder events on the coarse coordinates with the same coarse support (enumerate the unfixed digits of each coarse coordinate that  $E$  involves), and splitting every event in this way

does not change  $F$ . Exactly one piece of  $E$  holds at  $x$  if  $E$  does. The trace of the coarse support of  $E$  on  $V'$  is the set of  $B_{\ell,j}$  with  $(\ell, j) \in V$  and  $j < v_\ell(E)$ . Identify  $B_{\ell,j}$  with the pair  $(\ell, j)$  and put  $\hat{E} = \{(\ell, j) : i_0(\ell) \leq j < v_\ell(E)\}$ . By Lemma 7.2 and the remark on traces,

$$\|L'_V F\|^2 \leq \mathbb{E}_x[N_{\hat{\mathcal{H}}(x)}(V)^2], \quad \hat{\mathcal{H}}(x) = \{\hat{E} : E \in \mathcal{E}, x \in E\}.$$

Summing over selections and adding the nonnegative terms of all other sets  $V$  of pairs, the left side of the claim is at most  $\mathbb{E}_x[Q_{\mu'}(\hat{\mathcal{H}}(x))]$ . By Theorem 7.5(a) it remains to show  $\prod_{(\ell,j) \in \hat{E}} (1 + \mu'_{\ell,j}) \leq 2$  for every event. Fix  $\ell$  and write  $\lambda = \lambda_\ell$ ,  $v = v_\ell(E)$ ,  $i_0 = i_0(\ell)$ , and assume  $v > i_0$ . Here  $1 + \mu'_{\ell,i_0} = \lambda^{i_0+1}$  and  $1 + \mu'_{\ell,j} = 1 + \lambda^j(\lambda - 1)$  for  $j > i_0$ . The hypothesis gives  $\lambda^{2v} \leq 2$ , so  $\lambda^j \leq \lambda^v \leq \sqrt{2} < \lambda + 1$  for  $j < v$ , i.e.  $1 + \lambda^j(\lambda - 1) \leq \lambda^2$ . The factor of  $\ell$  is therefore at most  $\lambda^{i_0+1} \lambda^{2(v-i_0)} \leq \lambda^{2v}$ , and the product over  $\ell$  is at most  $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$ .  $\square$

**Corollary 7.10** (Modulus-weighted tail; proved). *In Theorem 7.9 let the  $\ell \in \mathcal{P}$  be primes and  $\lambda_\ell = 2^{\log \ell / (2\mathcal{L})}$ , and suppose  $\prod_\ell \ell^{v_\ell(E)} \leq T$  for every event. For a set  $U$  of digits put  $m_U = \prod_{\ell: U_\ell \neq \emptyset} \ell^{1+\max U_\ell}$ . Then for every  $\tau \geq 0$*

$$\sum_{U: m_U > e^\tau} \|F^{=U}\|^2 \leq 2^{-\tau/(2\mathcal{L})}.$$

*Proof.*  $\prod_\ell \lambda_\ell^{2v_\ell(E)} = 2^{\log \prod_\ell \ell^{v_\ell(E)} / \mathcal{L}} \leq 2$ , and the weight of  $U$  in Theorem 7.9 is  $2^{\log m_U / (2\mathcal{L})}$ .  $\square$

The cost of a junta is thus measured by the modulus  $m_U$  a function of the digits in  $U$  needs, not by the number of coordinates. For the Erdős–Straus events (digits in base  $\ell$ ) this removes both the binary width and the support size from the junta modulus.

## 8. THE RESIDUAL SYSTEM AND THE SANDWICH MINORANT

Throughout this section  $T$  is large,  $\beta = 1 + 1/\log \mathcal{L}$ ,  $\eta = \frac{3}{4} \log \beta$ , and  $(Q, r)$  is the square-class quarantine of Theorem 3.3;  $E_1, \dots, E_m$  ( $m \leq T^2$ ) are the events of the surviving atoms, in some order,  $A_i = \mathbf{1}_{E_i}$ ,  $F_{<i} = \prod_{j < i} (1 - A_j)$ ,  $F = F_{<m+1}$ ,  $S = \sum_i \mathbb{P}(E_i) \leq S_\beta$  and  $\delta = \mathbb{E}F$ . By (4) and Lemma 2.8, every subfamily of  $\{E_i\}$  satisfies the hypothesis of Lemma 2.8,  $\delta \geq e^{-4S_\beta/3}$ , and for  $T$  large

$$\eta \leq 0.19 \quad (\text{as soon as } \mathcal{L} \geq 33). \quad (9)$$

*Digits.* For  $\ell \in \mathcal{P}$  write  $x \in \Omega_\ell$  in base  $\ell$  as  $x = \sum_{i < f_\ell} \delta_{\ell,i} \ell^i$ . The digits  $\delta_{\ell,i}$  with  $i < a_\ell$  are those of  $r$ , hence constant on  $\Omega_\ell$ . The others are independent and uniform:  $\delta_{\ell,0}$  on  $\{1, \dots, \ell - 1\}$  if  $a_\ell = 0$ , and  $\delta_{\ell,i}$  on  $\{0, \dots, \ell - 1\}$  for  $\max(a_\ell, 1) \leq i < f_\ell$ ; indeed  $x \mapsto (\delta_{\ell,i})_{i \geq a_\ell}$  is a bijection from  $\Omega_\ell$  onto the product of these ranges. So the free digits are those with  $i \geq i_0(\ell) := a_\ell$ , and  $E_{M,D}$  is a digit-prefix event with  $v_\ell(E_{M,D}) = v_\ell(M)$  for  $\ell \in \text{supp}(M, D)$  and

$v_\ell(E_{M,D}) = 0$  otherwise. A function of the digits  $\delta_{\ell,i}$  with  $i \leq i_\ell$  depends only on  $X_\ell$  modulo  $\ell^{1+i_\ell}$ .

*Cells.* A *consistent unit cell* is a class  $b \bmod d$  with  $d$  composed of primes of  $\mathcal{P}$ ,  $d \mid \prod_{\ell \in \mathcal{P}} \ell^{f_\ell}$ ,  $\gcd(b, d) = 1$  and  $b \equiv r \pmod{\gcd(d, Q)}$ ; on  $\Omega$  it is the event  $\{X_\ell \equiv b \pmod{\ell^{v_\ell(d)}} \forall \ell \mid d\}$ , and on  $\mathbb{Z}$  the class itself. The events  $E_{M,D}$  are consistent unit cells (by survival). The intersection of two consistent unit cells is empty or a consistent unit cell whose modulus is the lcm of the two, both on  $\Omega$  and on  $\mathbb{Z}$ . A function on  $\Omega$  that depends only on  $(X_\ell \bmod \ell^{b_\ell})_\ell$  is a linear combination of consistent unit cells of modulus  $\prod_\ell \ell^{b_\ell}$  (the indicators of the points of its level sets).

**Lemma 8.1** (Haar means; proved). *Let  $Q' = Q\ell'$  with  $\ell'$  a prime not dividing  $Q \prod_{\ell \in \mathcal{P}} \ell$ , and  $r'$  the class modulo  $Q'$  with  $r' \equiv r \pmod{Q}$  and  $r' \equiv 1 \pmod{\ell'}$  (or  $Q' = Q$ ,  $r' = r$ ). Let  $\varphi$  be a combination of consistent unit cells with lcm of moduli  $D$ . With  $H$  as in Theorem 6.1 for  $Q'$ ,  $r'$  and  $D$ ,  $\mathbb{E}_H \varphi = \mathbb{E} \varphi$ . The same holds for  $\varphi\psi$ , where  $\psi$  is a character modulo a squarefree  $f \mid D$  coprime to  $Q$ , viewed on  $\Omega$  through  $X_\ell \bmod \ell$  for  $\ell \mid f$ .*

*Proof.* Let  $N_0 = \text{lcm}(Q', D)$ . By the Chinese remainder theorem the uniform measure on  $H$  is the product of the uniform measures on  $H_p = \{x \in (\mathbb{Z}/p^{v_p(N_0)})^\times : x \equiv r' \pmod{p^{v_p(Q')}}\}$ . For  $\ell \in \mathcal{P}$  with  $\ell \mid D$ ,  $v_\ell(Q') = a_\ell$  and  $v_\ell(N_0) \leq f_\ell$ , and reduction maps the uniform measure on  $\Omega_\ell$  to the uniform measure on  $H_\ell$  (all fibres have the same size). The other primes of  $N_0$  do not divide  $D$ , and  $\varphi, \psi$  do not depend on them.  $\square$

For  $j \leq m$  let  $F^{(j)}$  be the function of the digits not fixed by  $E_j$  obtained from  $F_{<j}$  by giving the digits fixed by  $E_j$  their values on  $E_j$ . Then  $A_j F_{<j} = A_j F^{(j)}$ . At  $\ell$ , the digits not fixed by  $E_j$  are those with  $i \geq i_0^{(j)}(\ell) := \max(a_\ell, v_\ell(E_j))$ , and  $F^{(j)}$  is the avoidance indicator of the restrictions of the events  $E_i$  ( $i < j$ ) compatible with  $E_j$ . Each restriction fixes the digits  $i_0^{(j)}(\ell) \leq i < v_\ell(E_i)$ , so it is a digit-prefix event for the filtration starting at  $i_0^{(j)}$ , with the same  $v_\ell(E_i)$  and  $\prod_\ell \ell^{v_\ell(E_i)} \leq M_i \leq T$ . (A restriction may fix nothing; then  $F^{(j)} \equiv 0$ .)

**Lemma 8.2** (Bazzi–Razborov sandwich [3, 37]; proved). *Let  $u_1, \dots, u_m$  be arbitrary real functions,  $v_i = \sum_{j < i} A_j u_j$ , and*

$$B = 1 - \sum_{i \leq m} A_i (1 - v_i)^2.$$

*Then  $B \leq F$  pointwise, and with  $e_j = F_{<j} - u_j$ ,*

$$F - B = \sum_i A_i \left( \sum_{j < i} A_j e_j \right)^2, \quad \mathbb{E}[F - B] \leq m^2 \sum_{j \leq m} \mathbb{E}[A_j e_j^2].$$

*If  $u_j$  is a function of the digits not fixed by  $E_j$ , then  $\mathbb{E}[A_j e_j^2] = \mathbb{P}(E_j) \mathbb{E}(F^{(j)} - u_j)^2$ .*



*Proof.* The identity  $1 - F = \sum_i A_i F_{<i}$  holds pointwise: the sum counts the first occurring event. Put  $f_{<i} = 1 - F_{<i} = \sum_{j<i} A_j F_{<j}$ . If  $F_{<i} = 1$ , then  $A_j = 0$  for all  $j < i$ , so  $v_i = 0$  and  $A_i(1 - v_i)^2 = A_i = A_i F_{<i}$ . If  $F_{<i} = 0$ , then  $f_{<i} = 1$ , and  $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$ . In both cases  $A_i(1 - v_i)^2 - A_i F_{<i} = A_i(f_{<i} - v_i)^2$ , and summing over  $i$ ,

$$F - B = \sum_i A_i [(1 - v_i)^2 - F_{<i}] = \sum_i A_i (f_{<i} - v_i)^2 \geq 0,$$

with  $f_{<i} - v_i = \sum_{j<i} A_j (F_{<j} - u_j)$ . By Cauchy–Schwarz,  $(\sum_{j<i} A_j e_j)^2 \leq m \sum_{j<i} A_j e_j^2$ . Drop  $A_i \leq 1$  and sum over  $i$ . For the last claim,  $A_j e_j^2 = A_j (F^{(j)} - u_j)^2$ , and  $F^{(j)} - u_j$  is independent of  $A_j$ .  $\square$

This is Razborov’s simplification [37] of Bazzi’s argument [3]. The choice of  $u_j$  as an  $\ell^2$ -approximation of the “conditional survival function”  $F^{(j)}$  is the one used there. We include the short proof for completeness. The identity and the inequality  $B \leq F$  use only that each  $A_i$  takes values in  $\{0, 1\}$ , so they also hold for the functions on  $\mathbb{Z}$  obtained from cell expansions.

*Choice of  $u_j$ .* Fix  $\tau \geq 0$ . For  $j \leq m$  let  $\mathcal{D}_j$  be the family of sets  $U$  of digits not fixed by  $E_j$  with  $m_U \leq e^\tau$ , where  $m_U = \prod_{\ell: U_\ell \neq \emptyset} \ell^{1+\max U_\ell}$ ; it is down-closed. Put

$$u_j = (F^{(j)})^{\mathcal{D}_j} = \sum_{U \in \mathcal{D}_j} (F^{(j)})^{=U}, \quad (10)$$

the Efron–Stein decomposition being taken over the digits not fixed by  $E_j$ .

**Lemma 8.3** (Approximation error; proved). *With  $u_j$  as in (10),  $\mathbb{E}[F - B] \leq m^2 S 2^{-\tau/(2\mathcal{L})}$ .*

*Proof.* By (7) and Corollary 7.10, applied to  $F^{(j)}$  with the filtration starting at  $i_0^{(j)}$  (the digits are indexed by their absolute position, so  $m_U$  is the modulus needed),  $\mathbb{E}(F^{(j)} - u_j)^2 \leq 2^{-\tau/(2\mathcal{L})}$ . Conclude with Lemma 8.2.  $\square$

**Lemma 8.4** (Cell structure of  $B$ ; proved). *With  $u_j$  as in (10),  $B$  is a finite real combination of consistent unit cells of modulus at most  $T^3 e^{2\tau}$ , composed of primes of  $\mathcal{P}$ . The identity  $B = 1 - \sum_i A_i (1 - v_i)^2$  also holds for the expanded cell combination evaluated at any integer  $n$ .*

*Proof.* By (7),  $u_j$  is a combination of the functions  $\mathbb{E}[F^{(j)} \mid \delta_W]$ ,  $W \in \mathcal{D}_j$ . Such a function depends only on  $X_\ell \bmod \ell^{1+\max W_\ell}$  for the  $\ell$  with  $W_\ell \neq \emptyset$ , so it is a combination of consistent unit cells of modulus  $m_W \leq e^\tau$ . Expanding,

$$B = 1 - \sum_i A_i + 2 \sum_i \sum_{j<i} A_i A_j u_j - \sum_i \sum_{j,j'<i} A_i A_j A_{j'} u_j u_{j'}.$$

Each term is a product of at most three events (moduli  $\leq T$ ) and at most two  $u$ ’s, so every cell is an intersection of consistent unit cells of total modulus at most  $T^3 e^{2\tau}$ . The last claim holds because the intersection identities hold on  $\mathbb{Z}$ .  $\square$

**Lemma 8.5** ( $\ell^1$ -tightness; proved). *If  $B \leq F$  pointwise,  $F \geq 0$  and  $\mathbb{E}[F - B] \leq \eta \mathbb{E}B$ , then  $\mathbb{E}|B| \leq (1 + 2\eta)\mathbb{E}B$ .*

*Proof.*  $B^- := \max(-B, 0) \leq F - B$  since  $F \geq 0$ , and  $\mathbb{E}|B| = \mathbb{E}B + 2\mathbb{E}B^-$ .  $\square$

**Lemma 8.6** (Twist; proved). *Assume  $\mathbb{E}[F - B] \leq \delta/100$ , and write  $B$  as in Lemma 8.4, with  $\mu = \mathbb{E}B$ . Let  $\psi$  be a real primitive character of conductor  $f > 1$  with  $\gcd(f, Q) = 1$  and  $f \mid d_i$  for some cell modulus  $d_i$ . Then  $\mu \geq 0.99\delta$  and  $|\mathbb{E}[B\psi]| \leq \mu/4$ , where  $\psi$  is viewed on  $\Omega$  through  $X_\ell \bmod \ell$ ,  $\ell \mid f$ .*

*Proof.*  $\mu = \mathbb{E}F - \mathbb{E}[F - B] \geq 0.99\delta$ . As  $2 \nmid f$ ,  $f$  is squarefree. Its prime factors divide  $d_i$ , so they lie in  $\mathcal{P}$ , and  $a_\ell = 0$  for them as  $\gcd(f, Q) = 1$ . So  $\psi = \prod_{\ell \mid f} \psi_\ell(X_\ell \bmod \ell)$  with nontrivial characters  $\psi_\ell$  of the uniformly distributed digits  $\delta_{\ell,0}$ . Now

$$|\mathbb{E}[B\psi]| \leq \mathbb{E}|B - F| + |\mathbb{E}[F\psi]| = \mathbb{E}[F - B] + |\mathbb{E}[F\psi]|.$$

Fix  $\ell_0 \mid f$  and let  $F'$  be the avoidance indicator of the events  $E_i$  with  $\ell_0 \notin \text{supp } E_i$ . Given the other coordinates  $X_{-\ell_0}$ , we have  $F = F' \cdot \mathbf{1}[X_{\ell_0} \notin \text{Forb}]$ , where Forb is the union of the classes  $X_{\ell_0} \equiv -4D_i \pmod{\ell_0^{v_{\ell_0}(M_i)}}$  over the events  $E_i \ni \ell_0$  whose other conditions hold. Write  $E_i \setminus \ell_0$  for the event given by those other conditions, and  $\pi_i$  for the  $\Omega_{\ell_0}$ -probability of the class at  $\ell_0$ , so that  $\mathbb{P}(E_i) = \mathbb{P}(E_i \setminus \ell_0) \pi_i$ . Since  $\mathbb{E}_{X_{\ell_0}} \psi_{\ell_0} = 0$ ,

$$\begin{aligned} |\mathbb{E}[F\psi]| &= \left| \mathbb{E} \left[ F' \prod_{\ell \neq \ell_0} \psi_\ell \mathbb{E}_{X_{\ell_0}} [\psi_{\ell_0} \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \right] \right| \\ &\leq \sum_{i: \ell_0 \in \text{supp } E_i} \mathbb{P}(F' = 1, E_i \setminus \ell_0) \pi_i. \end{aligned}$$

The event  $E_i \setminus \ell_0$  is determined by the  $s(E_i) - 1$  coordinates of  $\text{supp } E_i \setminus \{\ell_0\}$ , so Lemma 2.8, applied to the family defining  $F'$ , gives  $\mathbb{P}(E_i \setminus \ell_0 \mid F' = 1) \leq \beta^{s(E_i)-1} \mathbb{P}(E_i \setminus \ell_0)$ . Hence

$$|\mathbb{E}[F\psi]| \leq \beta^{-1} \sum_{i: \ell_0 \in \text{supp } E_i} \beta^{s(E_i)} \mathbb{P}(E_i) \mathbb{E}F' = \beta^{-1} \tilde{w}_{\ell_0} \mathbb{E}F' \leq \eta \mathbb{E}F'.$$

The same bound gives  $\mathbb{E}F' - \mathbb{E}F = \mathbb{E}[F' \mathbf{1}[X_{\ell_0} \in \text{Forb}]] \leq \eta \mathbb{E}F'$ , so  $\mathbb{E}F' \leq \delta/(1 - \eta)$ . By (9),  $|\mathbb{E}[B\psi]| \leq (0.01 + \eta/(1 - \eta))\delta \leq 0.245\delta \leq 0.245\mu/0.99 < \mu/4$ .  $\square$

## 9. PROOF OF THE MAIN THEOREMS

**Theorem 9.1** (Assembly; proved modulo Theorems 5.2 and 3.1). *Let  $T$  be large, let  $(Q, r)$  be the square-class quarantine of Theorem 3.3, put  $S_1 = C\mathcal{L}^3 \log \mathcal{L}$  (the bound for  $S_\beta$  in (4)) and*

$$\tau = 2\mathcal{L} \lceil \log_2(100 T^4 (S_1 + 1) e^{3S_1}) \rceil.$$

*Then there is a hard prime  $p$  with  $p > T$ ,  $W(p) > T$  and  $\log p \leq C_6 \log Z$ , where  $C_6$  is absolute and*

$$\log Z \leq \log Q + 4\tau + 6\mathcal{L} + 1.$$

*Proof.* Take the residual system of  $(Q, r)$ ;  $m \leq T^2$ . If  $m = 0$ , then  $F \equiv 1$ ; we put  $B = 1$  (one cell of modulus 1), so that  $\mu = A = 1$  and the twist condition is vacuous, and go directly to the transfer step below. Otherwise define  $B$  by Lemma 8.2 with  $u_j$  as in (10). By Lemma 8.3, the choice of  $\tau$ ,  $S \leq S_\beta \leq S_1$  and  $\delta \geq e^{-4S_\beta/3} \geq e^{-3S_1}$ ,

$$\mathbb{E}[F - B] \leq m^2 S 2^{-\tau/(2\mathcal{L})} \leq \frac{S e^{-3S_1}}{100(S_1 + 1)} \leq \frac{e^{-3S_1}}{100} \leq \frac{\delta}{100}.$$

Lemma 8.6 gives  $\mu = \mathbb{E}B \geq 0.99\delta > 0$ , and, as  $\mathbb{E}[F - B] \leq \mu/99$ , Lemma 8.5 gives  $\mathbb{E}|B| \leq 1.03\mu$ . By Lemma 8.4,  $B$  is a combination of consistent unit cells of moduli  $d_i \leq T^3 e^{2\tau}$ .

*Pointwise inequality on integers.* For every integer  $n$  the values  $A_i(n) \in \{0, 1\}$  satisfy the hypotheses of Lemma 8.2, and the cell expansion of  $B$  is valid on  $\mathbb{Z}$  (Lemma 8.4). So  $B(n) \leq F(n)$ , and by Lemma 2.5(i),  $B(n) \leq \mathbf{1}[W(n) > T]$  for every  $n \equiv r \pmod{Q}$ .

*Transfer.* Let  $R = \max(T, \max_i d_i)$ , choose a prime  $\ell_{\text{aux}} \in (R, 2R]$ , put  $Q' = Q\ell_{\text{aux}}$ , and let  $r'$  be the class modulo  $Q'$  with  $r' \equiv r \pmod{Q}$  and  $r' \equiv 1 \pmod{\ell_{\text{aux}}}$ . Then  $8 \mid Q'$ ,  $r' \equiv 1 \pmod{8}$ , and  $r'$  is a quadratic residue modulo every odd prime of  $Q'$ . As  $\ell_{\text{aux}} > \max d_i$  is prime,  $\gcd(d_i, Q') = \gcd(d_i, Q)$ , so the cells are consistent for  $(Q', r')$ , and no cell or event involves  $\ell_{\text{aux}}$ . The inequality  $B \leq \mathbf{1}[W > T]$  holds on  $n \equiv r' \pmod{Q'}$ . The functions  $B$  and  $|B|$  depend only on  $X \bmod D$ , so both are combinations of consistent unit cells (point cells modulo  $D$ ), and by Lemma 8.1,  $\mathbb{E}_H B = \mu$  and  $\mathbb{E}_H |B|/\mu \leq 1.03 \leq Z^{1/4}$ . A real primitive character of conductor  $f > 1$  coprime to  $Q'$  is coprime to  $Q$ , so Lemmas 8.6 and 8.1 give the twist condition. Here  $\log Z \leq \log Q + \log(2R) + \log \max d_i \leq \log Q + \log 2 + 2(3\mathcal{L} + 2\tau)$ . Theorem 6.1 with  $\log x = C_2(1 + \log 1.03) \log Z$  gives a prime  $p \equiv r' \pmod{Q'}$  with  $W(p) > T$  and  $\log p \leq C_6 \log Z$ . Since  $\ell_{\text{aux}} \mid p - 1$ ,  $p > \ell_{\text{aux}} > T$ ; and  $p \equiv r \pmod{Q}$  with  $840 \mid Q$ , so  $p$  is hard by Lemma 2.5(iii).  $\square$

The prime  $\ell_{\text{aux}}$  is needed for infinitely many *distinct* primes:  $W(p) > T$  alone does not force  $p > T$ .

*Proof of Theorem 1.1.* By (4),  $\tau \ll \mathcal{L}(\mathcal{L} + S_1) \ll \mathcal{L}^4 \log \mathcal{L}$  and  $\log Q \ll \mathcal{L}^3(\log \mathcal{L})^5$ . So  $\log Z \ll \mathcal{L}^4 \log \mathcal{L}$ , and Theorem 9.1 gives a hard prime  $p > T$  with  $W(p) > T$  and  $\log p \leq C\mathcal{L}^4 \log \mathcal{L}$ . As  $\mathcal{L} < \log p$ , this gives  $\mathcal{L}^4 \geq \log p / (C \log \log p)$ , i.e.  $W(p) > T \geq \exp(c(\log p)^{1/4}(\log \log p)^{-1/4})$ . Applying this to a sequence  $T_1 < T_2 < \dots$  with  $T_{j+1}$  larger than the prime found for  $T_j$  gives infinitely many distinct primes.  $\square$

**Remark 9.2** (Where the exponent comes from; Assessment). The certified bound is  $\log p \ll \log Z \ll \log Q + \tau$ . The quarantine term is  $\log Q \ll \mathcal{L}^3(\log \mathcal{L})^5$ , of the order of the Haar density (Theorems 3.3 and 4.4). The bottleneck is the junta term  $\tau \asymp \mathcal{L}(S_1 + \mathcal{L}) \asymp \mathcal{L}^4 \log \mathcal{L}$ : the cells of the sandwich have modulus about  $e^{\mathcal{L}S}$ . In the previous version both terms were

of order  $\mathcal{L}^5$  (up to  $\log \mathcal{L}$ ). Section 10 shows that the junta term cannot be reduced within the present architecture.

## 10. THE CEILING OF THE MINORANT ARCHITECTURE

The proof of Theorem 1.1 passes through a minorant  $B \leq F$  of positive Haar mean whose cells have modulus  $e^{O(\mathcal{L}S)}$ . Here we show that no minorant of smaller level exists, dense or sparse, on any fibre whose modulus satisfies  $\log Q \leq T^{0.05}$  (for larger  $Q$  the transfer itself is too expensive, Corollary 10.7).

*Setting.* Fix a modulus  $Q$  and a unit  $r$  modulo  $Q$ , and consider the fibre  $\{n \in \hat{\mathbb{Z}}^\times : n \equiv r \pmod{Q}\}$  with its Haar probability. Let  $F(n) = \mathbf{1}[W(n) > T]$ . A function on the fibre has *level* at most  $D$  if it is a linear combination of functions  $n \mapsto \phi(n \bmod qQ)$  with  $q \leq D$ . Every cell of modulus at most  $D$  and every character of conductor at most  $D$  (restricted to the fibre) has level at most  $D$ ; the minorant of Theorem 9.1 has level at most  $\max_i d_i \leq T^3 e^{2\tau}$ . All functions below depend on  $n$  modulo a fixed finite modulus, so all probability spaces are finite.

**Lemma 10.1** (Planting; proved). *Let  $b_1, \dots, b_N$  be independent bits with  $\mathbb{P}(b_i = 1) = p_i \in [0, 1)$ , and law  $\nu_0$ . Put  $r_i = p_i/(1 - p_i)$ ,  $R = \sum_i r_i$  and  $r^* = \max_i r_i$ . Let  $k \geq 0$  with*

$$R \geq (k + 1) + (2k + 1)r^*. \quad (11)$$

*Then there is a probability law  $\nu$  on  $\{0, 1\}^N$  with the same marginals as  $\nu_0$  on every set of at most  $k$  coordinates and  $\nu(b = 0) = 0$ .*

*Proof.* Write  $1_y$  for the configuration with ones exactly on  $y \subseteq [N]$ , and  $P_0 = \nu_0(1_\emptyset) = \prod (1 - p_i)$ , so  $\nu_0(1_y) = P_0 \prod_{i \in y} r_i$ . Let  $e_a$  denote the elementary symmetric functions. If  $e_{k+1}(r) = 0$ , at most  $k$  of the  $r_i$  are positive and  $R \leq kr^*$ , contradicting (11). For  $|J| = k + 1$  put  $w_J = \prod_{i \in J} r_i / e_{k+1}(r)$ , so  $\sum_J w_J = 1$ , and

$$\nu = \nu_0 + P_0 \sum_{|J|=k+1} w_J \sigma_J, \quad \sigma_J = \sum_{y \subseteq J} (-1)^{|y|+1} \delta_{1_y}.$$

*Marginals.* For  $|K| \leq k$  and  $z \subseteq K$ , the  $\sigma_J$ -mass of  $\{x : x_K = 1_z\}$  is  $[z \subseteq J] \sum_{y' \subseteq J \setminus K} (-1)^{|z|+|y'|+1} = 0$ , as  $J \setminus K \neq \emptyset$ . In particular ( $K = \emptyset$ )  $\nu$  has total mass 1. *Zero.*  $\nu(1_\emptyset) = P_0 - P_0 \sum_J w_J = 0$ . *Positivity.* Only  $y \subseteq J$  with  $|y| = j \geq 2$  even receives negative charge, of total  $P_0 \sum_{J \supseteq y} w_J = P_0 \prod_{i \in y} r_i e_{k+1-j}(s) / e_{k+1}(r)$ , where  $s = (r_i)_{i \notin y}$ . So it suffices that  $e_{k+1-j}(s) \leq e_{k+1}(r)$ . Now  $e_{k+1}(r) \geq e_{k+1}(s)$ , and  $e_a(s) \geq e_{a-1}(s)$  for  $1 \leq a \leq k+1$ : each  $a$ -set arises  $a$  times as  $I \cup \{i\}$  with  $|I| = a-1$ ,  $i \notin I$ , so  $a e_a(s) \geq e_{a-1}(s) (\sum s - (a-1)r^*)$ , and  $\sum s - (a-1)r^* \geq R - (j+a-1)r^* \geq R - (2k+1)r^* \geq k+1 \geq a$ . Chain  $a = k+2-j, \dots, k+1$ .  $\square$

This is the dual (linear programming) side of lower-bound sieves: minorants of degree  $k$  of  $\mathbf{1}[b = 0]$  with positive mean exist only if no such  $\nu$  exists (the duality itself is standard, cf. [41, Thm. 5]). It is close in spirit to work

on  $k$ -wise independent laws with extremal probabilities [4, 36]; we claim no novelty for it. In the Poisson-like model with  $R$  of order  $S$ , Bonferroni inequalities of order  $\asymp S$  do give positive minorants, so the order  $k \asymp R$  is right.

**Lemma 10.2** (Level barrier; proved). *Let independent coordinates be split into small ones  $X_s$  and big ones  $(X_b)_{b \in \mathfrak{B}}$ . Let  $\mathcal{F}$  be a family of events, each of the form  $E = \{X_s \in \Sigma_E\} \cap \{X_{b(E)} \in \Gamma_E\}$  with exactly one big coordinate  $b(E)$ , and let  $F$  be the indicator that no event of a family  $\mathcal{E} \supseteq \mathcal{F}$  occurs. For a value  $x_s$  put  $\Omega_b(x_s) = \bigcup_{E: b(E)=b, x_s \in \Sigma_E} \Gamma_E$ ,  $p_b(x_s) = \mathbb{P}(X_b \in \Omega_b(x_s))$ ,  $r_b = p_b/(1 - p_b)$  and  $R(x_s) = \sum_b r_b(x_s)$ . Suppose  $p_b(x_s) \leq p^* < 1$  for all  $b, x_s$ , put  $r^* = p^*/(1 - p^*)$ , and suppose  $R(x_s) \geq (k+1) + (2k+1)r^*$  for every  $x_s$ . Let  $B$  be a linear combination of functions each depending on  $X_s$  and on at most  $k$  big coordinates. If  $B \leq F$  pointwise, then  $\mathbb{E}B \leq 0$ .*

*Proof.* Define a law  $\nu$ : draw  $x_s$  from its true law; draw bits  $(b_b)_b$  from the law of Lemma 10.1 for the independent bits  $\mathbf{1}[X_b \in \Omega_b(x_s)]$  (bits with  $p_b = 0$  have  $w_J = 0$  and are never set); given the bits, draw each  $X_b$  independently from its true law conditioned on  $X_b \in \Omega_b(x_s)$  (bit 1) or  $X_b \notin \Omega_b(x_s)$  (bit 0). For a function  $\phi$  of  $x_s$  and of the big coordinates in a set  $K$  with  $|K| \leq k$ : given  $x_s$ , the bits on  $K$  are independent with the true marginals, so  $(X_b)_{b \in K}$  has its true product law, and  $\mathbb{E}_\nu \phi = \mathbb{E} \phi$ . Hence  $\mathbb{E}B = \mathbb{E}_\nu B \leq \mathbb{E}_\nu F$ . Under  $\nu$  some bit is 1, so some event of  $\mathcal{F}$  occurs and  $F = 0$ .  $\square$

**The ES instance.** Fix a small absolute  $\varepsilon \in (0, 1/20]$  (chosen in Lemma 10.4), and put  $X = T^\varepsilon$ ,  $V = X^{1/3}$ ,  $L = \log X = \varepsilon \mathcal{L}$  and  $y = \mathcal{L}^6$ . With  $D^*$  as in §4, let  $\mathcal{F}^*$  be the set of events  $E_{v\ell, D}$  with

$$n_D := D^* \in [V, X], \quad v \leq V \text{ squarefree and } y\text{-rough}, \\ \ell \in (T^{0.6}, T^{0.7}] \text{ prime}, \quad v\ell \equiv -1 \pmod{4n_D}.$$

The congruence gives  $v\ell \equiv 3 \pmod{4}$  and  $n_D \mid A_{v\ell}$ , i.e.  $D \mid A_{v\ell}^2$ ; and  $D \leq n_D^2 \leq T^{2\varepsilon}$ . On a fibre  $n \equiv r \pmod{Q}$  the big coordinates are the  $X_\ell = n \bmod \ell$  for primes  $\ell > T^{0.6}$  with  $\ell \nmid Q$ . Everything else is small: the residues of  $n$  modulo the primes  $\leq T^{0.6}$  and modulo 2, the coordinates fixed by the fibre (constants), and, for big  $\ell$ , the higher  $\ell$ -adic digits of  $n$  beyond  $n \bmod \ell$ . On the units these higher digits are uniform and independent of  $X_\ell$ , so all coordinates are independent, as Lemma 10.2 requires. A modulus  $q \leq D$  has at most  $\log D/(0.6\mathcal{L})$  distinct prime factors  $> T^{0.6}$ , and a function of  $n \bmod qQ$  depends on  $X_\ell$  only for these  $\ell$  (whatever their exponent in  $q$ ). So a function of level at most  $D$  is a combination of functions of the small coordinates and of at most  $k = \lfloor \log D/(0.6\mathcal{L}) \rfloor$  big ones.

**Lemma 10.3** (Unique small part; proved). *For each pair  $(\ell, D)$  there is at most one  $v$  with  $E_{v\ell, D} \in \mathcal{F}^*$ . Every event of  $\mathcal{F}^*$  has exactly one big coordinate (if  $\ell \nmid Q$ ),  $p_\ell(x) \leq p^* := T^{-1/2}$  for all  $\ell$  and  $x$ , and, for every*

small configuration  $x$ ,

$$\sum_{\ell > T^{0.6}} p_\ell(x) = \sum_v \sum_{\substack{D: n_D \in [V, X] \\ 4D \equiv -x \pmod{v}}} c(v, n_D), \quad c(v, n) = \sum_{\substack{\ell \in (T^{0.6}, T^{0.7}] \\ \ell \equiv -v^{-1} \pmod{4n}}} \frac{1}{\ell - 1},$$

with  $v$  over the  $y$ -rough squarefree  $v \leq V$ ,  $c(v, n) = 0$  if  $\gcd(v, 2n) > 1$ , and  $p_\ell(x)$  computed with  $X_\ell$  uniform on the units modulo  $\ell$ .

*Proof.*  $v \equiv -\ell^{-1} \pmod{4n_D}$  and  $1 \leq v \leq V \leq n_D < 4n_D$ , so  $v$  is determined by  $(\ell, n_D)$ . As  $v < T^{0.6} < \ell$ ,  $\ell$  is the only prime factor of  $v\ell$  above  $T^{0.6}$ . Given  $x$ , the events at  $\ell$  whose small part holds are the  $E_{v\ell, D}$  with  $x \equiv -4D \pmod{v}$ , and distinct  $D$  give distinct classes  $-4D \pmod{\ell}$ , as  $0 < |D - D'| < T^{2\varepsilon} < \ell$ . So  $p_\ell(x) = (\ell - 1)^{-1} \#\{D : v(\ell, D) \text{ exists, } x \equiv -4D \pmod{v}\}$ ; this is at most  $X(1 + L)/(\ell - 1) \leq T^{-1/2}$  by (F3). Sum over  $\ell$  and group by  $v$ .  $\square$

**Lemma 10.4** (Class-uniform primes; proved modulo Theorems 5.1 and 5.2). *There is an absolute  $\varepsilon_0 > 0$  such that for  $\varepsilon \leq \varepsilon_0$  and  $T$  large the following holds. Apply Theorem 5.2 with  $Q_G = 4X$ , let  $q_1$  be the conductor of the exceptional character if there is one, and put  $\mathcal{N}_{\text{exc}} = \{n : q_1 \mid 4n\}$  if  $q_1$  exists and  $q_1 > \mathcal{L}^{1.9}$ , and  $\mathcal{N}_{\text{exc}} = \emptyset$  otherwise. Then  $c(v, n) \geq 0.13/\varphi(4n)$  for every  $n \in [V, X] \setminus \mathcal{N}_{\text{exc}}$  and every  $v \leq V$  with  $\gcd(v, 2n) = 1$ .*

*Proof.* Put  $q = 4n \leq Q_G$  and  $b \equiv -v^{-1} \pmod{q}$ . Each  $\chi \pmod{q}$  is induced by a primitive  $\chi^*$ , distinct  $\chi$  give distinct  $\chi^*$ , and  $|\vartheta(x; \chi) - \vartheta(x; \chi^*)| \leq \log q$ . By orthogonality and Theorem 5.2, for  $x \in [T^{0.6}, T^{0.7}]$ ,

$$\left| \vartheta(x; q, b) - \frac{x}{\varphi(q)} + [q_1 \mid q] \chi_1(b) \frac{x^{\beta_1}}{\beta_1 \varphi(q)} \right| \leq \frac{\text{Err}_G(x)}{\varphi(q)} + \log q,$$

where  $\text{Err}_G(x)$  is the right-hand side of Theorem 5.2 (times  $C_G$ ), and the bracket is present only in the exceptional case. The range condition  $Q_G^{6c} \leq x$  holds for  $\varepsilon \leq 0.05/c$ , and  $\log x / \log Q_G \geq 0.6/(\varepsilon + o(1))$ . Without an exceptional zero,  $\text{Err}_G(x) \leq C_G x (e^{-0.6/(\kappa(\varepsilon + o(1)))} + o(1))$ . With one,  $(1 - \beta_1) \log x \leq \log x / (\kappa \log Q_G) \leq 0.7/(\kappa(\varepsilon + o(1)))$ , so  $\text{Err}_G(x) \leq C_G x (0.7/(\kappa\varepsilon))(e^{-0.6/(\varepsilon + o(1))} + o(1))$ . Both are at most  $x/16$  for  $\varepsilon$  small, and  $\log q = o(x/\varphi(q))$ . If  $q_1 \mid q$  and  $q_1 \leq \mathcal{L}^{1.9}$ , Theorem 5.1 gives  $1 - \beta_1 \gg \mathcal{L}^{-0.95}(\log \mathcal{L})^{-2}$ , so  $x^{\beta_1 - 1} = o(1)$ ; if  $q_1 \mid q$  and  $q_1 > \mathcal{L}^{1.9}$ , then  $n \in \mathcal{N}_{\text{exc}}$ . So  $\vartheta(t; q, b) \in [\frac{7}{8} - o(1), \frac{9}{8} + o(1)] t/\varphi(q)$  on  $[T^{0.6}, T^{0.7}]$ , and partial summation against  $1/(t \log t)$  gives  $c(v, n) \geq \sum 1/\ell \geq \frac{7}{8} \log \frac{7}{6} / \varphi(q) - O(1/(\mathcal{L}\varphi(q))) \geq 0.13/\varphi(q)$ .  $\square$

If  $q_1 > \mathcal{L}^{1.9}$ , put  $q'' = q_1 / \gcd(q_1, 4) \geq q_1/4$ ; then  $n \in \mathcal{N}_{\text{exc}}$  if and only if  $q'' \mid n$ , and by the divisor bound  $\tau(q'')/q'' \leq q''^{-0.95} \leq 4\mathcal{L}^{-1.8}$  for  $T$  large.

**Lemma 10.5** (Class-uniform  $D$ ; proved). *For  $T$  large, every odd squarefree  $y$ -rough  $v \leq V$  and every unit  $a$  modulo  $v$ ,*

$$W(v, a) := \sum_{\substack{D: n_D \in [V, X] \setminus \mathcal{N}_{\text{exc}} \\ D \equiv a \pmod{v}}} \frac{1}{\varphi(4n_D)} \geq \frac{L^2}{200v}.$$

*Proof.* Write  $D = \kappa t^2$  with  $\kappa$  squarefree, so  $n_D = \kappa t$ . Keep only  $t \leq X^{1/6}$  with  $(t, v) = 1$  and  $\kappa \in (K_1, K_2]$ ,  $K_1 = X^{2/3}$ ,  $K_2 = X/t$ ; then  $n_D \in [V, X]$ , and  $D \equiv a \pmod{v}$  means  $\kappa \equiv b := at^{-2} \pmod{v}$ . As  $\varphi(4n) \leq 2n$ ,  $W(v, a) \geq H_{\text{all}} - H_{\text{exc}}$ , where  $H_{\text{all}}$  and  $H_{\text{exc}}$  are the sums of  $1/(2\kappa t)$  over the kept pairs, with all  $n$  and with  $n \in \mathcal{N}_{\text{exc}}$  respectively. For a unit  $b$  modulo  $v$ , the number of squarefree  $\kappa \leq K$  with  $\kappa \equiv b \pmod{v}$  is  $\sigma_v K/v + O(\sqrt{K})$  with  $\sigma_v = \sum_{(d,v)=1} \mu(d)/d^2 \geq 6/\pi^2$  (write  $\mu^2(\kappa) = \sum_{d^2|\kappa} \mu(d)$ ). Partial summation gives  $\sum_{\kappa \in (K_1, K_2], \kappa \equiv b} 1/\kappa \geq (\sigma_v/v) \log(X^{1/3}/t) - O(K_1^{-1/2})$ . For  $y$ -rough  $v$ ,  $\sum_{t \leq X^{1/6}, (t,v)=1} t^{-1} \log(X^{1/3}/t) \geq L^2/24 - O(\omega(v)L^2/y) - O(L) \geq L^2/48$ . So  $H_{\text{all}} \geq \frac{1}{2} \cdot \frac{6}{\pi^2} \cdot \frac{L^2}{48v} - O(LX^{-1/3})$ . If  $\mathcal{N}_{\text{exc}} \neq \emptyset$ :  $q'' \mid \kappa t$  forces  $q''/(q'', t) \mid \kappa$ , which with  $\kappa \equiv b \pmod{v}$  is one class modulo  $vq''/(q'', t)$  or empty (empty if a prime of  $v$  divides  $q''/(q'', t)$ , since  $b$  is a unit modulo  $v$ ; otherwise the Chinese remainder theorem gives one class), so by (F4) the  $1/\kappa$ -sum is at most  $1/K_1 + (q'', t)L/(vq'')$ . Since  $\sum_{t \leq X} (q'', t)/t \leq \tau(q'')(1+L)$ , we get  $H_{\text{exc}} \leq L^2 \tau(q'')/(vq'') + O(LX^{-2/3}) \leq 4L^2 \mathcal{L}^{-1.8}/v$ . As  $X^{-1/3} \leq 1/v$ ,  $W(v, a) \geq (L^2/v)(6/(96\pi^2) - o(1)) \geq L^2/(200v)$ .  $\square$

**Theorem 10.6** (No positive low-level minorant; proved modulo Theorems 5.1, 5.2 and the fundamental lemma). *There are absolute constants  $c, \varepsilon > 0$  such that for  $T$  large, every modulus  $Q$  with  $\log Q \leq T^{0.05}$  and every unit  $r$  modulo  $Q$  the following holds. If  $B$  is a function on the fibre  $n \equiv r \pmod{Q}$  of level at most  $D$ , with  $B \leq F$  pointwise and  $\log D \leq c\mathcal{L}^4/\log \mathcal{L}$ , then  $\mathbb{E}B \leq 0$ .*

*Proof.* Use  $\mathcal{F}^*$  without the events whose  $\ell$  divides  $Q$ ; each of its events has exactly one big coordinate, and  $p^* \leq T^{-1/2}$  (Lemma 10.3). Fix a small configuration  $x$ ; its entries are units. By Lemma 10.3,  $R(x) \geq \sum_{\ell} p_{\ell}(x) \geq \sum_v \sum_{D \equiv -x/4 \pmod{v}, n_D \notin \mathcal{N}_{\text{exc}}} c(v, n_D) - \omega_{\text{big}}(Q)p^*$ , where  $\omega_{\text{big}}(Q) \leq \log Q/(0.6\mathcal{L}) \leq T^{0.05}$  counts the prime factors of  $Q$  above  $T^{0.6}$ . For  $D \equiv -x/4 \pmod{v}$ ,  $D$  is a unit modulo  $v$ , so  $\gcd(v, 2n_D) = 1$ , and Lemmas 10.4 and 10.5 give

$$R(x) \geq 0.13 \cdot \frac{L^2}{200} \sum_{\substack{v \leq V \\ y\text{-rough, squarefree}}} \frac{1}{v} - T^{0.05} T^{-1/2} \geq \mu^* := c_9 \varepsilon^3 \frac{\mathcal{L}^3}{\log \mathcal{L}},$$

where the  $v$ -sum is  $\geq c_3 \log V/\log y$  by the fundamental lemma (as in Lemma 4.5; squarefreeness costs  $O(\mathcal{L}/y)$ ). With  $k = \lfloor \log D/(0.6\mathcal{L}) \rfloor$  we have  $(k+1) + (2k+1)r^* \leq (k+1)(1+4p^*) \leq \mu^*$  if  $c \leq 0.3c_9\varepsilon^3$ . Lemma 10.2, applied with  $\mathcal{E}$  the set of all events with  $M \leq T$ , gives  $\mathbb{E}B \leq 0$ .  $\square$

The mechanism: conditioned on the coordinates below  $T^{0.6}$ , what remains is a sieve on the big primes with removed classes of total density  $\gg \mathcal{L}^3/\log \mathcal{L}$  (the bulk of the ES mass comes from moduli with one large prime factor), so a sieve of dimension  $\kappa \asymp \mathcal{L}^3/\log \mathcal{L}$  with all moduli above  $T^{0.6}$ . Lower-bound sieves of dimension  $\kappa$  need level about  $z^\kappa$ , here  $e^{\asymp \mathcal{L}^4/\log \mathcal{L}}$ . Lemma 10.1 is the elementary form of this, and the uniqueness family makes  $R(x)$  large for every small configuration, not only on average.



**Corollary 10.7** (The  $1/4$  ceiling; proved implication). *Consider any argument that produces a hard prime  $p$  with  $W(p) > T$  from a function  $B \leq F$  on a fibre  $n \equiv r \pmod{Q}$ , of level at most  $Z$  and with  $\mathbb{E}B > 0$ , through a transfer theorem whose hypothesis requires  $\log x \geq C \log Z$  for the search bound  $x$  (with  $Z \geq Q$ ). Every such argument has  $\log x \gg \mathcal{L}^4 / \log \mathcal{L}$ . Theorem 6.1, and hence the proof of Theorem 1.1, is of this type, whatever the minorant (sandwich, Selberg, Bonferroni, dense or sparse, any value of  $A$ ). So such arguments cannot give  $W(p) \geq \exp((\log p)^{1/4+\delta})$  for any  $\delta > 0$ , and Theorem 1.1 is optimal among them up to a factor  $(\log \log p)^{1/2}$  in  $\log W(p)$ . Modulo [42], Proposition 11.3 improves  $\mathcal{L}^4 / \log \mathcal{L}$  to  $\mathcal{L}^4$  and the factor to  $(\log \log p)^{1/4}$ .*

*Proof.* If  $\log Q > T^{0.05}$ , then  $\log x \geq C \log Z \geq C \log Q > e^{0.05\mathcal{L}}$ . Otherwise Theorem 10.6 forces  $\log Z > c\mathcal{L}^4 / \log \mathcal{L}$ . In both cases  $\log x \gg \mathcal{L}^4 / \log \mathcal{L}$ , so the certified statement “some hard  $p \leq x$  has  $W(p) > T$ ” holds at best with  $\mathcal{L} \ll (\log x \log \log x)^{1/4}$ .  $\square$

*Scope.* Corollary 10.7 says nothing about the true size of the least such prime; nothing about prime input other than low-conductor minorants (bilinear or Type II information, parity-sensitive input, or mixtures of majorants and minorants); nothing about transfers whose positivity comes from an exceptional-character term rather than from  $\mathbb{E}B > 0$  (such transfers can only work at Siegel scales); and nothing about functions that are not of bounded level in the above sense. Section 12 extends the obstruction to minorants of unbounded level, to Siegel-model main terms and to equidistribution hypotheses, for linear certificates with full-orbit uniform accounting; Section 13 gives a hypothesis outside that class which reaches  $1/3$ .

**Corollary 10.8** (The energy tail cannot be improved; proved modulo Theorems 5.1, 5.2, 3.1 and the fundamental lemma). *On the fibre of Theorem 3.3, there is no bound  $\mathbb{E}(F^{(j)} - u_j)^2 \leq 2^{-\tau/\rho}$  for all  $j$  and all  $\tau$ , with  $u_j$  as in (10) and  $\rho \leq c'\mathcal{L}/(\log \mathcal{L})^2$  ( $c'$  small absolute). So the weights of Theorem 7.9 are optimal for this purpose up to a factor  $(\log \mathcal{L})^2$ .*

*Proof.* Such a bound gives  $\mathbb{E}[F - B] \leq m^2 S 2^{-\tau/\rho} \leq \delta/100$  for some  $\tau \ll \rho(S_1 + \mathcal{L}) \ll \rho \mathcal{L}^3 \log \mathcal{L}$ , so  $\mathbb{E}B \geq 0.99\delta > 0$  (Lemma 8.6), and  $B$  has level at most  $T^3 e^{2\tau}$  (Lemma 8.4). For  $c'$  small this contradicts Theorem 10.6.  $\square$

**Remark 10.9** (The heuristic exponent  $1/3$ ; Assessment). By Theorems 3.3 and 4.4,  $\log(1/\delta^*(T)) = \mathcal{L}^{3+o(1)}$ . If the least prime in the avoiding set behaved like the least prime in a set of Haar density  $\delta^*$  (as for a single progression, by Linnik’s theorem), one would expect a hard prime with  $W(p) > T$  and  $\log p \ll \mathcal{L}^{3+o(1)}$ , i.e.  $\log W(p) \approx (\log p)^{1/3}$  infinitely often. This is a heuristic, not a claim. By Corollary 10.7 it cannot be reached by a level- $Z$  minorant with a transfer costing  $\log x \gg \log Z$ : one would need prime-detecting input for a sifted set of sieve dimension  $\asymp \mathcal{L}^3$  at level well below its sieving limit, that is, information beyond minorants of bounded level, such as bilinear



(Type II) estimates or other parity-sensitive input adapted to the ES system. We do not know how to provide it. Section 12 makes this precise for linear certificates, and Section 13 shows that a Linnik-type hypothesis for sifted sets (Theorem 13.1) gives the exponent  $1/3$  up to logarithms.

11. ONE-BIG-PRIME ATOMS: THE HAAR EXPONENT WITHOUT LOSS, TYPICAL SIZE, AND THE CEILING AT LEVEL  $\mathcal{L}^4$

This section imports one input from the companion note [42], which proves the exceptional-set bound  $E(N) \ll N \exp(-c(\log N)^{3/4})$ . That note is *internally proved, not externally refereed*; results below that use it are labelled “modulo [42]”. Its analytic inputs are the Bombieri–Vinogradov theorem, the Brun–Titchmarsh inequality and Shiu’s theorem, and its constants are not asserted to be effective.

*The atoms of [42].* Fix  $\varkappa = 1/480$  (any fixed  $\varkappa \in (0, 1/240)$  would do). For  $X$  large put  $t = \log X$ ,  $K_X = \lfloor X^\varkappa \rfloor$ ,  $\mathcal{K} = \{k \leq K_X : k \equiv 1 \pmod{4}\}$  and  $L_K = \text{lcm } \mathcal{K}$ . The family  $\mathcal{A}_X$  of [42, (4)] consists of quadruples  $(k, \ell, u, v)$  with  $k \in \mathcal{K}$ ,  $\ell$  a prime in a dyadic block  $I_j \subseteq (X^{1/2}, X]$ ,  $H < u, v \leq z_j = x_j^{1/6}$  (with  $H = K_X^{10}$  and  $x_j$  the left end of  $I_j$ ),  $(u, v) = (uv, k) = 1$ , a bound on  $\omega(uv)$ , and  $4uv \mid k\ell + 1$ ; the atom is the class  $E_A = \{n \equiv -uv^{-1} \pmod{k\ell}\}$ , and  $H_X(n) = \sum_A \mathbf{1}_{E_A}(n)$ . With  $M = k\ell \equiv 3 \pmod{4}$  and  $w = (M + 1)/(4uv)$  we have  $uvw = A_M$ , so  $E_A$  is one of the multiplier congruences in (2): if  $n \in E_A$  then  $W(n) \leq k\ell \leq K_X X$ . We use three facts from [42]:

- (A1) [42, Lemma 2.2]: at a fixed  $\ell$  the atoms have pairwise distinct residues modulo  $\ell$ , and there are at most  $z_j^2 \leq \ell^{1/3}$  of them; the residue  $-uv^{-1} \pmod{\ell}$  is a unit, since  $uv < \ell$ . An atom is *active* on the class  $c = n \pmod{L_K}$  iff  $k \mid u + cv$ .
- (A2) [42, Cor. 4.3 and Lemma 3.2]: with  $f_c(\ell)$  the number of atoms at  $\ell$  active on  $c$  and  $\mu_c = \sum_\ell f_c(\ell)/\ell$ , there are constants  $a, C_u > 0$  with  $a t^3 \leq \mu_c \leq C_u t^3$  for every unit  $c$  modulo  $L_K$  and all large  $X$ .
- (A3) [42, Lemma 8.1 and Thm. 8.2]: with  $y = Bt^3$ ,  $r$  the least even integer  $\geq D_B t^3$  ( $B, D_B$  as there;  $D_B$  may be enlarged),  $S_y(n) = \mathbf{1}[(n, P_y) = 1]$  and  $Q_r(h) = \sum_{j \leq r} (-1)^j \binom{h}{j}$ , the function  $\nu_X = S_y Q_r(H_X)$  is  $\geq 0$  on  $\mathbb{Z}$ ,  $Q_r(0) = 1$ ,  $Q_r(h) \leq \mathbf{1}[h = 0] + \binom{h}{r+1}$ , and  $\nu_X$  is a combination  $\sum_i c_i \mathbf{1}[n \equiv a_i \pmod{q_i}]$  with  $\log \max q_i \leq C' t^4$  and  $\sum_i |c_i| \leq e^{C' t^4}$ ; moreover  $\sum_{n \leq N} \nu_X(n) \ll N e^{-c_a t^3}$  whenever  $\log N \geq C_0 t^4$ .

Throughout,  $X = T^{1/(1+\varkappa)}$ , so that  $K_X X \leq T$  and  $t \asymp \mathcal{L}$ ; and  $\mathbb{P}_*, \mathbb{E}_*$  denote Haar measure on  $\hat{\mathbb{Z}}^\times$ . Under  $\mathbb{P}_*$  the coordinate  $c = n \pmod{L_K}$  is a unit, and the residues  $n \pmod{\ell}$ ,  $\ell \in (X^{1/2}, X]$  (all coprime to  $L_K$ ), are independent of  $c$  and of each other, each uniform on  $(\mathbb{Z}/\ell)^\times$ . By (A1), given  $c$ ,  $H_X = \sum_\ell I_\ell$  with independent Bernoulli variables  $I_\ell$  of mean  $f_c(\ell)/(\ell - 1)$ .

**Proposition 11.1** (Haar lower bound without loss; proved modulo [42]). *For large  $T$ ,  $\mathbb{P}_*(H_X = 0) \leq e^{-at^3}$  and  $\delta^*(T) \leq e^{-at^3}$ . Hence  $\log(1/\delta^*(T)) \gg \mathcal{L}^3$ , and with Theorem 3.3,  $\mathcal{L}^3 \ll \log(1/\delta^*(T)) \ll \mathcal{L}^3(\log \mathcal{L})^5$ .*

*Proof.* Given  $c$ ,  $\mathbb{P}_*(H_X = 0 \mid c) = \prod_\ell (1 - f_c(\ell)/(\ell - 1)) \leq e^{-\mu_c} \leq e^{-at^3}$  by (A2). This holds for every unit  $c$ , hence also after conditioning on  $n \equiv 1 \pmod{24}$ , which restricts only  $c \pmod{3}$  (note  $9 \in \mathcal{K}$ ) and the independent coordinate  $n \pmod{8}$ . If  $W(n) > T$  then  $n$  lies in no atom, since atoms are multiplier congruences of modulus  $\leq K_X X \leq T$ ; so  $\delta^*(T) \leq \mathbb{P}_*(H_X = 0 \mid n \equiv 1 \pmod{24}) \leq e^{-at^3}$ .  $\square$

This removes the factor  $1/\log \mathcal{L}$  of Theorem 4.4, at the price of the inputs of [42] and ineffective constants; Theorem 4.4 uses only the fundamental lemma. The reason is structural: each atom of [42] has one large prime and a small coordinate  $c$  that is revealed, so the events are conditionally independent and an exact product replaces the Janson-type inequality.

**Theorem 11.2** (Typical size; proved modulo [42] and Page’s theorem). *There are absolute  $c, c_1 > 0$  such that, uniformly for  $x \geq 3$  and  $3 \leq T \leq \exp(c_1(\log x)^{1/4})$ ,*

$$\#\{p \leq x \text{ prime} : W(p) > T\} \ll \pi(x) e^{-c(\log T)^3}.$$

Here “Page’s theorem” is the classical prime number theorem in progressions, uniform for  $q \leq \exp(c_2\sqrt{\log x})$ , with the possible exceptional term [8, Ch. 20]: for  $(a, q) = 1$ ,

$$\psi(x; q, a) = \frac{x}{\varphi(q)} - \mathbf{1}[q_1 \mid q] \frac{\chi_1(a)x^{\beta_1}}{\varphi(q)\beta_1} + O(xe^{-c_3\sqrt{\log x}}), \quad (12)$$

where  $\chi_1$  (primitive modulo  $q_1$ ) is the possible exceptional real character for this range. Any version with this range and error suffices. The Siegel–Walfisz theorem would not: the moduli below have size  $\exp(C(\log \log x)^{4/3})$ .

*Proof.* We may assume  $T \geq T_0$ . For a prime  $p > y$  with  $W(p) > T$  we have  $S_y(p) = 1$  and  $H_X(p) = 0$ , so  $\nu_X(p) = 1$ ; hence the count is at most  $y + \sum_{p \leq x} \nu_X(p)$ , and  $y = Bt^3$  is negligible. Choose  $c_1$  so that  $\log x \geq C_0 t^4$ .

*Case A:*  $c_a t^3 \geq 2 \log \log x$ . As  $\nu_X \geq 0$ , (A3) gives  $\sum_{p \leq x} \nu_X(p) \leq \sum_{n \leq x} \nu_X(n) \ll xe^{-c_a t^3} \leq xe^{-c_a t^3/2} / \log x$ .

*Case B:*  $c_a t^3 < 2 \log \log x$ . Then  $t^4 \ll (\log \log x)^{4/3}$ , so every modulus  $q_i$  and the sum  $\sum_i |c_i|$  are at most  $\exp(O((\log \log x)^{4/3}))$ , inside the range of (12) for large  $x$ . Since  $0 \leq \nu_X \leq \sum |c_i|$ ,

$$\begin{aligned} \sum_{p \leq x} \nu_X(p) &\leq \sqrt{x} \sum_i |c_i| + \frac{2}{\log x} \sum_{n \leq x} \nu_X(n) \Lambda(n), \\ \sum_{n \leq x} \nu_X(n) \Lambda(n) &= \sum_i c_i \psi(x; q_i, a_i). \end{aligned}$$

Terms with  $(a_i, q_i) > 1$  count only powers of the primes dividing  $(a_i, q_i)$  and contribute  $O(\sum |c_i| \log q_i (\log x)^2)$ . For the others insert (12). Put  $\mathbb{E}_* \nu_X = \sum_{(a_i, q_i)=1} c_i / \varphi(q_i)$  and  $\tilde{\chi}_1(n) = \chi_1(n \bmod q_1)$  on  $\hat{\mathbb{Z}}^\times$ . Since  $\int_{n \equiv a \pmod{q}} \tilde{\chi}_1 d\mathbb{P}_*$  is  $\chi_1(a)/\varphi(q)$  if  $q_1 \mid q$  and 0 otherwise ( $\chi_1$  is primitive), the main terms add up to  $x \mathbb{E}_*[\nu_X(1 - \epsilon \tilde{\chi}_1)]$  with  $\epsilon = x^{\beta_1-1}/\beta_1 \in [0, 2]$ . As  $\nu_X \geq 0$  on  $\hat{\mathbb{Z}}^\times$ , this is at most  $3x \mathbb{E}_* \nu_X$ . So  $\sum_{p \leq x} \nu_X(p) \leq (6x/\log x) \mathbb{E}_* \nu_X + O(xe^{-c_3 \sqrt{\log x}/2})$ . On units  $S_y = 1$ . Given  $c$ ,  $\mathbb{E}_*[(H_X)_m \mid c] = m! e_m((f_c(\ell)/(\ell-1))_\ell) \leq (2\mu_c)^m \leq (2C_u t^3)^m$  for the falling factorials, so by (A3) and Proposition 11.1,

$$\mathbb{E}_* \nu_X \leq \mathbb{P}_*(H_X = 0) + \frac{\mathbb{E}_*(H_X)_{r+1}}{(r+1)!} \leq e^{-at^3} + \left(\frac{2eC_u t^3}{r+1}\right)^{r+1} \leq 2e^{-at^3}$$

once  $D_B \geq \max(2e^2 C_u, a)$ . The error term is  $\ll \pi(x)e^{-at^3}$  because  $e^{at^3} \leq (\log x)^{2a/c_a}$ . In both cases the count is  $\ll \pi(x)e^{-\min(a, c_a/2)t^3}$ , and  $t \asymp \mathcal{L}$ .  $\square$

The normalisation by  $\pi(x)$  rather than  $x$  matters when  $\mathcal{L}^3 \lesssim \log \log x$ . At  $T = \exp(c_1(\log x)^{1/4})$  the theorem gives the prime case of the exceptional-set bound of [42]; it is a reproof, not an improvement. It is the opposite inequality to Theorem 1.1: an upper bound for the number of primes with  $W(p) > T$  needs a *majorant*, and for majorants the exceptional character costs only the factor  $1 + \epsilon \leq 3$ ; Theorem 1.1 needs a *minorant*. Heuristically (Assessment) the count is  $\pi(x)e^{-\mathcal{L}^{3+o(1)}}$  up to  $\mathcal{L} \approx (\log x)^{1/3}$ ; a matching lower bound in the range  $\log T \leq c(\log x)^{1/4}/(\log \log x)^{O(1)}$  would follow from the main term of Theorem 6.1 except when an exceptional character divides the quarantine modulus; we have not pursued this.

The same atoms sharpen the ceiling of Theorem 10.6.

**Proposition 11.3** (Ceiling at level  $c\mathcal{L}^4$ ; proved modulo [42]). *There are absolute  $c, T_0 > 0$  such that for  $T \geq T_0$ , every modulus  $Q$  with  $\log Q \leq T^{0.05}$  and every unit  $r$  modulo  $Q$ : if  $B$  is a function on the fibre  $n \equiv r \pmod{Q}$  of level at most  $D$ , with  $B \leq F$  pointwise and  $\log D \leq c\mathcal{L}^4$ , then  $\mathbb{E}B \leq 0$ .*

*Proof.* Apply Lemma 10.2 with big coordinates  $X_\ell = n \bmod \ell$  for the primes  $\ell \in (X^{1/2}, X]$  with  $\ell \nmid Q$ , and all other coordinates small (including  $c = n \bmod L_K$ , the residues at all other primes, and the higher  $\ell$ -adic digits at big  $\ell$ , as in §10). As there, a function of level  $\leq D$  reads at most  $k = \lfloor 2 \log D/t \rfloor$  big coordinates, since each big prime exceeds  $X^{1/2}$ . Let  $\mathcal{F}$  be the set of atoms with  $\ell \nmid Q$ ; each reads  $c$  (small, as  $k \mid L_K$ ) and one big  $X_\ell$ , and  $\mathcal{F} \subseteq \mathcal{E}$ , the set of all multiplier congruences with  $M \leq T$ . Fix a small configuration; its  $c$  is a unit. By (A1),  $p_\ell = f_c(\ell)/(\ell-1) \leq \ell^{1/3}/(\ell-1) \leq 2X^{-1/3} =: p^*$ . At most  $2 \log Q/t \leq T^{0.05}$  big primes divide  $Q$ , so by (A2)

$$R \geq \sum_{\ell \nmid Q} p_\ell \geq \mu_c - T^{0.05} p^* \geq at^3/2$$

for  $T \geq T_0$ , for every small configuration. Since  $r^* \leq 4X^{-1/3}$ ,  $(k+1) + (2k+1)r^* \leq 2(k+1) \leq at^3/2$  once  $\log D \leq c\mathcal{L}^4$  with  $c$  small. Lemma 10.2 gives  $\mathbb{E}B \leq 0$ .  $\square$

Compared with Theorem 10.6, the factor  $1/\log \mathcal{L}$  disappears, and the inputs (Gallagher’s theorem, Landau–Page, the fundamental lemma) are replaced by those of [42] (with an ineffective  $T_0$ ). In Corollary 10.7 one may therefore replace  $\mathcal{L}^4/\log \mathcal{L}$  by  $\mathcal{L}^4$ , and Theorem 1.1 is optimal among the arguments described there up to a factor  $(\log \log p)^{1/4}$  in  $\log W(p)$ .

**Remark 11.4** (One sieve limit behind two ceilings; proved parts and Assessment). The atoms of [42] form a system with one big prime per event, big cost  $\asymp \mathcal{L}$  and conditional mass  $P \asymp \mathcal{L}^3$  uniformly in the small coordinates. For such a system, order- $k$  certificates are trivial below  $k \asymp P$  on *both* sides (proved, no novelty claimed): a minorant of  $\mathbf{1}[\text{no event}]$  depending on at most  $k \leq 0.6P - 1$  big coordinates has conditional mean  $\leq 0$  (Lemma 10.1); a majorant of order  $k$  saves at most  $k \log(C(P + 4k)/k) + O(k + \log k)$  in  $\log(1/\mathbb{E})$  (the binomial extrapolation of [40, Lemma 8.3], whose core is [36, 4]); and the Bonferroni truncations  $Q_k(H)$  of order  $k \geq e^2 P$  achieve both (as minorant for odd  $k$ , as majorant for even  $k$ ). So the critical level is  $\asymp \mathcal{L} \cdot \mathcal{L}^3 = \mathcal{L}^4$ . A budget  $\lambda \asymp \log N$  (rounding to integers) then exploits  $\mathcal{L} \asymp \lambda^{1/4}$  and saves  $\asymp \lambda^{3/4}$ : this is the exceptional-set exponent  $3/4$  of [42], which is sharp for coefficient-sum congruence majorants [40, Thm. 10.14]. A budget  $\lambda \asymp \log x$  (transfer to primes  $\leq x$ ) exploits  $\mathcal{L} \lesssim \lambda^{1/4}$ : the exponent  $1/4$  of Theorem 1.1, sharp in the sense of Proposition 11.3. Within the stated scopes of these two ceilings this is a conjunction of proved statements; as a description of “the” mechanism it is an Assessment. The full avoidance indicator is not a one-big-prime system: on the minorant side the subfamily is used only for the obstruction (as  $F \leq \mathbf{1}[\text{no event of } \mathcal{F}]$ ), while the positive minorant of Theorem 9.1 is built differently.

## 12. BEYOND BOUNDED LEVEL: A WIENER-NORM BARRIER FOR LINEAR CERTIFICATES

Theorem 10.6 concerns minorants of bounded level. Here we show that the planted law behind it is exponentially pseudorandom against residue classes and characters of *every* modulus, and deduce that a large class of linear certificates, whatever the prime input, cannot beat the exponent  $1/4$ . The results are from the working note [32] (internally reviewed); we give the two short key proofs and state the rest with pointers.

*Setting.* Fix  $Q$  with  $\log Q \leq T^{0.05}$ , a unit  $r$ , the fibre  $H = \{n \equiv r \pmod{Q}\}$  with Haar probability  $\mathbb{P} = \mathbb{P}_H$ , and  $F = \mathbf{1}[W > T]$ . In this section the big coordinate  $X_\ell$  ( $\ell > T^{0.6}$  prime,  $\ell \nmid Q$ ) is the whole  $\ell$ -adic component of  $n$  (modulo a power of  $\ell$  large enough for all functions considered); the events of  $\mathcal{F}^*$  read it only modulo  $\ell$ , so the quantities  $p_\ell(x)$  and the bound  $R(x) \geq \mu^*$  of the proof of Theorem 10.6 are unchanged, with  $p^* \leq T^{-1/2}$ . Put  $k + 1 = \lfloor \mu^*/2 \rfloor$ ; then  $R(x) \geq (k + 1) + (2k + 1)r^*$  for every small configuration  $x$ . Let  $\nu$  be the planted law of the proof of Lemma 10.2 for  $\mathcal{F}^*$  and this  $k$ , and  $\rho = \nu - \mathbb{P}$ . Then  $F = 0$   $\nu$ -almost surely.

A *reduced product* is a function  $h = h_s(x_s) \prod_{b \in I} h_b(X_b)$ ,  $I$  a finite set of big coordinates, with  $|h_s|, |h_b| \leq 1$  and  $|\mathbb{E}h_b| \leq 1/4$  for  $b \in I$ . Restricted to  $H$ , the indicator of a residue class, a Dirichlet character and an additive character  $n \mapsto e(an/q)$  of any modulus are reduced products (up to a unimodular constant): the local factor at a big  $b \mid q$  is a class indicator of mean  $\leq 1/(b-1)$ , a non-trivial character of mean 0, or a non-trivial additive character of mean  $c_{b^v}(a')/\varphi(b^v)$ , of modulus  $\leq 1/(b-1)$ ; trivial factors are dropped from  $I$ .

**Lemma 12.1** (Pseudorandomness of the planted law; proved). *For every reduced product  $h$ ,  $|\mathbb{E}_\rho h| \leq (4r^*)^{k+1}$ , and  $\mathbb{E}_\rho h = 0$  if  $|I| \leq k$ .*

*Proof.* Fix  $x_s$  and use the notation of Lemma 10.1 for the bits  $\mathbf{1}[X_b \in \Omega_b(x_s)]$ . Given the bits the big coordinates are independent; put  $\beta_b = \mathbb{E}[h_b \mid X_b \in \Omega_b]$  and  $\gamma_b = \mathbb{E}[h_b \mid X_b \notin \Omega_b]$  ( $\beta_b = \gamma_b = 1$  for  $b \notin I$ ). The conditional perturbation is  $P_0 \sum_{|J|=k+1} w_J \sigma_J$ , and

$$\begin{aligned} \mathbb{E}_{\sigma_J} \left[ \prod_{b \in I} h_b \right] &= \sum_{y \subseteq J} (-1)^{|y|+1} \prod_{b \in y} \beta_b \prod_{b \in J \setminus y} \gamma_b \prod_{b \in I \setminus J} \gamma_b \\ &= - \prod_{b \in J} (\gamma_b - \beta_b) \prod_{b \in I \setminus J} \gamma_b. \end{aligned}$$

For  $b \in J \setminus I$  the factor  $\gamma_b - \beta_b$  vanishes, so only  $J \subseteq I$  contribute; none if  $|I| \leq k$ . Now  $|\gamma_b - \beta_b| \leq 2$ , and  $\mathbb{E}h_b = (1 - p_b)\gamma_b + p_b\beta_b$  gives  $|\gamma_b| \leq (1/4 + p_b)/(1 - p_b) < 1/2$ . Next  $w_J \leq r^{*k+1}/e_{k+1}(r)$ , and  $e_{k+1}(r) \geq 1$ : the chain in the proof of Lemma 10.1 with  $s = r$  gives  $a e_a(r) \geq (k+1)e_{a-1}(r)$  for  $a \leq k+1$ . With  $m = |I|$ ,  $\sum_{J \subseteq I, |J|=k+1} \prod_{b \in I \setminus J} |\gamma_b| \leq \binom{m}{k+1} 2^{-(m-k-1)} \leq 2^{k+1}$ . As  $P_0 \leq 1$  and  $|h_s| \leq 1$ , the conditional bound is  $r^{*k+1} 2^{k+1} 2^{k+1}$ ; average over  $x_s$ .  $\square$

**Theorem 12.2** (Wiener-norm barrier; proved modulo Theorems 5.1, 5.2 and the fundamental lemma). *There are absolute  $c, T_0 > 0$  such that for  $T \geq T_0$ , every  $Q$  with  $\log Q \leq T^{0.05}$  and every unit  $r$ : if  $B \leq F$  on  $H$  and  $B = \sum_i c_i h_i$  is any finite combination of reduced products, then*

$$\mathbb{E}_H B \leq \eta \sum_i |c_i|, \quad \eta = \exp(-c\mathcal{L}^4/\log \mathcal{L}).$$

*Only the  $h_i$  involving at least  $k+1 \asymp \mathcal{L}^3/\log \mathcal{L}$  big primes, hence of modulus  $> T^{0.6(k+1)}$ , contribute.*

*Proof.*  $\mathbb{E}_H B = \mathbb{E}_\nu B - \mathbb{E}_\rho B \leq \mathbb{E}_\nu F - \mathbb{E}_\rho B = -\mathbb{E}_\rho B \leq \sum_i |c_i| (4r^*)^{k+1}$  by Lemma 12.1, and  $(4r^*)^{k+1} \leq (8T^{-1/2})^{\mu^*/2-1} \leq \exp(-c\mathcal{L}\mu^*)$ .  $\square$

In particular  $\|F\|_\times \geq \mathbb{E}_H F/\eta$  for the  $\ell^1$  norm  $\|\cdot\|_\times$  over such representations: the avoidance indicator costs  $e^{c\mathcal{L}^4/\log \mathcal{L}}$  times its mean. Theorem 10.6 is the case  $|I| \leq k$ . (With the atoms of §11 in place of  $\mathcal{F}^*$ , the same argument gives  $\eta = e^{-c\mathcal{L}^4}$  modulo [42]; we do not use this.)

*Linear certificates* [32, Def. 2.1]. Let  $\mathfrak{h}$  be a family of reduced products on  $H$  with moduli in a finite set,  $N_x$  the number of primes  $p \leq x$  in  $H$  coprime

to these moduli, and  $m_x$  their counting measure. A nonnegative measure  $m$  on  $H$  of mass  $N_x$  is  $\mathfrak{h}$ -consistent at accuracy  $\mathfrak{E}$  if  $|\int h dm - N_x \mathbb{E}_H h| \leq \mathfrak{E}$  for all  $h \in \mathfrak{h}$ . A *linear certificate* concludes “ $\int F dm_x > 0$ ” from the facts that  $m_x$  is nonnegative of mass  $N_x$  and  $\mathfrak{h}$ -consistent at accuracy  $\mathfrak{E}$ , and from nothing else (in particular not from its atomicity, integrality or support); it is valid iff every consistent  $m$  has  $\int F dm > 0$ . Every minorant transfer is one: from  $B = \sum c_i h_i \leq F$  it concludes  $\sum_p F(p) \geq N_x \mathbb{E}_H B - \mathfrak{E} \sum |c_i|$ . By Lemma 12.1 the diffuse measure  $N_x \nu$  is consistent at every accuracy  $\mathfrak{E} \geq N_x (4r^*)^{k+1}$ , for every family  $\mathfrak{h}$ , and has  $\int F d(N_x \nu) = 0$ ; so valid linear certificates need  $\mathfrak{E} < N_x \eta$  [32, Thm. 2.2]. On the other hand, for a modulus  $q > x$  no bound that is uniform over a translation orbit can be small: if  $\mathfrak{h}$  contains all classes in the fibre modulo  $q$ , then  $\mathfrak{E} \geq 1/2$ ; if it contains all Dirichlet characters (or all additive characters) modulo  $q$ , then  $\mathfrak{E}^2 \geq N_x/2$  ([32, Lemma 2.3], by orthogonality, as soon as  $N_x$  is at most half the number of units of the fibre modulo  $q$ ).

**Corollary 12.3** (Full-orbit uniform linear certificates are capped at  $1/4$ ; proved implication, same inputs). *Let a linear certificate use a family  $\mathfrak{h}$  of residue classes, Dirichlet characters or additive characters which contains, with each member of modulus  $> x$ , its whole translation orbit, with one error bound per orbit. If it certifies a prime  $p \leq x$  in  $H$  with  $W(p) > T$ , then  $\log x \gg \mathcal{L}^4 / \log \mathcal{L}$ . Hence no such certificate, whatever true prime input it uses (Bombieri–Vinogradov, Elliott–Halberstam, GRH for the characters it uses, Siegel-zero main terms as below), gives  $W(p) \geq \exp((\log p)^{1/4+\delta})$ .*

The proof [32, Cor. 2.4] splits  $B$  into members with at most  $k$  big primes, on which  $\mathbb{E}_\rho = 0$ , and deep members, whose moduli exceed  $x^2$  if  $\log x \leq c\mathcal{L}^4 / \log \mathcal{L}$ ; each deep orbit costs at least  $1/2$  (classes) or  $\sqrt{N_x}/2$  (characters), while by Theorem 12.2 it contributes at most  $\eta$  per unit of coefficient to the mean, so positivity needs  $N_x \eta > 1/2$ . One-sided accounting (dropping deep classes with positive coefficients) and Hölder-averaged accounting over an orbit do not help [32, Cor. 2.4 and Prop. 2.5]. *Siegel zeros* [32, Thm. 3.1]: for any real character  $\chi_1$  and  $\epsilon \in [0, 1]$ , the reweighted fake  $(1 - \epsilon\chi_1)\nu$  is nonnegative, has  $F = 0$  almost surely, and agrees with the Siegel-model law  $(1 - \epsilon\chi_1)\mathbb{P}$  to within  $2(4r^*)^{k+1}$  on every class, Dirichlet character and additive character; and if  $B$  has level  $\leq T^{0.6(k-s)}$ ,  $s$  the number of big primes dividing the conductor of  $\chi_1$ , then  $\mathbb{E}_{(1-\epsilon\chi_1)\mathbb{P}} B \leq 0$  exactly. So an exceptional term in the prime law does not rescue a minorant transfer of this kind.

**Proposition 12.4** (Equidistribution hypotheses cap at  $1/4$ ; proved implication, same inputs). *Let  $\mathcal{I}$  be any set of Haar-centred error bounds for the primes  $p \leq x$  in  $H$  in residue classes, Dirichlet characters or additive characters of moduli  $q \leq x$ , individually or on average over  $q$ , with error bounds  $\geq 2 \log x$  per statement (for example Bombieri–Vinogradov, the Elliott–Halberstam conjecture at any level  $\theta < 1$ , GRH restricted to moduli  $\leq x$ ). A linear certificate using  $\mathcal{I}$  that produces a prime  $p \leq x$  in  $H$  with  $W(p) > T$  needs  $\log x \geq c\mathcal{L}^4 / \log \mathcal{L}$ .*



*Proof.* Let  $\log x < 0.6\mathcal{L}(k+1)$ . A modulus  $q \leq x$  has at most  $k$  big prime factors, so every function in  $\mathcal{I}$  restricted to  $H$  involves at most  $k$  big coordinates, and  $\mathbb{E}_\nu = \mathbb{E}_H$  on it (Lemma 12.1). So  $m = N_x \nu$  takes exactly the values  $N_x \mathbb{E}_H(\cdot)$ , which differ from the centring  $N_{x,q} \mathbb{E}_H(\cdot)$  by at most  $N_x - N_{x,q} \leq \omega(q) \leq 2 \log x$  ( $N_{x,q}$  counts the primes coprime to  $q$ ). For averaged statements the extra error  $\sum_{q \leq x^\theta} 2 \log x$  is below the allowed  $x(\log x)^{-A}$ . Thus  $m$  satisfies  $\mathcal{I}$ , and  $\int F dm = 0$ .  $\square$

Raising the level of distribution only changes  $c$ : beyond level  $x$  the primes are not equidistributed in the orbit-uniform sense above. Full GRH is *not* covered by Proposition 12.4: it also asserts  $|\sum_{p \leq x} \chi(p)| \ll \sqrt{x} \log^2(qx)$  for moduli  $q > x$ , and whether the planted measure satisfies these bounds is open; Corollary 12.3 covers such characters only when they are used orbit-uniformly.

*Scope* [32, §6]. Not covered: certificates that use the atomicity, integrality or support of the prime measure (for instance that classes modulo  $q > x$  whose least positive member exceeds  $x$  are empty, i.e. minorants valid only on  $[1, x]$  or only on primes; single moduli  $\leq x^{1/5}/(QT)$  are excluded in [32, Prop. 5.2]); non-periodic families inside the avoiding set; non-linear arguments (pair correlations, density increments); and whether Type II information can help. On the integer side, the Type I sums produced by Vaughan- or Heath-Brown-type identities are blocked only under full-orbit uniform accounting [32, Prop. 4.1], which is not the natural accounting for integers. Our reading (Assessment) is that  $1/4$  is a sieve-*dimension* barrier (dimension  $\kappa \asymp \mathcal{L}^3/\log \mathcal{L}$ , sifting limit  $\beta_\kappa \asymp \kappa$ ,  $\log z \asymp \mathcal{L}$ ), not a parity barrier.

### 13. THE EXPONENT $1/3$ UNDER A LINNIK-TYPE HYPOTHESIS FOR SIFTED SETS

By §12, the heuristic exponent  $1/3$  (Remark 10.9) is out of reach of linear certificates with standard prime input. Here we identify a natural hypothesis that gives it. The material is from [33] (internally reviewed).

A *unit-class sieve system* with moduli  $\leq T$  is a finite family  $\mathcal{E}$  of events  $\{n \equiv a_E \pmod{m_E}\}$  with  $2 \leq m_E \leq T$  and  $(a_E, m_E) = 1$ ; no condition is imposed on their number, overlaps or dimension. Its *avoiding set* is  $S(\mathcal{E}) = \{n \in \hat{\mathbb{Z}}^\times : n \not\equiv a_E \pmod{m_E} \forall E\}$ , and  $\delta(\mathcal{E})$  is its Haar measure (the proportion of units modulo  $\text{lcm}(m_E)$  in no event).

**Hypothesis LS(C)** (Linnik's theorem for sifted sets; a conjecture). For every  $T \geq 3$  and every unit-class sieve system  $\mathcal{E}$  with moduli  $\leq T$  and  $\delta(\mathcal{E}) > 0$  there is a prime  $p > T$  avoiding  $\mathcal{E}$  with

$$\log p \leq C(\log T + \log(1/\delta(\mathcal{E}))).$$

For a single residue class (all other classes modulo  $q$  removed,  $T = q$ ,  $\delta = 1/\varphi(q)$ ) this is Linnik's theorem. LS is the log-scale form, not the Cramér form, of the model in which a sifted set behaves like a residue class of



the same density, for the purpose of containing one prime. The term  $C \log T$  cannot be dropped: shifted Jacobsthal configurations (one class modulo each prime  $\leq z$ ) have  $\delta \asymp 1/\log z$  and least avoiding integers of size  $z^{1+o(1)}$ . The restriction to moduli  $\leq T$  matters: for an arbitrary set of density  $\delta$  modulo  $Q$  no bound  $p \ll \delta^{-1}(\log Q)^{O(1)}$  holds (take the units in  $(Q(1-\delta), Q)$ ).

**Theorem 13.1** (LS implies exponent  $1/3$ ; proved modulo Theorem 3.1, conditional on  $LS(C)$ ). *Assume  $LS(C)$ . For every large  $T$  there is a hard prime  $p > T$  with  $W(p) > T$  and  $\log p \ll_C \mathcal{L}^3(\log \mathcal{L})^5$ . Consequently, for infinitely many hard primes,*

$$W(p) \geq \exp(c_C(\log p)^{1/3}(\log \log p)^{-5/3}).$$

$LS(C)$  is used only for one system  $\mathcal{E}_T$  for each  $T$ .

*Proof.* Let  $\mathcal{E}_T$  consist of the events  $n \equiv -4D \pmod{M}$  for all atoms  $(M, D)$  with  $M \leq T$  (unit classes, as  $\gcd(4D, M) = 1$ ), and the events  $n \equiv c \pmod{840}$  for the unit classes  $c$  that are not squares modulo 840. Its moduli are  $\leq T$ . Let  $(Q, r)$  be the quarantine of Theorem 3.3. By Lemma 2.5(i) the units  $n \equiv r \pmod{Q}$  with  $F(n) = 1$  have  $W(n) > T$ , so by Lemma 2.1 they avoid every atom; as  $840 \mid Q$  and  $r$  is a square modulo 840 (Lemma 2.5(iii)), they lie in  $S(\mathcal{E}_T)$ . The class  $r \bmod Q$  has Haar measure  $1/\varphi(Q)$  and within it  $\mathbb{E}F = \delta \geq e^{-4S_\beta/3}$ , so by (4)

$$\log \frac{1}{\delta(\mathcal{E}_T)} \leq \log \varphi(Q) + \frac{4}{3}S_\beta \ll \mathcal{L}^3(\log \mathcal{L})^5.$$

$LS(C)$  gives a prime  $p > T$  avoiding  $\mathcal{E}_T$  with  $\log p \ll_C \mathcal{L}^3(\log \mathcal{L})^5$ . It is a square modulo 840, hence hard, and avoids every atom with  $M \leq T$ , so  $W(p) > T$  by Lemma 2.1. As  $p > T$ , letting  $T \rightarrow \infty$  gives infinitely many primes, and  $\mathcal{L} \gg (\log p)^{1/3}(\log \log p)^{-5/3}$ .  $\square$

More generally, LS with  $C(\cdot)$  replaced by an increasing function  $\Phi(\log T + \log(1/\delta))$  gives  $\log p \leq \Phi(C_0 \mathcal{L}^3(\log \mathcal{L})^5)$ ; for instance  $\Phi(\lambda) = \lambda^\theta$  gives the exponent  $1/(3\theta)$ . LS is used far below the sieve limit: the system has dimension  $\kappa \asymp \mathcal{L}^3/\log \mathcal{L}$  on the big primes, LS asks for  $\log x \asymp \log(1/\delta) \asymp \kappa \log \mathcal{L}$  (up to logarithms), while a lower-bound sieve needs  $\log x \gg \kappa \log T$ . This ratio  $\log T$  versus  $\log \log T$  per unit of dimension is the gap between  $1/4$  and  $1/3$ . By Proposition 12.4 and Corollary 12.3, LS cannot be derived through linear certificates from equidistribution hypotheses; it is a beyond-the-barrier hypothesis. The planted measures of §12 do not refute it: they are reweightings of Haar measure onto the complement of the avoiding set, not sequences of primes.

*Other strengths* [33, §§2,5]. (i) A Cramér form ( $\log p \leq \log(1/\delta) + A \log T$ ) implies  $LS(\max(1, A))$ ; a log-scale counting form  $\#\{T < p \leq x : p \in S(\mathcal{E})\} \geq c_0 \delta^C \pi(x)$  for  $\log x \geq C(\log T + \log(1/\delta))$  implies  $LS(C+1)$ . (ii) Main-term forms are too strong: in the integer analogue (events  $n \equiv 0 \pmod{\ell}$ ,  $\ell \leq z$ , at  $x = z^u$ ) the count is  $(e^\gamma \omega(u) + o(1))\delta x$  with Buchstab's function  $\omega$ , not  $\sim \delta x$  (proved, classical); for primes a uniform  $\kappa$ -tuple heuristic

predicts a compounding deficit  $(e^{\gamma\omega(u)})^\kappa$  (Assessment). (iii) LS holds, by the Brun–Titchmarsh inequality, for systems of total event mass at most  $(1 - \varepsilon)/2$  (proved); the ES system is very far from this case.

*Hypotheses for product sets do not suffice.* A product system has only prime-power moduli, so that  $S(\mathcal{E}) = \prod_\ell A_\ell$  is a CRT product with  $\delta = \prod_\ell |A_\ell|/\varphi(\ell^{e_\ell})$ , a singular series. Let  $\text{HL}_{\text{prod}}(C)$  be  $\text{LS}(C)$  restricted to product systems; it has the shape of a Hardy–Littlewood or Bateman–Horn statement, but is a least-prime hypothesis for growing systems, not a standard fixed-polynomial statement. Let  $F_T = \{n \in \mathbb{Z}^\times : W(n) > T\}$ .

**Proposition 13.2** (Product subsets of the avoiding set are thin [33, Prop. 4.1, Cor. 4.2]). (a) (Proved.) *There are product sets  $S \subseteq F_T$ , inside the hard classes, with  $\log(1/\delta(S)) \leq T^{1/2+o(1)}$ .* (b) (Proved modulo the Barban–Davenport–Halberstam theorem [8, Ch. 29]; ineffective.) *Every product set  $S \subseteq F_T$  has  $\log(1/\delta(S)) \gg T^{1/2}(\log T)^{-8}$ . Consequently  $\text{HL}_{\text{prod}}(C)$  implies  $W(p) \geq (\log p)^{2-o(1)}$  for infinitely many hard  $p$ , and no application of  $\text{HL}_{\text{prod}}$  to a product subset of  $F_T$  certifies more than  $W(p) \geq (\log p)^{2+o(1)}$ .*

*Proof of (a).* Fix  $\varepsilon > 0$  and  $y = T^{1/2+\varepsilon}$ . Take  $A_2 = \{1 \bmod 8\}$  and, for odd  $\ell \leq y$ , the quadratic residues (lifted to  $\ell^{e_\ell}$ ). An atom  $(M, D)$  all of whose primes are  $\leq y$  is avoided:  $(n/M) = 1$  for  $n \in S$  while  $(-4D/M) = -1$  (Lemma 2.2). Every other atom has exactly one prime  $\ell > y$ , to the first power; let  $B_\ell$  be the set of its classes  $-4D \bmod \ell$  and  $A_\ell = (\mathbb{Z}/\ell)^\times \setminus B_\ell$  for  $y < \ell \leq T$ . Then  $S$  avoids all atoms, and  $|B_\ell| \leq \sum_{v \leq T/\ell} \tau(A_{v\ell}^2) \leq T^{1/2-\varepsilon+o(1)} \leq \ell/2$ , so  $\log(1/\delta(S)) \leq \pi(y) \log 2 + O(1) + \sum_{\ell > y} 2|B_\ell|/(\ell-1) \leq T^{1/2+\varepsilon+o(1)}$ .  $\square$

Part (b) [33, Prop. 4.1(b)] uses only the atoms  $(\ell_1 \ell_2, q)$  with primes  $\ell_1 \equiv 1$ ,  $\ell_2 \equiv 3 \pmod{4}$  in  $(\sqrt{T}/2, \sqrt{T}]$  and a prime  $q \mid (\ell_1 \ell_2 + 1)/4$ : since  $S$  is a product, each such atom must be blocked at  $\ell_1$  or at  $\ell_2$ , which forces  $\gg T^{1/2}(\log T)^{-2}$  blocked classes per prime on average once the primes  $\ell$  are equidistributed modulo  $4q$  on average over  $q \leq \sqrt{T}(\log T)^{-6}$ . So the ES avoiding set, of density  $e^{-\mathcal{L}^{3+o(1)}}$ , exceeds its best product subset by a factor  $e^{T^{1/2+o(1)}}$ : hypotheses that see only prime-by-prime local structure are useless here, and the hypothesis must apply to non-product sifted sets, as LS does.

#### 14. SUMMARY OF STATUS

- Proved: Lemmas 2.1, 2.2, 2.3, 2.5, 2.8 (with the classical Lemma 2.7), 2.9, Lemma 3.2(B); the Janson-type inequality (Lemmas 4.1, 4.2, Theorem 4.3) and Lemmas 4.6, 4.7; the energy bound (Lemmas 7.2–7.4, Theorems 7.5 and 7.9, Corollaries 7.6 and 7.10, Remark 7.7, and Remark 7.8 except the cited switching-lemma constant); Lemmas 8.1, 8.2, 8.3, 8.4, 8.5, 8.6; the planting and level-barrier Lemmas 10.1, 10.2, and Lemmas 10.3, 10.5.

- Proved modulo the Nair–Tenenbaum theorem (Theorem 3.1): part (A) of Lemma 3.2 and the Haar upper bound Theorem 3.3.
- Proved modulo the fundamental lemma of sieve theory: Lemma 4.5 and the Haar lower bound Theorem 4.4. With the previous item: Theorem 1.3 (the Haar exponent is 3).
- Proved modulo Gallagher’s theorem (Theorems 5.2 and 5.1, quoted from the draft [26]): the linear transfer Theorem 6.1 and Lemma 10.4.
- Proved modulo Gallagher’s theorem and Theorem 3.1: Theorem 9.1 and Theorem 1.1.
- Proved modulo Gallagher’s theorem and the fundamental lemma: Theorem 10.6 (= Theorem 1.4). Proved implications: Corollary 10.7 (within its stated scope) and Corollary 10.8 (which also uses Theorem 3.1).
- Evidence, not used: the computer check of Lemma 2.2 for  $M \leq 2 \cdot 10^5$ ; the exact check of Theorem 4.3 on small instances; the Monte Carlo fit of  $\delta^*(T)$  for  $T \leq 32767$ .
- Assessment: Remarks 6.2, 9.2 and 10.9.
- Not claimed: anything about the Erdős–Straus conjecture; the heuristic exponent  $1/3$ ; anything about arguments outside the scope of Corollary 10.7; any numerical instance; priority for Theorem 7.5 or Theorem 4.3 (literature searches partial).
- The Elsholtz–Tao estimates, Håstad’s switching lemma and the Thorner–Zaman theorem are no longer used; they are cited only for comparison.

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